MEWRE RENEWABLE PROJECT MANAGER · TRIAL CRASH COURSE

From zero to your first week at MEWRE

29 short lessons in 4 parts, about 3 hours on paper. Electricity from first principles, the grid, Kuwait's power sector and its actors, solar, CSP and batteries, grid connection, project delivery, contracts, money and how the ministry works.

How to study on the plane

1. One part at a time (about 35 to 45 minutes each), then a 5 to 10 minute break: stand up, look away from the page.
2. For every lesson: read one section, **look away and say it out loud** in two or three sentences, then check. Do not copy text.
3. Read the drawing and its caption: the caption tells you what to notice. (On the website the same drawings are interactive.)
4. Answer the self-check questions **before** turning to the answers at the end of the part. A wrong answer now is what makes it stick.
5. Learn the "Say it at MEWRE" lines: they are what you will actually use in meetings.

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7. Earned value: are we on time and on budget? · 3 min · quick
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Part 4: Contracts, money, the ministry, and rehearsal · about 31 min

1. Contract strategies and the FIDIC books · 7 min
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3. IPPs and PPAs: when a private company builds and sells power · 6 min
4. Money over time: NPV, IRR and LCOE · 4 min · quick
5. How Kuwait tenders a project · 4 min · quick
6. How a ministry works, and writing for it · 6 min

Kuwait facts come from the course's source lessons and keep their confidence: H official, M trade press, L weak source; anything marked unverified must be checked before you quote it. Trial starts Sunday, 15 November 2026.

PART 1 OF 4

Electricity from zero to the grid

By the end you can: Say in your own words what voltage, current, DC and AC, coils, capacitors and reactance are, then explain reactive power and three-phase and why the grid cares.

8 lessons · about 54 minutes

Lesson 1 · Part 1 · full lesson · about 8 min

Voltage, current and resistance: pressure, flow and the narrow pipe

Why you care at MEWRE. Every drawing, datasheet and grid study you will see at MEWRE is written in volts, amps and ohms. Get these three words solid today and everything else this week (power, AC, reactive power, transformers) is built on top of them.

The picture first

Think of a closed loop of water pipe, completely full of water, with a pump in it. This is the one analogy we will use all day.

- The **water** is electric **charge**. It is already in the pipe everywhere; the pump does not create it, it just pushes it round.
- The **pressure difference** the pump makes between its outlet and its inlet is **voltage**.
- The **flow rate** (litres per second past any point) is **current**.
- A **narrow section of pipe** that makes it hard to push water through is **resistance**.

More pressure, more flow. A narrower pipe, less flow for the same pressure. That is the whole idea; the rest is units.

The water loop: what voltage, current, resistance, coils and capacitors do

Pump = voltage. Flow = current. Narrow pipe = resistance. Heavy wheel = coil. Rubber membrane = capacitor.

In the loop

Narrow pipe + Heavy wheel + Membrane

Pump pressure (voltage) 240 V

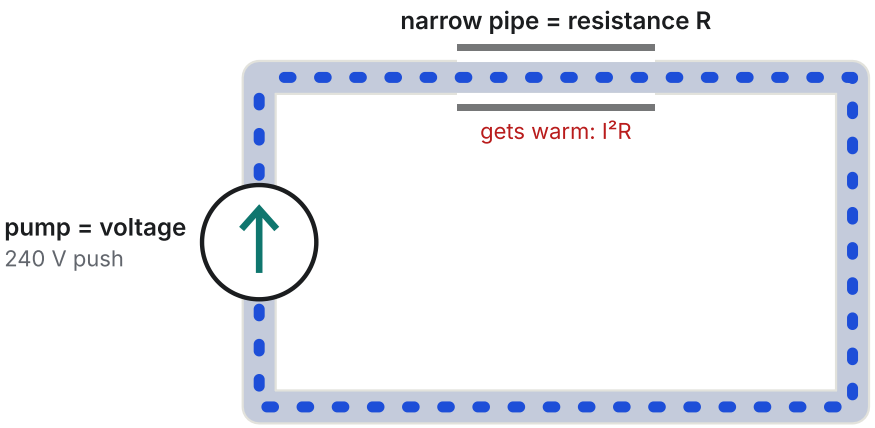
Narrowness (resistance) 24 Ω

Switch the pump off

Flow now (cu... 10 A

Steady flow V... 10 A

Heat in the pi... 2,400 W



flow over the last 6 s

Electric symbols: pump ↔ battery or generator, narrow pipe ↔ resistor (zig-zag), wheel ↔ coil (loops), membrane tank ↔ capacitor (two plates).

More pump pressure, more flow; a narrower pipe, less flow: I = V / R. The narrow section heats up (I²R), which is why cables have current ratings and why the grid moves power at high voltage, low current. Switch off and the flow stops at once. Where the picture breaks: real wires need a complete loop or nothing flows, and the electric push travels almost instantly even though the charges themselves crawl.

Raise the pump pressure and the flow speeds up; narrow the pipe and it slows. Watch the heat number: double the flow and the heat goes up four times.

What is actually happening

A copper wire is full of electrons that are free to move. A battery or a generator pushes them by creating a difference in electrical "pressure" between two points.

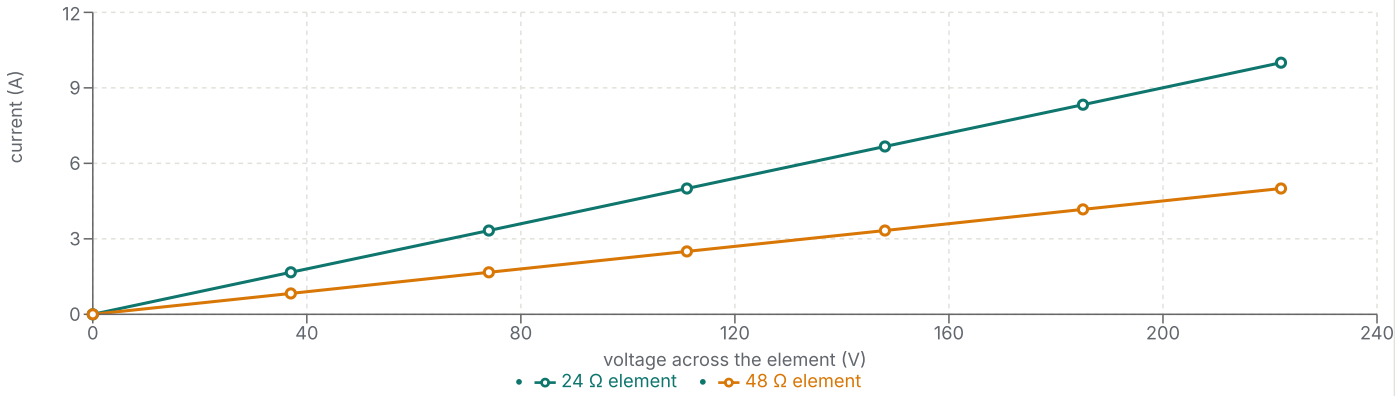
- **Charge** is the "stuff" that moves, measured in coulombs. In wires it is carried by electrons.
- **Voltage** (V, volts) is the push: the energy each unit of charge gains or loses between **two points**. A voltage is always *between* two things (phase and neutral, a line and earth, the two ends of a cable). "The voltage at a point" only means something against an agreed reference, usually earth. Its proper name is **potential difference**.
- **Current** (I, amperes or amps, A) is the flow: how much charge passes a point each second. It is measured *through* something, at one place in the circuit.
- **Resistance** (R, ohms, Ω) is how hard a material makes that flow. Copper and aluminium are low; air, glass and plastic insulation are extremely high.

The three are tied together by **Ohm's law**:

$$V = I \times R$$

Kuwaiti homes are supplied at **240 V** (the 415/240 V low-voltage system). A heater element of 24 Ω plugged into a 240 V outlet draws $I = V/R = 240/24 = 10$ A. A 48 Ω element on the same outlet draws only 5 A.

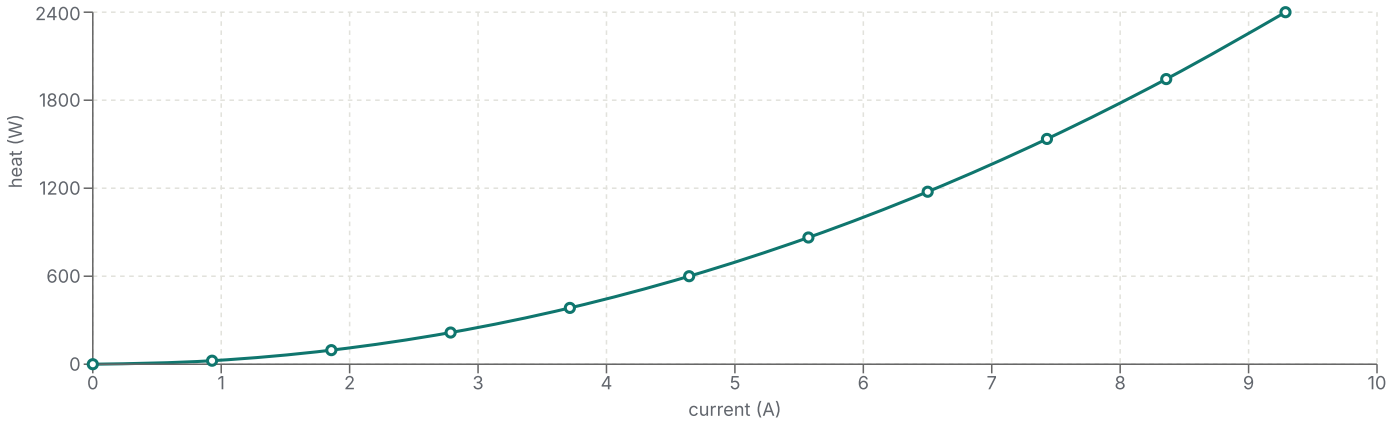
Current drawn from a household voltage by two heater elements



Both lines are straight: current rises in proportion to voltage. The lower resistance (wider pipe) gets twice the current at every voltage; at 240 V that is 10 A against 5 A.

Where the energy goes: heat. When current is pushed through a resistance, the electrical energy turns into heat. The heat rate is I^2R: double the current and the heat goes up **four** times. That is how a kettle works, and also why cables get warm, why cables have a maximum current rating, and why the grid sends power at very high voltage (for the same power, higher voltage means less current, so far less heat lost in the lines; more on this tomorrow).

Heat produced in the 24 Ω element as the current rises (I^2R)



Not a straight line but a curve that bends upward: 5 A gives 600 W, 10 A gives 2,400 W. Double the current, four times the heat; this curve is why cables have current limits.

Where the water picture breaks:

- **Cut a water pipe and water pours out. Cut a wire and the current simply stops.** Electricity needs a complete loop, out and back. No closed circuit, no current.
- **Current is not used up.** The same amps that go into a heater come out of it. What gets used is energy: the charge leaves at lower "pressure" than it arrived.
- **The electrons crawl** (well under a millimetre per second in a typical wire), but the push travels through the full pipe almost instantly, which is why a lamp lights the moment you flip the switch.

In one sentence: voltage is the push between two points, current is the flow that push causes, and resistance decides how much flow you get for a given push.

From the power station to your socket

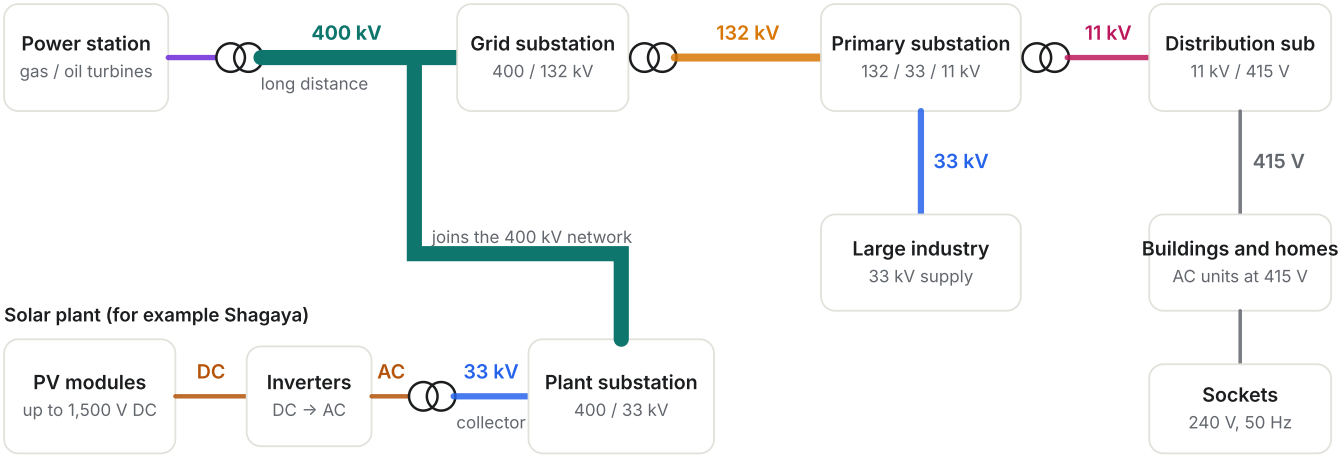
Because heat loss grows with current squared, the grid moves power at very high voltage and steps it down only near the people who use it. Every step happens in a **substation** through a **transformer** (the two-circle symbol below). Read the drawing left to right:

- **Power station**: the generator makes about 11 to 21 kV; a step-up transformer next to it raises that to transmission voltage.
- **400 kV transmission**: the backbone, over long distances. A 1,100 MW plant at 400 kV needs about 1,600 A per phase.
- **Grid substation, 400/132 kV**, then **132 kV** sub-transmission to the cities and big industry.
- **Primary substation, 132/33/11 kV**: 33 kV for districts and large customers, 11 kV along the streets.
- **Distribution substation, 11 kV/415 V**: the last step, in every neighbourhood. 415 V feeds air-conditioning units; one phase to neutral gives **240 V** at the socket.
- **A solar plant goes the other way**: DC at the modules, AC at a few hundred volts out of the inverters, **33 kV** across the site, then **33/400 kV** at the plant substation into the network.

Click any line or box: the panel shows what that level does and how much current the same power needs there.

The grid by voltage level: from power station to socket

Click any line or box. Thicker line = higher voltage. Power steps up to travel far, then steps down to be used.



Transmission (EHV): 400 kV Kuwait: confirmed

The backbone: carries bulk power over long distances between power stations and grid substations.

Confirmed for Kuwait (for example the Shagaya to Al-Wafra 400 kV line). Older EHV circuits are quoted as 300 or 275 kV; sources differ.

Power to deliver 100 MW

Current at 400 kV
144 A

Heat loss vs 400 kV, same wire
1×

Same power, different voltage: current is power divided by voltage, and the heat lost in a wire grows with current squared. That is why power leaves the station at 400 kV (100 MW needs about 144 A) and is only brought down to 240 V at the last few metres (the same 100 MW at 240 V would need over 400,000 A). Each step down happens in a substation through a transformer, the two-circle symbol. A solar plant does the reverse: low DC voltage at the modules, stepped up twice to reach the 400 kV network. Only 400 kV is confirmed for Kuwait in our sources; the other levels are commonly cited and worth confirming with MEWRE.

Move the power slider, then click 400 kV and 240 V: the same 100 MW needs about 144 A on the transmission line and over 400,000 A at socket voltage. That is the whole reason for the voltage ladder.

✳ Numbers you will hear

- **240 V** at a Kuwaiti socket and **415 V** between phases on the low-voltage network (Track 0, lesson 3). Grid levels: **400 kV** transmission (confirmed for Kuwait), with **132 kV** sub-transmission and **33 kV and 11 kV** distribution commonly cited (confirm with MEWRE; older EHV circuits are quoted as 300 or 275 kV). 1 kV = 1,000 V.
- Currents: a few amps for a laptop, **10 to 30 A** for an air conditioner or heater, hundreds to a few thousand amps in the cables and busbars of a PV plant or substation.
- Resistance of a long cable is small (fractions of an ohm), but with large currents the I^2R heat is large.
- **Triple the voltage, one third the current, one ninth the heat loss** for the same power through the same wire. 400 kV against 132 kV is almost exactly this.

Say it at MEWRE

- Between which two points is that voltage measured: line to line, or line to earth?
- What current does that cable carry at full output, and what is its rated current after derating for our site?
- Where is the heat going? That loss is I squared R, so a small increase in current costs a lot.
- In plain words: voltage is the push, current is the flow, resistance is what limits the flow.

The classic trap.

Treating voltage and current as the same thing, or saying current 'flows through' a voltage. Voltage is measured across two points and is the cause; current is measured through one point and is the effect, and how much current you get depends on the resistance in the loop. A high voltage with nothing connected drives no current at all.

Check yourself (answers at the end of Part 1)

Q1.

A heater element of 48 Ω is plugged into a 240 V Kuwaiti socket. What current does it draw?

Your answer: _____ A

Q2.

The current in a cable doubles while its resistance stays the same. What happens to the heat produced in the cable?

A. It doubles

B. It increases four times

C. It stays the same

D. It halves

Q3.

Why does the grid move bulk power at 400 kV rather than at, say, 33 kV?

A. For the same power, higher voltage means much less current, so far less I^2R heat lost in the lines

B. Higher voltage makes the electrons move faster, so power arrives sooner

C. Higher voltage increases the resistance of the line, which protects it

D. Customers need 400 kV at their sockets

Lesson 2 · Part 1 · full lesson · about 6 min

Power and energy: MW is speed, MWh is distance

Why you care at MEWRE. Plants are sized in MW, PPAs pay per MWh, and peak demand in Kuwait is quoted in GW. Mixing the two is the single most common mistake in briefings, and the first thing a senior will catch.

The picture first

A car. Its **speed** (km/h) is how fast it is going right now. The **distance** (km) is how far it has gone. Drive at 100 km/h for 2 hours and you have covered 200 km.

- **Power** is the speed: how fast energy is being delivered *now*. Measured in **watts** (W), kW, MW, GW.
- **Energy** is the distance: how much has been delivered *in total* over some time. Measured in **watt-hours** (Wh), kWh, MWh, GWh.

In the water picture: power is how much work the flowing water is doing each second (pressure times flow); energy is the total work done over the afternoon.

What is actually happening

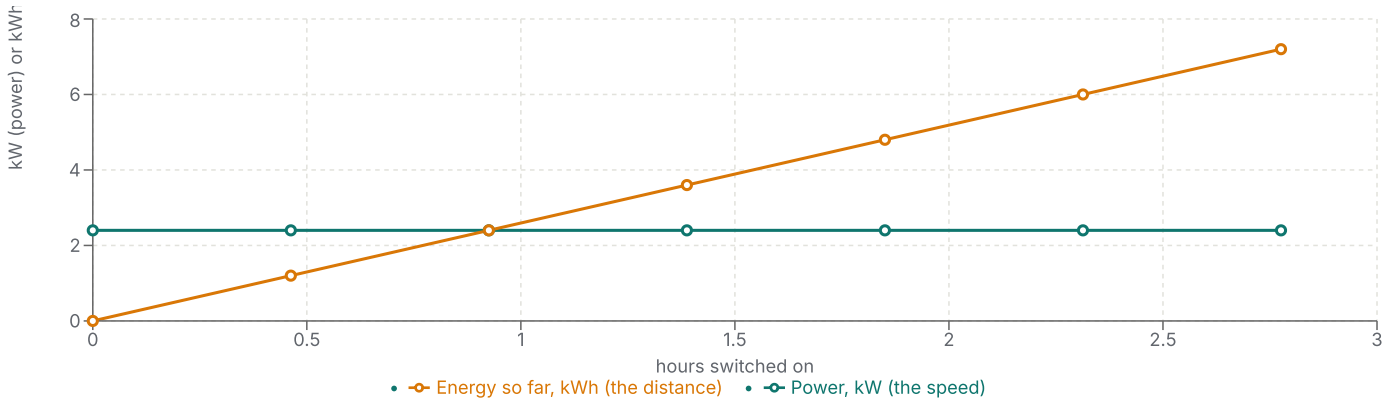
Power = voltage × current. Push (V) times flow (I) gives the rate at which energy is delivered:

$$P = V \times I$$

The 24 Ω heater from the last explainer, at 240 V and 10 A, uses $240 \times 10 = 2{,}400$ W $= 2.4$ kW. (Check: $I^2R = 10^2 \times 24 = 2{,}400$ W, the same answer.) For AC this simple product holds for heaters; motors and other "reactive" loads need one extra factor, which comes on Day 2.

Energy = power × time. Run that heater for 3 hours and it uses $2.4 \times 3 = 7.2$ kWh. That is what the electricity meter counts.

The 2.4 kW heater over 3 hours: power stays flat, energy piles up



Power is a level (2.4 kW the whole time, like a steady 100 km/h); energy is a running total that climbs to 7.2 kWh after 3 hours, like the odometer.

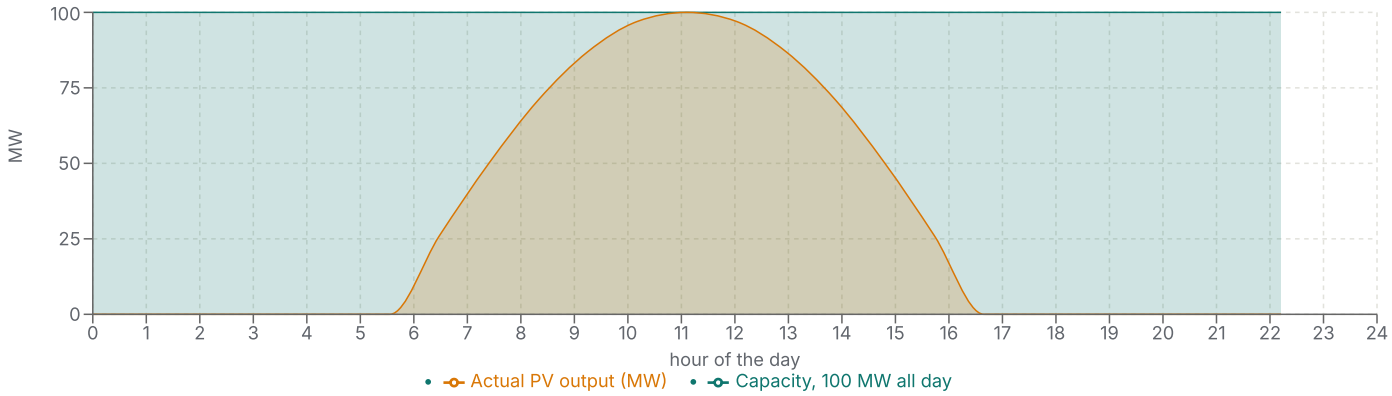
The prefixes go up in steps of 1,000:

Power (rate)	Energy (amount)
1 kW = 1,000 W	1 kWh = 1 kW for 1 hour
1 MW = 1,000 kW	1 MWh = 1,000 kWh
1 GW = 1,000 MW	1 GWh = 1,000 MWh; 1 TWh = 1,000 GWh

A 1 MW plant running flat out for 1 hour delivers 1 MWh. For 10 hours, 10 MWh.

Capacity vs energy. A plant's **capacity** (MW) is the most it can deliver at once: it decides the size of cables, transformers and the grid connection. Its **energy** (MWh per year) is what it actually produces: it decides revenue and how much fuel it saves. A PV plant has a big capacity but makes nothing at night, so its yearly energy is much less than capacity × 8,760 hours.

A 100 MW PV plant on an illustrative clear day: capacity is the box, energy is the hill



Illustrative shape, not measured data. The full box is 100 MW × 24 h = 2,400 MWh; the hill under the PV curve holds only about 760 MWh (about 32 % that day). The ratio of hill to box is the capacity factor.

Capacity factor links the two:

$$\mathrm{CF} = \frac{\text{energy actually delivered}}{\text{capacity} \times \text{hours}}$$

A 1,100 MW PV plant at a 26 % capacity factor produces $1{,}100 \times 8{,}760 \times 0.26 \approx 2{,}505{,}000$ MWh, about 2.5 TWh, per year.

In one sentence: power (MW) is how fast energy flows right now, energy (MWh) is how much has flowed over time, and capacity factor tells you how hard a plant's MW are actually used.

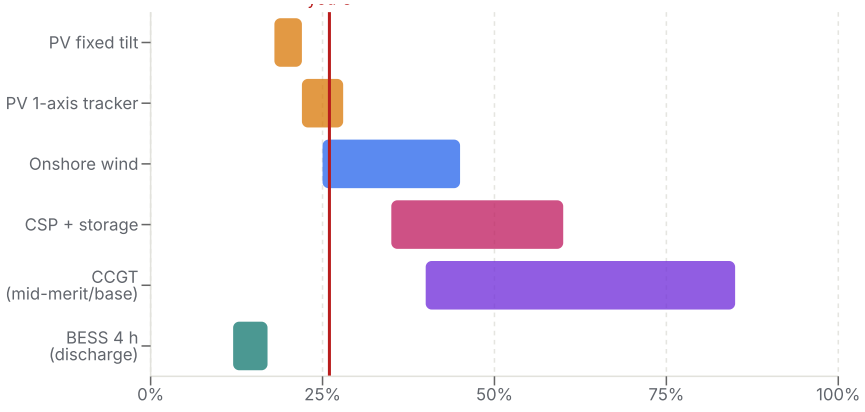
Capacity factor

CF = energy delivered ÷ (rated power × hours in period)

Rated capacity	1100	MW
Energy in period	2505360	MWh
Hours in period	8760	h

Year | Leap year | 30-day month

Capacity factor **26.0 %**	Full-load hours **2,278** h
Average output **286** MW	Specific yield **2,278** kWh/kW



What to notice: a PV plant's CF is limited by the sun, not by reliability — 25 % is excellent for PV but poor for gas. Compare capacity factors only within a technology; for firm supply look at when the energy arrives, not just how much.

Ranges are indicative global/regional values (AC basis for PV). Kuwait desert PV with trackers typically sits at the upper end of the PV range.

A 1,100 MW PV plant producing 2.5 TWh a year: a 26 % capacity factor, about 2,280 full-load hours. Compare it with the typical ranges for wind, CSP and gas.

Why it matters at MEWRE

- **Tariffs are per unit of energy** (fils/kWh, or per MWh) because energy is what the buyer receives and the fuel it replaces.
- **Plants, grid connections and peak demand are in MW** because cables, transformers and generators are limited by how much they carry *at once*.
- **Delay damages** in contracts may be per MW per day; **performance guarantees** are in MWh or %. Each line of a contract tells you which one it means; read the unit before the number.

🇰🇼 **Numbers you will hear**

- Kuwait's **summer peak demand is about 17.6 GW** (MEWRE figures: 17,640 MW in 2024, 17,610 MW in 2025, 17,540 MW in 2026; Track 0 lesson 5). That is power, at one moment on a hot afternoon.
- **Shagaya Phase III Zone 1 is about 1,100 MW (AC)**; at a 26 % capacity factor (the Track 0 worked example) that is about **2.5 TWh per year**.
- Typical capacity factors (Track 0 lesson 5): **PV in Kuwait about 19 to 32 %** (fixed tilt lower, trackers higher, AC basis), **CSP with storage 35 to 50 %**, **efficient gas combined cycle 50 to 85 %**.
- The citizen tariff is **2 fils/kWh** (Track 0 lesson 1): that is a price per unit of energy.
- One year has **8,760 hours**.

Say it at MEWRE

- Is that figure MW or MWh? Capacity or annual energy?
- What capacity factor does that energy estimate assume, and on AC or DC basis?
- The tariff pays for MWh delivered, but the grid connection must be sized for the MW peak.
- Over a year, how many full-load hours is that? It should be roughly capacity factor times 8,760.

The classic trap.

Saying 'MW per hour' or 'the plant produces 1,100 MW a year'. MW is already a rate, like km/h; it does not need 'per hour'. A 1,100 MW plant produces about 2.5 million MWh a year at a 26 % capacity factor, not 1,100 MW. (The only valid use of 'MW per minute' is a ramp rate: how quickly output changes.)

Check yourself (answers at the end of Part 1)

Q1.

A 50 MW battery discharges at full power for 4 hours. How much energy does it deliver?

Your answer: _____ MWh

Q2.

A 100 MW PV plant delivers 219,000 MWh in a year (8,760 hours). What is its capacity factor?

Your answer: _____ %

Q3.

Which quantity should set the rating of the transformer that connects a PV plant to the grid?

A. The plant's maximum output in MW (more exactly MVA), the most it carries at once

B. The plant's annual energy in MWh

C. The tariff in fils/kWh

D. The capacity factor

Lesson 3 · Part 1 · full lesson · about 7 min

DC and AC: one-way flow and back-and-forth

Why you care at MEWRE. Solar modules and batteries make DC; the grid runs on AC. Every PV and BESS plant MEWRE buys is, at its heart, a machine for turning one into the other, and the plant's size is quoted on both sides.

The picture first

Back to the water loop.

- **DC (direct current)**: the pump pushes steadily in one direction. Water goes round and round the same way, at a steady rate.
- **AC (alternating current)**: the pump is a piston that pushes, then pulls, then pushes again. The water sloshes forward and back in the pipe and never really "goes round" at all.

Can sloshing water do useful work? Yes. Put a paddle in the pipe that gets warm from friction, or a mechanism that is driven both ways: work is done on every stroke, forward and back. A heater does not care which way the current goes.

What is actually happening

DC has a fixed polarity: one terminal is always positive. Batteries, phone chargers and **PV modules** produce DC. A single module gives a few tens of volts; modules connected in series ("strings") reach up to 1,500 V DC in a utility plant.

AC reverses direction regularly. In Kuwait it does so at **50 Hz**: 50 complete cycles every second, so one cycle takes **20 ms**, and the current changes direction 100 times a second. The voltage follows a smooth **sine wave**, which is what a spinning generator naturally makes.

Why the grid uses AC: transformers. A transformer can step AC voltage up or down with no moving parts and over 99 % efficiency, but it only works with a voltage that keeps changing (the next explainer shows why). That lets the grid generate at a convenient voltage, step up to 132 or 400 kV to send power far with little loss, then step down to 33 kV, 11 kV and finally 415/240 V for homes. Doing the same with DC needs power electronics, which is why DC is used only for special links (very long or subsea connections, called HVDC).

Inverters are the bridge. A PV plant's modules make DC; its **inverters** chop and shape that into 50 Hz AC synchronised with the grid. A battery plant does the same in both directions (its converter is often called a PCS, power conversion system). That is why a PV plant has two sizes: **MWp on the DC side** (modules) and **MW on the AC side** (inverters and grid connection).

Where DC ends and AC begins in a solar and battery plant

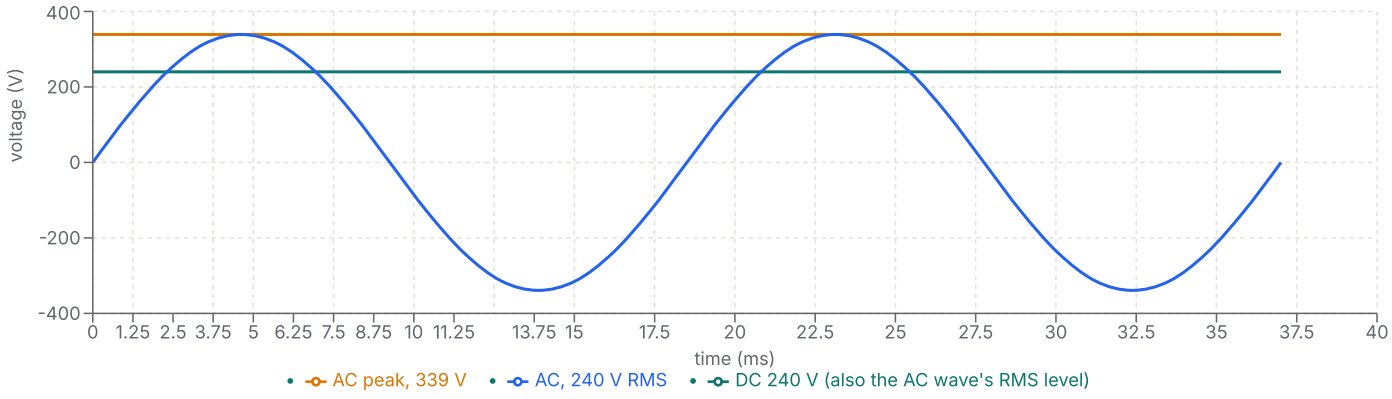
1. **1 PV modules: DC**
   A few tens of volts each; strings in series reach up to 1,500 V DC. Rated in MWp
2. **2 Inverter: DC to AC**
   Chops and shapes DC into 50 Hz AC in step with the grid. Rated in MW (AC)
3. **3 Step-up transformer: AC**
   A few hundred volts AC raised to 33 kV across the site
4. **4 Plant substation and grid: AC**
   33 kV raised to 400 kV; the grid is AC from here to every socket
5. **5 Battery: DC, through a two-way converter (PCS)**
   Charges from the AC grid (AC to DC) and discharges back into it (DC to AC)

DC lives only in the modules and the battery cells. Everything after the inverter or PCS is AC: the converter is the border post, and the plant has one size on each side of it.

RMS: one number for a wave. The AC voltage is constantly changing, so which number is "240 V"? It is the **RMS** (root mean square) value: the DC voltage that would heat a kettle exactly as much. The wave actually swings higher: its **peak** is $\sqrt{2} \approx 1.41$ times the RMS, so a 240 V socket peaks at $240 \times 1.414 \approx 339$ V, then swings to −339 V. Every voltage and current you will be quoted (400 kV, 33 kV, 1,500 A) is RMS unless it says otherwise.

In one sentence: DC pushes one way all the time, AC pushes back and forth 50 times a second, and the grid uses AC because transformers can easily change its voltage.

240 V DC against 240 V RMS AC at 50 Hz (two cycles, 40 ms)



One AC cycle every 20 ms. The top flat line marks the 339 V peak the wave touches; the 240 V line is both the DC supply and the AC wave's RMS level. The wave crosses zero 100 times a second yet heats a kettle exactly as much as that flat 240 V line: that is what 240 V RMS means.

Why it matters at MEWRE

- **Two sizes, one plant.** A tracker plant may have about 1,430 MWp of modules behind 1,100 MW of inverter and grid capacity (a DC/AC ratio of about 1.3; Track 0 lesson 5). PPAs and grid connections are in **AC MW**; module purchases and yield models are in **DC MWp**. Always ask which.
- **Inverters are where the plant meets the grid.** Grid-code behaviour (voltage support, frequency response, fault ride-through) is mostly inverter software and rating. Batteries connect through the same kind of converter.

- **Insulation is designed for peaks, heating for RMS.** A 240 V cable's insulation must survive 339 V every cycle, plus surges.

DC/AC ratio and inverter clipping

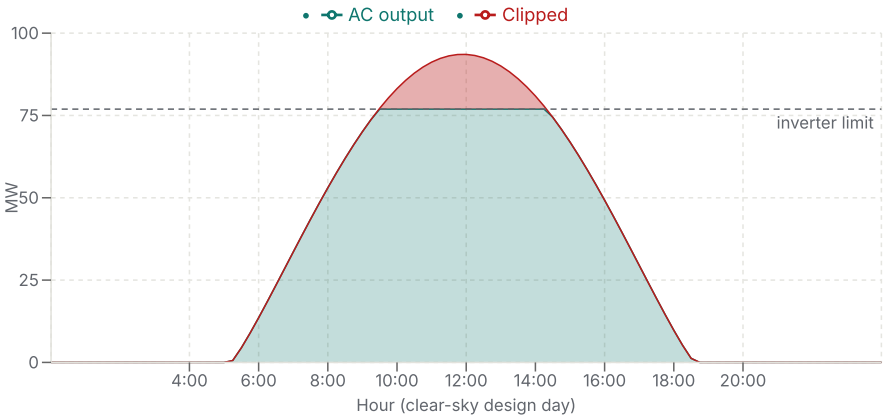
100 MWp DC array; inverter AC rating = DC ÷ ratio

DC/AC ratio 1.30

Inverter + MV cost 60 USD/kWac

Energy value 17 USD/MWh

AC capacity **76.9** MWac	Clipping (clear day) **7.0 %**
Clipping (annual e... **3.7 %**	Clipped energy **8,590** MWh/yr
Lost revenue **146** kUSD/yr	Inverter capex sav... **1.38** USD m vs ratio 1.00



What to notice: clipping only shaves the top of clear, cool middays, so a ratio of 1.2–1.35 loses little energy while cutting inverter, transformer and grid-connection capacity. Beyond ~1.4 the clipped energy grows quickly.

Annual estimate uses 12 representative monthly days scaled from indicative Kuwait GHI; a full 8760-h simulation (PVsyst) is needed for design.

100 MWp of DC modules behind about 77 MW of AC inverters (ratio 1.3): the DC curve would rise above the AC limit, so the inverters shave (clip) only the top of clear middays. Slide the ratio to see why one plant is quoted with two sizes.

✳ Numbers you will hear

- **50 Hz** in Kuwait (one cycle = 20 ms). Some countries, such as the USA, use 60 Hz.
- **RMS × 1.414 = peak.** 240 V RMS peaks at 339 V.
- PV strings: **up to 1,500 V DC**. Inverter AC output is typically a few hundred volts, then a transformer raises it to **33 kV** inside the plant.
- **DC/AC ratio** of a PV plant: typically **1.2 to 1.4** (MWp of modules per MW of inverter).
- Kuwait's chain: generation → **400 kV** transmission (confirmed) → **132 kV** → **33 / 11 kV** distribution → **415 / 240 V** to customers. Only 400 kV is confirmed in our sources; the lower levels are commonly cited (Track 0 m06 lesson 1).

Say it at MEWRE

- Is that capacity in MWp DC or in MW AC at the point of connection?
- What is the DC/AC ratio, and how much clipping does the yield model assume at the inverters?
- Is that voltage RMS or peak? Insulation and arresters are checked against peaks.
- In plain words: DC pushes one way, AC sloshes back and forth fifty times a second, and inverters turn the first into the second.

The classic trap.

Thinking AC does no work because 'it just goes back and forth' and averages to zero. The current averages to zero, but the heating and the power do not: energy is delivered on every half-cycle, whichever way the current flows. That is exactly why we quote RMS values, which measure the heating effect.

Check yourself (answers at the end of Part 1)

Q1.

A socket supplies 240 V RMS. What is the peak voltage of the wave, to the nearest volt?

Your answer: _____ V

Q2.

What is the main reason power grids use AC rather than DC?

A. Transformers can step AC voltage up and down simply and efficiently

B. AC carries more energy per amp than DC

C. PV modules and batteries produce AC

D. AC cannot cause electric shocks

Q3.

A PV plant is described as "1,430 MWp and 1,100 MW". What do the two numbers mean?

A. 1,430 MWp is the DC module capacity; 1,100 MW is the AC capacity of the inverters and grid connection

B. 1,430 MWp is peak summer output; 1,100 MW is winter output

C. 1,430 MWp is the AC capacity; 1,100 MW is after transformer losses

D. They are the same quantity measured by two different consultants

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Lesson 4 · Part 1 · full lesson · about 8 min

Magnetism and coils: current that resists change

Why you care at MEWRE. Generators, transformers, motors and overhead lines all work through magnetic fields, and Kuwait's summer load is millions of air-conditioning motors. This one idea explains why current 'lags', which is half of reactive power.

The picture first

Put a **heavy water wheel** in our pipe loop, one that the water must turn as it flows past.

- When the pump starts, the wheel is hard to get moving: the flow builds up **slowly**, not instantly.
- When the pump stops, the spinning wheel **keeps pushing** water round for a while.
- Once the flow is steady, the wheel just spins along and barely gets in the way.

The wheel does not resist flow. It resists **changes** in flow. It soaks up energy while speeding up and gives it back while slowing down. A coil of wire, an **inductor**, behaves exactly like this towards current.

The water loop: what voltage, current, resistance, coils and capacitors do

Pump = voltage. Flow = current. Narrow pipe = resistance. Heavy wheel = coil. Rubber membrane = capacitor.

In the loop

Narrow pipe + Heavy wheel + Membrane

Pump pressure (voltage) 240 V

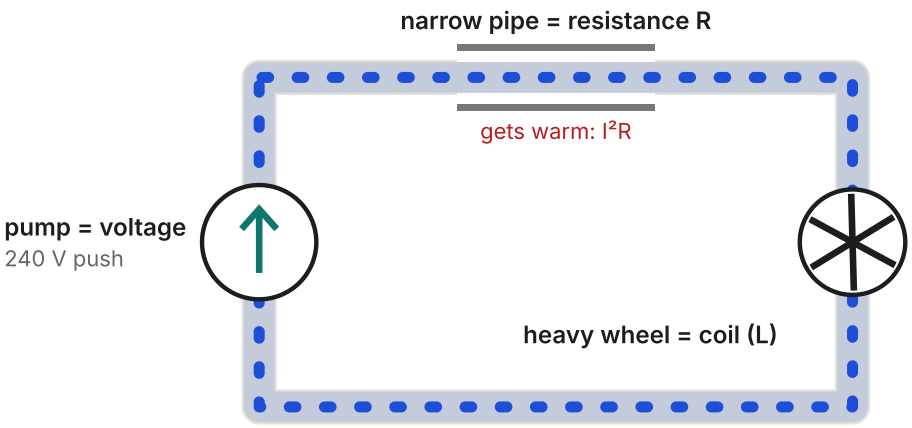
Narrowness (resistance) 24 Ω

Switch the pump off

Flow now (cu... 10 A

Steady flow V... 10 A

Heat in the pi... 2,400 W



flow over the last 6 s

Electric symbols: pump ↔ battery or generator, narrow pipe ↔ resistor (zig-zag), wheel ↔ coil (loops), membrane tank ↔ capacitor (two plates).

Switch on: the heavy wheel has to be spun up, so the flow builds slowly instead of jumping. Switch off: the spinning wheel keeps pushing water round for a while. A coil resists changes in current, not current itself. In AC the pump reverses 50 times a second, the wheel is always being pushed against its motion, and the current ends up lagging the voltage. Where the picture breaks: real wires need a complete loop or nothing flows, and the electric push travels almost instantly even though the charges themselves crawl.

Switch the pump off and on and watch the trace: with the heavy wheel the flow builds and dies away slowly instead of jumping. That delay is inductance.

What is actually happening

Two facts of nature, both discovered in the 1800s, run the entire power system:

1. **Current makes a magnetic field.** Any wire carrying current is surrounded by a magnetic field. Wind the wire into a coil, especially around an iron core, and the field becomes strong and concentrated.
2. **A changing magnetic field makes a voltage** (Faraday's law of **induction**). If the magnetic field through a coil changes, a voltage appears across the coil. A steady field does nothing; only change counts.

The two facts as one chain of cause and effect

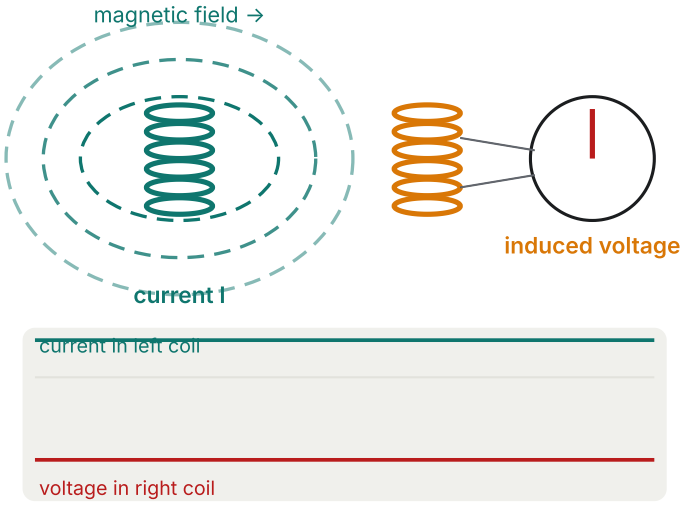
1. **1 Current flows in a wire or coil**
   Steady or changing
2. → **2 A magnetic field forms around it**
   Fact 1. An iron core concentrates it
3. → **3 The current changes (AC, switching on or off)**
   So the field grows, shrinks or reverses
4. → **4 A voltage is induced**
   Fact 2 (Faraday). In a second coil: a transformer. In the same coil: self-inductance
5. → **5 No change, no voltage**
   Steady DC gives a steady field and induces nothing

Read it left to right: current makes field, changing field makes voltage. Generators, transformers, motors and the lag of a coil's current all come from this one chain.

Magnetism and induction: the idea behind generators, motors and transformers

Left coil: current makes a magnetic field. Right coil: a voltage appears only while that field is changing

Current in the left coil: Steady DC | Switch on/off | AC (slow motion) — Switch off



Steady DC: a strong field, but the meter stays at zero, because only a changing field induces voltage. Switch on or off: a kick on the meter while the current changes, then nothing. AC: the current never stops changing, so the meter swings all the time, a quarter cycle out of step with the current. A generator makes the change by spinning a magnet, a transformer by feeding AC into one coil, and a coil's own induced voltage is what makes it resist changes in current (inductance).

Try steady DC first: a strong field, no voltage on the right. Then switch on and off, then AC: the meter moves only while the field is changing.

Everything follows from these:

- **Generator**: a spinning magnet (driven by a turbine) sweeps its field past coils, so the field through them keeps changing and a voltage is induced. One turn of a simple two-pole machine gives one AC cycle, so it turns at 3,000 rpm to make 50 Hz.
- **Transformer**: AC in the first coil makes a constantly changing field in an iron core; that changing field induces a voltage in a second coil. More turns on the second coil, higher voltage. No moving parts, but it **needs change**,

which is why it works with AC and not with steady DC.

- **Motor**: the reverse of a generator. Current in a magnetic field feels a force, and the shaft turns.

Magnetism at every step from power station to air conditioner

1. **1 Generator** — Turbine spins a magnet inside coils; the changing field induces 50 Hz AC →
2. **2 Step-up transformer** — Changing field in an iron core induces a higher voltage (to 132 or 400 kV) →
3. **3 Overhead line** — Current makes a field around the conductors: the line behaves like a long coil →
4. **4 Step-down transformers** — Same principle in reverse: 33 kV, 11 kV, then 415/240 V →
5. **5 AC compressor motor** — Current in a magnetic field makes force and turns the shaft; current lags voltage

Every box depends on a magnetic field, and every one of them absorbs reactive power. That is why the grid as a whole draws lagging current.

Inside a transformer: two coils on one iron core

AC in the first coil makes a changing magnetic flux; the flux passes through the second coil and induces a voltage there

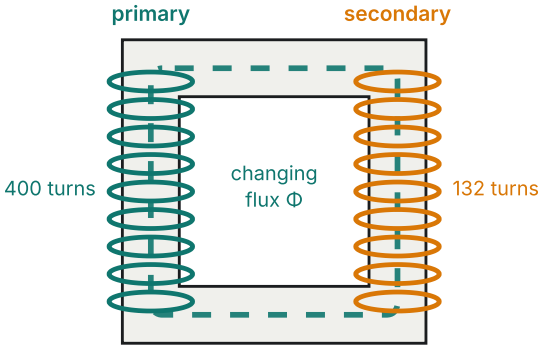
Transformer

400/132 kV | 400/33 kV | 11 kV/415 V

Power through it 500 MVA

Supply: AC (50 Hz) | Steady DC

Primary voltage	Secondary voltage
400 kV	132 kV
Primary current	Secondary current
722 A	2.19 kA



voltage ratio = turns ratio = 3.03 : 1

Voltage goes down by the turns ratio and current goes up by the same ratio, so power in equals power out (minus well under 1 % loss in a big unit). For example, on a 300 MVA, 400/33 kV unit the 33 kV side carries over 5,000 A while the 400 kV side carries about 430 A. Switch to steady DC: the flux stops changing and the secondary goes dead, which is why the grid is AC. Turns shown are illustrative; real windings use the same ratio with far more turns.

Two coils, one iron core, no moving parts. Change the preset and watch voltage fall by the turns ratio while current rises by the same ratio. Then try steady DC.

Inductance. A coil's own changing field induces a voltage in the coil itself that **opposes the change in current**. Try to increase the current and the coil pushes back; try to cut it and the coil pushes to keep it going (which is why switching off a big motor or transformer can draw a spark, and why breakers must handle arcs). The coil stores energy in its magnetic field and returns it. This property is **inductance** (L, henries, H).

In AC the current lags. AC is changing all the time, so a coil is pushing back all the time. The result: the current reaches its peak **after** the voltage. In a pure coil it lags by a quarter of a cycle (90°, 5 ms at 50 Hz). Energy flows into the field for a quarter-cycle and back out the next, so a pure coil uses **no** energy on average. Real motors are part resistance (useful work) and part coil, so their current lags by less, typically 25 to 40°.

Why reactive power exists: voltage, current and power over time

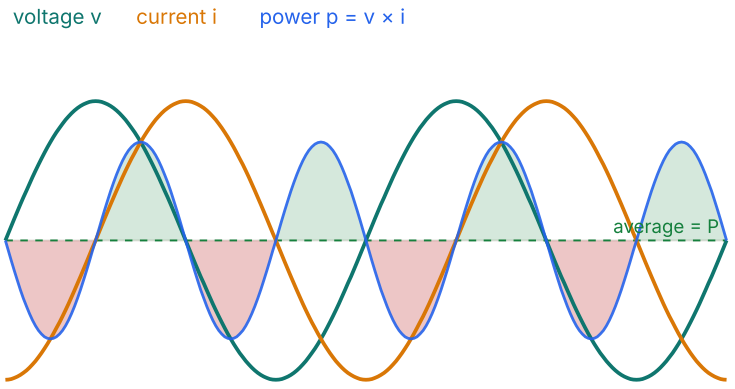
Same voltage and same current size; only the timing of the current changes

Load

Heater | AC compressor | **Pure coil** | Capacitor bank

Current lags voltage by 90 °

Real power P (avg) **0** pu	Reactive power Q **1** pu
Apparent power S **1.00** pu	Power factor **0.00**
Energy back to grid **100** % of forward	Current for same P ∞ × unity-PF



2 cycles = 40 ms at 50 Hz

The current is lagging (inductive: motors, transformers). Green area is energy delivered to the load, red is energy handed back to the grid in the same cycle. With a heater there is no red: every joule does work. With a coil (90°) green and red cancel: the current is full size but the average power is zero, so the cables carry current, get hot and lose voltage while delivering nothing. Utility loads sit in between: at power factor 0.8 the grid must push 25 % more current for the same useful power. That extra current, not the returned energy, is what reactive power costs: transformer and cable capacity, losses and voltage drop.

A pure coil: current peaks a quarter-cycle after voltage. Power (p = v × i) is positive half the time and negative half the time, so the net energy delivered is zero. Then pick 'AC compressor' to see a real motor.

Memory trick, first half: **ELI**. In an inductor (L), voltage (E) comes before current (I).

Where the wheel breaks: a coil's energy is stored in a magnetic field around it, not in moving mass, and that field reaches out into space. Another coil nearby can pick it up, which is the whole basis of the transformer. A water wheel has no equivalent.

In one sentence: current makes a magnetic field, a changing field makes a voltage, and a coil uses this to resist any change in current, so in AC its current lags behind its voltage.

✳ **Numbers you will hear**

- **Motor power factor** about **0.8 to 0.9 lagging** (current lags by roughly 25 to 37°).
- **Transformers**: 400/132 kV, 132/33 kV, 33/11 kV, 11 kV/415 V in the grid; 33 kV/(a few hundred volts) at each PV inverter station. Large transformers are about **99.5 to 99.8 % efficient** (Track 0 lesson 4).
- **Generators**: a two-pole machine turns at **3,000 rpm** for 50 Hz; a four-pole machine at 1,500 rpm.
- Lag in time at 50 Hz: **90° = 5 ms**, a quarter of the 20 ms cycle.

Say it at MEWRE

- The compressors are inductive, so they draw lagging current and absorb reactive power.
- Is that transformer rating at our site ambient, given it will run in 50 °C summers?
- In plain words: a coil resists changes in current, like a heavy wheel resists changes in speed.
- Who supplies the magnetising reactive power for all these motors and transformers at peak?

The classic trap.

Thinking a coil 'uses up' power the way a resistor does. A pure coil stores energy in its field and hands it back every half-cycle, so it uses no energy on average. What it does cost is extra current, which heats cables and transformers and has to be supplied by someone.

Check yourself (answers at the end of Part 1)

Q1.

Why does a transformer need AC to work?

A. It works by induction, which needs a magnetic field that keeps changing, and only a changing current makes one

B. DC is too low a voltage for transformers

C. The iron core would rust with DC

D. DC flows only in the second winding

Q2.

In a pure coil connected to a 50 Hz supply, how does the current relate to the voltage?

A. Current lags voltage by a quarter-cycle (90°)

B. Current leads voltage by a quarter-cycle (90°)

C. Current is exactly in step with voltage

D. There is no current at all

Q3.

On a summer afternoon most of Kuwait's load is air-conditioning compressors. What does that mean for the current the grid supplies?

A. It lags the voltage, so the grid must supply reactive power as well as real power

B. It leads the voltage, so the load supplies reactive power to the grid

C. It is in phase, because motors are resistive

D. It is DC, because compressors run at constant speed

Lesson 5 · Part 1 · full lesson · about 7 min

Capacitors: storing charge and resisting changes in voltage

Why you care at MEWRE. Capacitor banks correct power factor in substations, long underground cables behave like giant capacitors, and every inverter has a bank of them inside. Capacitors are the mirror image of coils, and together they make up reactive power.

The picture first

Stretch a **rubber membrane** across our pipe, sealing it completely.

- Start the pump: water flows for a moment on both sides as the membrane **stretches**. No water crosses it, but water moves.
- As it stretches, it pushes back harder, until the flow stops. The stretched membrane now **holds pressure**.
- Now run the pump back and forth (AC): the membrane flexes one way, then the other, and water keeps sloshing on both sides. **Steady flow is blocked; back-and-forth flow gets through.**

A coil (the heavy wheel) resists changes in **flow**. The membrane resists changes in **pressure**: you cannot change the pressure across it instantly, because it must stretch first. That is a **capacitor**, and pressure is voltage.

The water loop: what voltage, current, resistance, coils and capacitors do

Pump = voltage. Flow = current. Narrow pipe = resistance. Heavy wheel = coil. Rubber membrane = capacitor.

In the loop

Narrow pipe + Heavy wheel + Membrane

Pump pressure (voltage) 240 V

Narrowness (resistance) 24 Ω

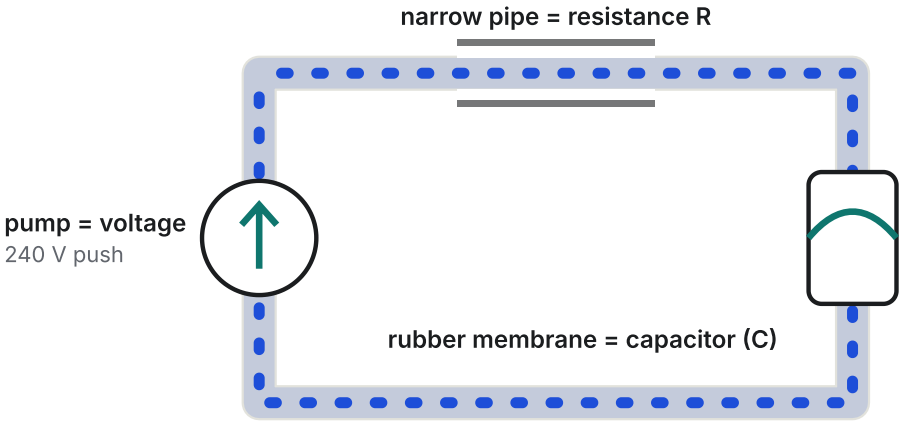
Switch the pump off

Flow now (cu... 0.000006 A

Steady flow V... 0 (blocked)

Heat in the pi... 0 W

Membrane pr... 240 V



flow over the last 6 s

Electric symbols: pump ↔ battery or generator, narrow pipe ↔ resistor (zig-zag), wheel ↔ coil (loops), membrane tank ↔ capacitor (two plates).

Switch on: water rushes in and stretches the membrane, then the flow stops because the stretched membrane pushes back as hard as the pump. No steady flow passes a capacitor: it blocks DC. Switch off and the membrane pushes the water back the other way. In AC the pump keeps reversing, the membrane keeps filling and emptying, and the current leads the voltage. Where the picture breaks: real wires need a complete loop or nothing flows, and the electric push travels almost instantly even though the charges themselves crawl.

Switch the pump on: a rush of flow that dies away as the membrane stretches. Switch it off: the membrane pushes the water back. No steady flow ever passes: a capacitor blocks DC.

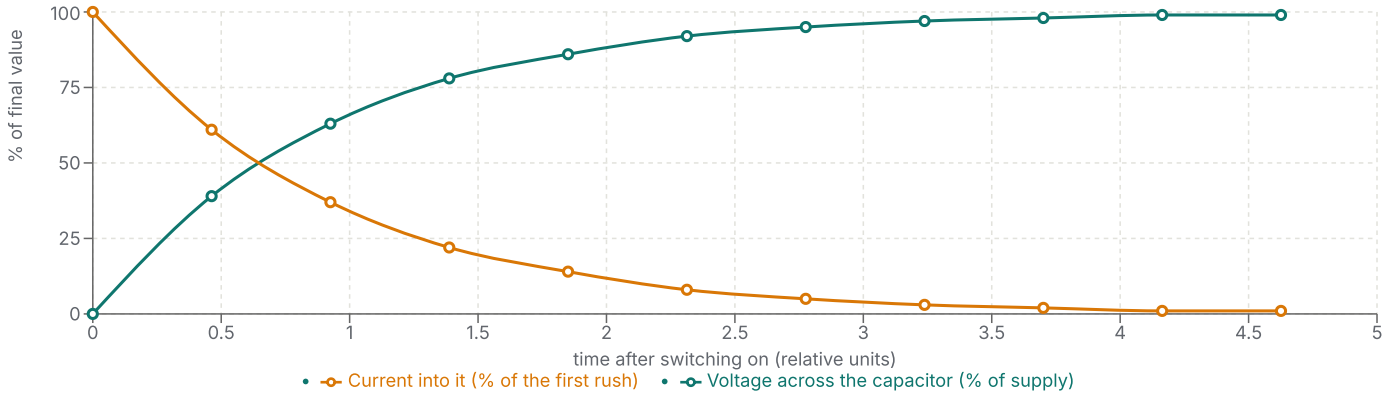
What is actually happening

A capacitor is **two conducting plates separated by an insulator**. Put a voltage across it and electrons pile up on one plate and leave the other. The plates now hold opposite charges and an **electric field** fills the gap between them. Energy is stored in that field.

- **Capacitance** (C, farads, F) is how much charge the plates hold per volt. Bigger plates, closer together, more capacitance.
- The voltage across a capacitor **cannot jump**: it changes only as charge flows in or out. So a capacitor **resists changes in voltage**, the mirror image of a coil resisting changes in current.

- **DC is blocked** once the plates are charged. **AC passes**, because the plates keep charging one way, then the other.

Switching a capacitor onto a steady (DC) voltage through a resistor



The voltage climbs smoothly from zero: it cannot jump. The current does the opposite: a full rush at the first instant that dies away to nothing as the plates fill. Current first, voltage after: the seed of 'current leads'.

In AC the current leads. Current flows most when the voltage is changing fastest (as it crosses zero) and stops when the voltage peaks (the plates are full). So the current peaks **before** the voltage, by a quarter-cycle (90°) in a pure capacitor. Like a coil, it borrows energy for a quarter-cycle and returns it the next, using none on average.

Why reactive power exists: voltage, current and power over time

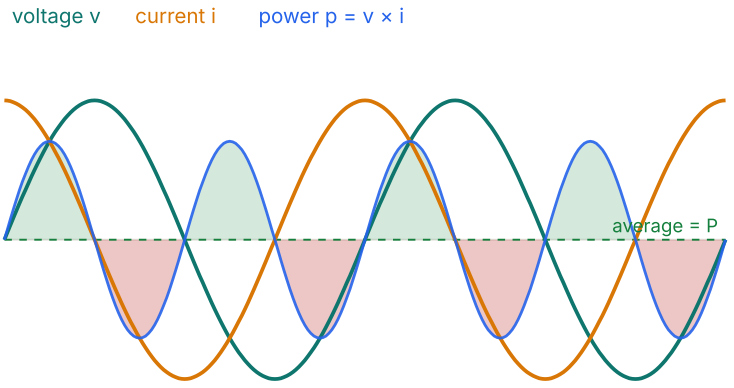
Same voltage and same current size; only the timing of the current changes

Load

Heater | AC compressor | Pure coil | Capacitor bank

Current lags voltage by -90 °

Real power P (avg)	Reactive power Q
0 pu	−1 pu
Apparent power S	Power factor
1.00 pu	0.00
Energy back to grid	Current for same P
100 % of forward	∞ × unity-PF



2 cycles = 40 ms at 50 Hz

The current is leading (capacitive: cables, capacitor banks). Green area is energy delivered to the load, red is energy handed back to the grid in the same cycle. With a heater there is no red: every joule does work. With a coil (90°) green and red cancel: the current is full size but the average power is zero, so the cables carry current, get hot and lose voltage while delivering nothing. Utility loads sit in between: at power factor 0.8 the grid must push 25 % more current for the same useful power. That extra current, not the returned energy, is what reactive power costs: transformer and cable capacity, losses and voltage drop.

A pure capacitor: current peaks a quarter-cycle before voltage. Compare with the coil (pick 'Pure coil'): the same shape shifted the other way. That is why a capacitor bank can cancel a motor's lagging current.

Memory trick, second half: **ICE**. In a capacitor (C), current (I) comes before voltage (E). Together: **ELI the ICE man**.

The coil and the capacitor side by side, as mirror images:

	Coil (inductor, L)	Capacitor (C)
Water picture	heavy wheel	rubber membrane
Resists sudden change in	current	voltage
Stores energy in	a magnetic field	an electric field
On steady DC	after the start, just a piece of wire	after charging, blocks completely
In AC, current	lags voltage by 90° (ELI)	leads voltage by 90° (ICE)
Energy used on average	none	none

Where capacitors show up on the job:

- **Capacitor banks** in substations and factories. Their leading current cancels the lagging current of motors, so the grid carries less current. This is **power-factor correction**.
- **Underground cables**: the conductor and the earthed metal sheath around it are two "plates" with insulation between, kilometres long. A long cable is a big capacitor; even with no load it draws **charging current** and generates reactive power. Large PV plants with many kilometres of 33 kV cable see this at night.
- **Inverters**: a **DC link** capacitor bank inside every PV inverter and battery converter smooths the DC voltage while the electronics chop it into AC.

How a capacitor bank corrects power factor

1. **Motors draw lagging current** — Part of their current only builds magnetic fields
2. → **Capacitor bank draws leading current** — Installed next to the motors, at the same substation or factory
3. → **The two cancel locally** — Energy sloshes between bank and motors instead of across the network
4. → **Network carries less current** — Same real power, lower current: less I^2R heat and less voltage drop

The bank does not add energy; it supplies the motors' back-and-forth need on the spot, so the cables upstream are freed for useful current.

Where the membrane breaks: it suggests a capacitor "fills up" with extra charge. It does not. As much charge leaves one plate as arrives on the other; it stores **separated** charge, not more of it. And a word clash: engineers write charge as Q, but on Day 2 **Q means reactive power**. Context tells you which.

In one sentence: a capacitor is two plates with a gap that store charge in an electric field, it resists sudden changes in voltage, and in AC its current leads its voltage, the opposite of a coil.

✳ **Numbers you will hear**

- **Capacitor bank** sizes in substations: a few **Mvar** per step at 11 or 33 kV (for example a 4 Mvar bank in 2 × 2 Mvar steps in the Track 0 lesson 2 worked example).

- A bank's output falls with the **square of voltage**: a 10 Mvar bank at 0.92 of rated voltage gives only about **8.5 Mvar** ($0.92^2 = 0.846$), just when the grid needs support most (Track 0 lesson 2).
- **Cable charging** (typical, design-specific): roughly **1 to 2 Mvar per km** for a 132 kV cable and **around 10 Mvar per km** at 400 kV. Overhead lines have far less capacitance per km than cables.
- Inverter **DC links**: capacitor banks working at the full DC voltage (up to 1,500 V DC on modern PV plants).

Say it at MEWRE

- How much charging reactive power do the collector cables generate at night, and who absorbs it?
- Are the capacitor banks switched in steps, and has a harmonic resonance study been done?
- Would a STATCOM or the inverters do this better than fixed capacitors, given their output falls with voltage?
- In plain words: a capacitor resists changes in voltage the way a coil resists changes in current, so its current leads instead of lags.

The classic trap.

Treating a capacitor like a small battery that stores energy for later use. A power capacitor stores only a little energy and trades it back and forth with the grid every half-cycle; it delivers reactive power, not energy. Storing energy for the evening peak is a battery's job.

Check yourself (answers at the end of Part 1)

Q1.

In a pure capacitor on an AC supply, when is the current largest?

A. When the voltage is crossing zero, changing fastest, so current leads voltage

B. When the voltage is at its peak, so current is in phase

C. A quarter-cycle after the voltage peak, so current lags

D. Never: a capacitor blocks AC

Q2.

A factory full of motors installs a capacitor bank. What does the bank do?

A. Its leading current cancels part of the motors' lagging current, so the supply carries less current for the same real power

B. It stores energy in the daytime and releases it in the evening

C. It increases the motors' real power output

D. It converts the AC supply to DC for the motors

Q3.

A capacitor bank is rated 10 Mvar at rated voltage. On a summer evening the voltage sags to 0.9 of rated. Roughly how many Mvar does it deliver?

Your answer: _____ Mvar

Lesson 6 · Part 1 · full lesson · about 6 min

Reactance and impedance: the AC version of resistance

Why you care at MEWRE. Transformer datasheets quote impedance in %, line studies quote reactance in ohms, and grid engineers will tell you 'the line is mostly reactive'. This closes Day 1 and is the bridge to reactive power and voltage control on Day 2.

The picture first

Our pipe loop now has all three things in it, driven by the back-and-forth (AC) pump:

- The **narrow pipe** (resistor) fights the flow with friction. The energy becomes heat and is gone.
- The **heavy wheel** (coil) fights every change in flow, but stores the energy and gives it back.
- The **rubber membrane** (capacitor) fights every change in pressure, and also gives the energy back.

All three limit how much water sloshes for a given pump stroke. The narrow pipe does it by **wasting** energy; the wheel and membrane do it by **borrowing** energy and returning it. In electrical terms: resistance for the first, **reactance** for the other two.

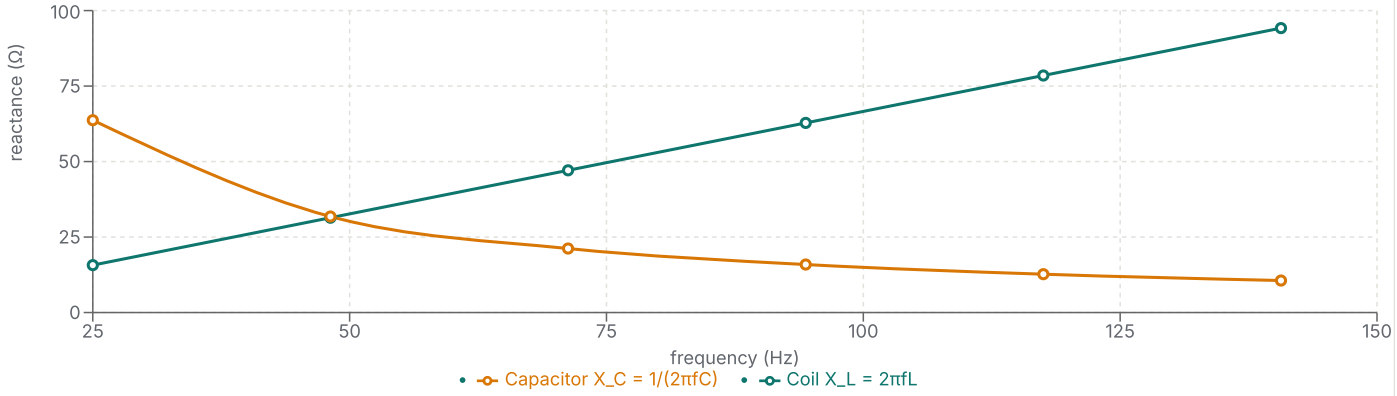
What is actually happening

Reactance (X, in ohms like resistance) is how strongly a coil or capacitor opposes AC. It limits current just as resistance does, but it does not turn energy into heat: it stores it and gives it back every half-cycle.

Unlike resistance, reactance **depends on frequency**:

- **Coil**: $X_L = 2\pi f L$. Faster changes, harder push-back, so X rises with frequency. At zero frequency (DC) a coil is just a piece of wire.
- **Capacitor**: $X_C = 1/(2\pi f C)$. Faster back-and-forth, more current flows, so X falls with frequency. At DC a capacitor blocks completely.

Reactance of a 0.1 H coil and a 100 µF capacitor against frequency



The coil's reactance climbs in a straight line with frequency; the capacitor's falls. These two values happen to cross near 50 Hz, where they would cancel (resonance), which is why capacitor banks need a harmonic study.

Impedance (Z, ohms) is the total opposition to AC, resistance and reactance together. It is what sets the current, so Ohm's law for AC becomes $V = I \times Z$. The catch: R and X do **not** simply add, because they act a quarter-cycle apart. For a resistor and a coil in series:

$$Z = \sqrt{R^2 + X^2}$$

Example: a 30 Ω resistor in series with a coil of 40 Ω reactance has $Z = \sqrt{30^2 + 40^2} = 50$ Ω, not 70 Ω. On 240 V it draws $240/50 = 4.8$ A (the resistor alone would draw 8 A), and the current lags the voltage. A coil's reactance and a capacitor's reactance work in **opposite** directions, so in a circuit they partly cancel.

The impedance triangle: why Z is not R + X

Resistance and reactance act at right angles (a quarter cycle apart), so they add like the sides of a right triangle

Resistance R 30 Ω

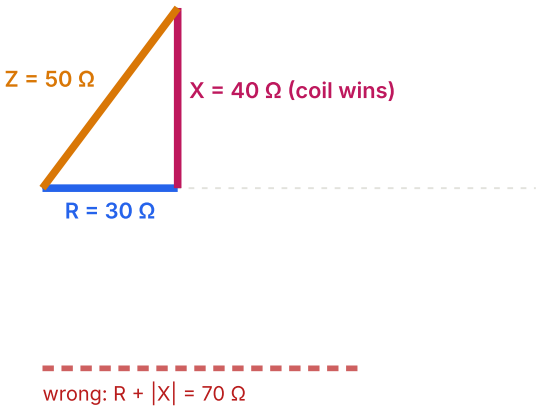
Coil reactance at 50 Hz 40 Ω

Capacitor reactance at 50 Hz 0 Ω

Frequency 50 Hz

X_L − X_C (net X)	Impedance |Z|
40 Ω	50 Ω

Current at 240 V	Angle φ
4.8 A	53.1° lag



Coil reactance points up, capacitor reactance points down, so they cancel each other. Raise the frequency: the coil's X grows and the capacitor's shrinks. If they cancel exactly (resonance) only R limits the current. The angle of the triangle is the angle between voltage and current, and cos φ = R/Z is the power factor.

R = 30 Ω and X = 40 Ω give Z = 50 Ω, not 70 Ω: they add at right angles. Add capacitor reactance and watch it cancel the coil; raise the frequency and watch the two move in opposite directions.

ELI the ICE man, recap:

Element	Opposes	Current vs voltage	Energy	On DC	As frequency rises
Resistor (R)	flow	in step	turned into heat	same R	no change
Coil (X_L)	change in current	lags (ELI)	stored and returned	just a wire	X_L rises
Capacitor (X_C)	change in voltage	leads (ICE)	stored and returned	blocks	X_C falls

Why transmission lines are mostly reactive. An overhead line is a long loop of conductor with a magnetic field around it: a long, thin coil. At 400 kV its reactance is typically **10 to 40 times** its resistance. So when power moves along a line, the voltage drop along it is set mostly by the **reactance** and by the reactive current flowing through it. That is the link you will meet tomorrow: **moving reactive power is what moves voltage** on the grid.

Where the water picture breaks: with water it is hard to see why the wheel and membrane should care how fast the pump strokes. With electricity they do, strongly, and that frequency dependence is what makes reactance different from resistance.

In one sentence: reactance is the opposition of coils and capacitors to AC, it stores energy instead of burning it and changes with frequency, and impedance is resistance and reactance combined, which sets how much current flows.

✳ **Numbers you will hear**

- Overhead line **reactance about 0.3 Ω per km** at 50 Hz (a Track 0 protection example uses an impedance of 0.28 Ω/km); resistance much smaller.
- **X/R of 10 to 40** on 400 kV lines (Track 0 lesson 2): mostly reactive.

- **Transformer impedance** is quoted in % of its rating, typically **8 to 14 %** for MV and HV units: almost all reactance. It limits fault current and causes voltage drop under load.
- At 50 Hz: $X_L = 314 \times L$ ohms (L in henries) and $X_C = 1/(314 \times C)$ ohms (C in farads).
- Series circuit: $Z = \sqrt{R^2 + X^2}$; 3, 4, 5 and 30, 40, 50 are the friendly examples.

Say it at MEWRE

- What is the X/R ratio at the connection point? That tells me how much reactive power will move the voltage.
- Is the transformer impedance on its own MVA base, and what does it do to fault current and voltage drop?
- Has anyone checked for resonance between the capacitor banks or cable capacitance and the network reactance?
- In plain words: reactance is resistance that gives the energy back, and impedance is the total that sets the current.

The classic trap.

Adding resistance and reactance as plain numbers (30 Ω + 40 Ω = 70 Ω). They act a quarter-cycle apart, so they combine like the sides of a right-angled triangle: 30 and 40 give 50 Ω. A second trap is assuming reactance is fixed: it changes with frequency, so a network that is fine at 50 Hz can resonate at a harmonic.

Check yourself (answers at the end of Part 1)

Q1.

A load has resistance 6 Ω in series with inductive reactance 8 Ω. It is connected to 240 V AC. What current flows?

Your answer: _____ A

Q2.

The frequency of a supply doubles. What happens to the reactance of a coil and of a capacitor?

A. The coil's reactance doubles; the capacitor's halves

B. Both double

C. Both halve

D. Neither changes; reactance is fixed like resistance

Q3.

A 400 kV line has a reactance about 20 times its resistance. What mainly controls the voltage along it?

A. The reactive power flowing through it, because the reactance dominates the voltage drop

B. The real power alone, because only real power does work

C. The resistance, because it turns energy into heat

D. Nothing: line voltage is fixed by the transformers

Lesson 7 · Part 1 · full lesson · about 6 min

Reactive power: the foam on the beer

Why you care at MEWRE. Every PV and BESS plant MEWRE connects must supply or absorb reactive power at the grid connection, and in a Kuwaiti summer millions of air-conditioning motors drag voltage down. If you can explain reactive power simply, you can follow every grid-code, inverter-sizing and voltage discussion in the room.

The picture first

Order a beer. The glass holds **beer** and **foam**. You pay for the whole glass and the barman has to fill the whole glass, but only the beer quenches your thirst.

- **Real power P** (MW) is the beer: it turns motors, cools rooms, lights lamps.
- **Reactive power Q** (Mvar) is the foam: it is needed by some equipment, it takes up space, it does no useful work.
- **Apparent power S** (MVA) is the whole glass: what the cables and transformers must carry.

Cables and transformers are rated in **MVA**, the size of the glass. A transformer full of foam has less room for beer.

What is actually happening

In AC the voltage swings back and forth 50 times a second. Some equipment needs a **magnetic field** to work: motors, transformers, overhead lines. Building that field takes energy for a quarter of each cycle, and the field gives the energy back in the next quarter. Capacitors and long underground cables do the same with an **electric field**.

That energy sloshing back and forth every cycle is reactive power. The slosh shows up as a **timing shift** between voltage and current:

- **Lagging**: current peaks *after* voltage. Coils and motors. They *absorb* reactive power.
- **Leading**: current peaks *before* voltage. Capacitors and cables. They *supply* reactive power.

Memory trick: **ELI the ICE man**. In an inductor (L), voltage (E) comes before current (I); in a capacitor (C), current (I) comes before voltage (E).

Slide the angle below and watch the red area: energy handed back to the grid every cycle.

Why reactive power exists: voltage, current and power over time

Same voltage and same current size; only the timing of the current changes

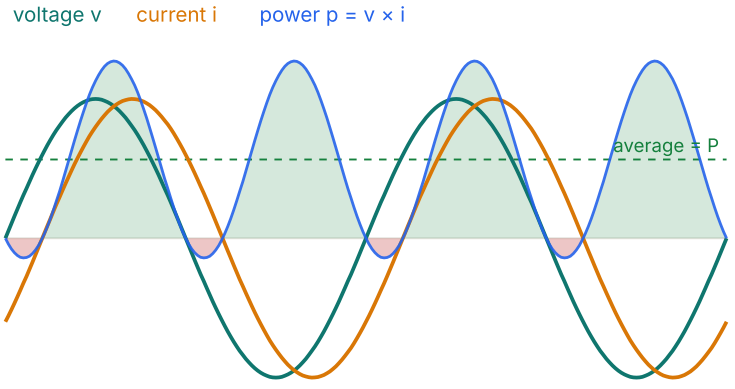
Load

Heater | AC compressor | Pure coil | Capacitor bank

Current lags voltage by 37 °



Real power P (avg) **0.8** pu	Reactive power Q **0.6** pu
Apparent power S **1.00** pu	Power factor **0.80**
Energy back to grid **3.3** % of forward	Current for same P **1.25** × unity-PF



2 cycles = 40 ms at 50 Hz

The current is lagging (inductive: motors, transformers). Green area is energy delivered to the load, red is energy handed back to the grid in the same cycle. With a heater there is no red: every joule does work. With a coil (90°) green and red cancel: the current is full size but the average power is zero, so the cables carry current, get hot and lose voltage while delivering nothing. Utility loads sit in between: at power factor 0.8 the grid must push 25 % more current for the same useful power. That extra current, not the returned energy, is what reactive power costs: transformer and cable capacity, losses and voltage drop.

An air-conditioning compressor at power factor 0.8: the current is 25 % larger than the useful power needs.

Why the grid cares: voltage

Here is the part most engineers miss. On transmission lines, **moving reactive power is what moves voltage**. Pull a lot of Q across a long line and the voltage at the far end sags; supply Q close to where it is used and the voltage holds.

Why pulling reactive power across a line sags the voltage

1. **1 Summer afternoon: AC motors absorb Q**
   Lagging current, millions of compressors
2. **2 Q is pulled from distant generators**
   Extra current flows along long transmission lines
3. **3 The line's reactance drops voltage**
   Lines are mostly reactive, so Q flow sets the drop
4. **4 Voltage sags at the load end**
   Motors draw even more current; equipment is stressed
5. **5 Fix: supply Q near the loads**
   Capacitor banks, SVCs, STATCOMs, PV and BESS inverters

Follow the chain from the motors to the sagging voltage, then see where the fix breaks it: a local Q source means the reactive current no longer has to cross the line.

Why reactive power moves voltage

A line from a strong bus to a load. On high-voltage lines X is much larger than R, so reactive flow sets most of the drop

Real power to the load P `0.60` pu

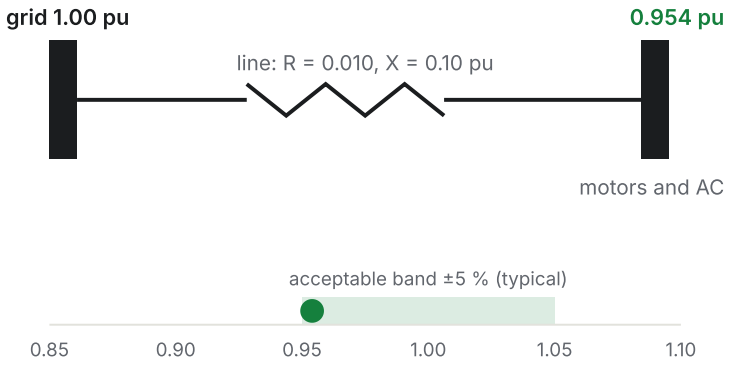
Reactive power drawn Q `0.40` pu

Line X/R ratio `10`

Capacitor bank at the load `0.00` pu

Drop from P (R·P)	Drop from Q (X·Q)
0.6 %	4 %

Receiving voltage	Q still sent over th...
0.954 pu	**0.40** pu



Push Q up and watch the receiving voltage fall much faster than when you push P up: with X ten times R, every unit of reactive power drops voltage ten times as much as a unit of real power. Now add capacitors at the load: they supply the Q locally, less Q crosses the line, and the voltage recovers while the load is unchanged. PV inverters and STATCOMs do the same job. $\Delta V \approx (R \cdot P + X \cdot Q)/V$ is an approximation for small drops.

Push Q up, then P up: reactive power drops the voltage about ten times faster on a line with X ten times R. Then add the capacitor bank and watch the voltage come back.

So the grid needs reactive power **sources near the loads**: generators, capacitor banks, SVCs and STATCOMs, and today **inverters of PV and BESS plants**. That is why the grid code makes a PV plant supply or absorb Q at its connection point, and why inverters are rated in **MVA larger than the plant's MW**: they need headroom to deliver Q even at full sunshine.

Power triangle and power-factor correction

Constant real power P; capacitor bank supplies reactive power locally

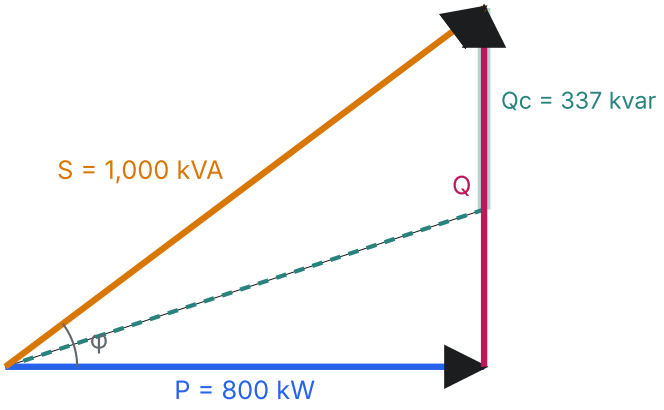
Real power P 800 kW

Power factor (initial) 0.80

Load is Lagging (inductive) Leading

Target PF 0.95

Reactive Q **600** kvar	Apparent S **1,000** kVA
Angle φ **36.9** °	Capacitor Qc **337** kvar
S after **842** kVA	Current reduction **15.8 %**



What to notice: correcting PF from 0.80 to 0.95 needs Qc = P(tan φ_1 – tan φ_2). The real power is unchanged, but apparent power — and therefore current, cable and transformer loading — falls. Going all the way to unity costs disproportionately more kvar for the last few points.

The same idea as a triangle: P along the bottom, Q up the side, S the long side. Correct the power factor and S, the current, shrinks while P stays the same.

✳ Numbers you will hear

- **Power factor** $= P/S$. Industrial and AC loads: about 0.8 to 0.9 lagging. Utilities want it near 0.95 or better.
- **Grid codes** typically ask a PV plant for reactive capability around **0.95 lagging to 0.95 leading** at the point of interconnection, at full output. Kuwait's exact requirement comes from MEWRE's grid code and the PPA: ask for it, do not assume it.
- **Inverter rating**: a 100 MW plant at PF 0.95 needs about $100/0.95 \approx 105$ MVA of inverter capacity, more once you allow for the 50 °C derating.
- $S^2 = P^2 + Q^2$. At PF 0.8: P = 0.8, Q = 0.6, S = 1.0.

Say it at MEWRE

- What reactive capability does the grid code require at the point of interconnection, and does it hold at 50 °C ambient?
- Is the inverter MVA rating sized for that, or are we relying on a STATCOM?
- Who sets the voltage or reactive power setpoint: our power plant controller or the control centre?
- Has the reactive capability been demonstrated in the commissioning tests, or only on paper?

The classic trap.

Thinking reactive power is wasted energy. Very little energy actually flows back (about 3 % per cycle at power factor 0.8). The real cost is the extra current: it fills cables and transformers, heats them and drops voltage. And it is not useless: motors and transformers cannot work without it, and the grid needs it to hold voltage.

Check yourself (answers at the end of Part 1)

Q1.

An air-conditioning compressor draws its current **after** the voltage peaks. What is it doing?

A. Absorbing reactive power (lagging)

B. Supplying reactive power (leading)

C. Drawing only real power

D. Returning real power to the grid

Q2.

A load uses 8 MW at power factor 0.8. What apparent power S (MVA) must the transformer carry?

Your answer: _____ MVA

Q3.

Why does MEWRE's grid code ask PV plants for reactive power capability?

A. To help hold the grid voltage, since reactive power flow largely sets voltage on transmission lines

B. To increase the energy the plant sells

C. To reduce soiling losses on the modules

D. Because inverters cannot work at unity power factor

Lesson 8 · Part 1 · full lesson · about 6 min

Three-phase: three waves taking turns

Why you care at MEWRE. Every voltage on a MEWRE single-line diagram (400 kV, 132 kV, 33 kV) is a three-phase voltage, and every current check on a drawing uses a √3. Once three-phase clicks, you can sanity-check a cable or a breaker rating in your head in a design review.

The picture first

Think of an engine. A one-cylinder engine (a lawnmower) delivers a push, then nothing, then a push: it shakes. A **three-cylinder** engine has its cylinders firing in turn, so one is always pushing and the crankshaft turns much more smoothly.

Three-phase is the same trick with electricity. Instead of one AC wave that rises, falls through zero and rises again, the generator makes **three waves of the same size, each one-third of a cycle (120°) behind the previous one**. When one wave is passing through zero, the other two are doing the work.

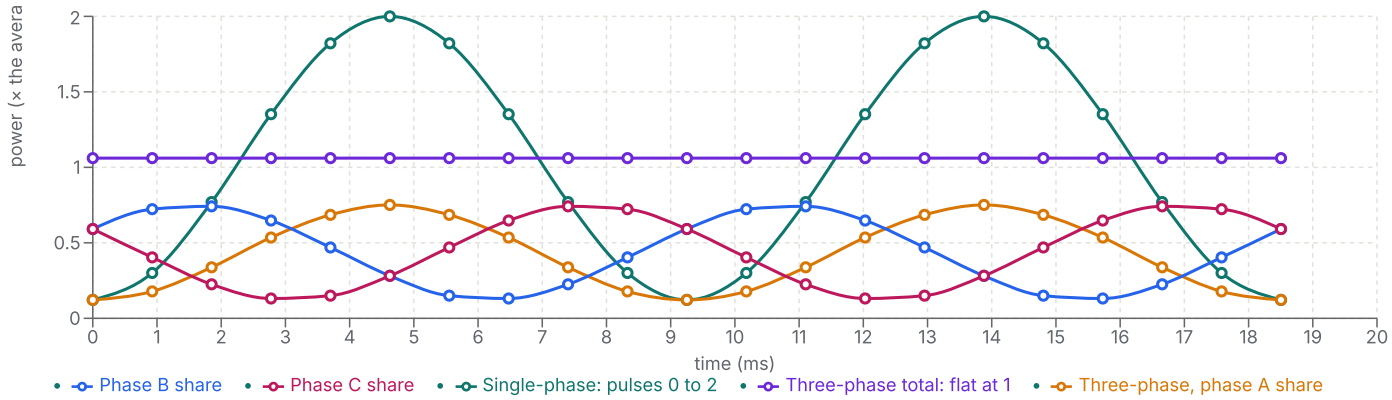
Where the analogy breaks: a three-cylinder engine still has a little ripple in its torque. A balanced three-phase supply delivers **perfectly constant** total power, not just smoother power.

What is actually happening

The generator has three coils (windings) spaced evenly around it. As the magnet spins past them, each coil makes its own sine wave, 120° apart. The three outputs are the **phases**, often called L1, L2, L3 (or A, B, C, or red, yellow, blue). Three things fall out of this:

- **Constant power.** Add up the power in the three phases at any instant and the dips cancel. Big generators and motors feel a steady push instead of a 100 Hz shake.

Power delivered over one 20 ms cycle: single-phase against balanced three-phase (same average, unity power factor)



The single-phase supply hits zero twice per cycle (a 100 Hz pulse). The three phase shares rise and fall in turn, and whenever one dips the other two make up for it, so their total is a flat line: the engine with no shake.

- **Three wires instead of six.** Three separate single-phase circuits would need six wires (out and back for each). In three-phase, when the three currents are equal (**balanced**), they add up to zero at every instant, so the "return" currents cancel and no return wire is needed. That is why a transmission line has three conductors (plus earth wires on top for lightning, which carry no load current).
- **A rotating magnetic field.** Feed three coils spaced around a motor with three currents peaking in turn, and the magnetic field turns by itself, at a speed set by the frequency. Every large motor (including the compressors in Kuwait's air-conditioning load) works this way: it starts and runs without the extra starting tricks a single-phase motor needs.

Line vs phase voltage. In the common **star** connection each coil sits between one line and a shared middle point (the **neutral**). The voltage from a line to neutral is the **phase voltage**; the voltage between two lines is the **line voltage**, and it is $\sqrt{3} \approx 1.73$ times bigger (the two waves are 120° apart, so their difference is bigger than either one, but not double). A generic example: European low voltage is **400/230 V** (230 V line to neutral, $230 \times 1.73 \approx 400$ V between lines). Track 0 describes Kuwaiti homes as fed at **415/240 V**. **Equipment is always labelled with the line voltage**: "33 kV" means 33 kV between phases and 19.05 kV from each phase to earth.

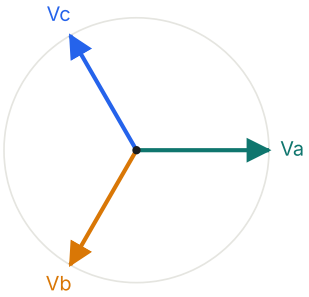
Three-phase voltages

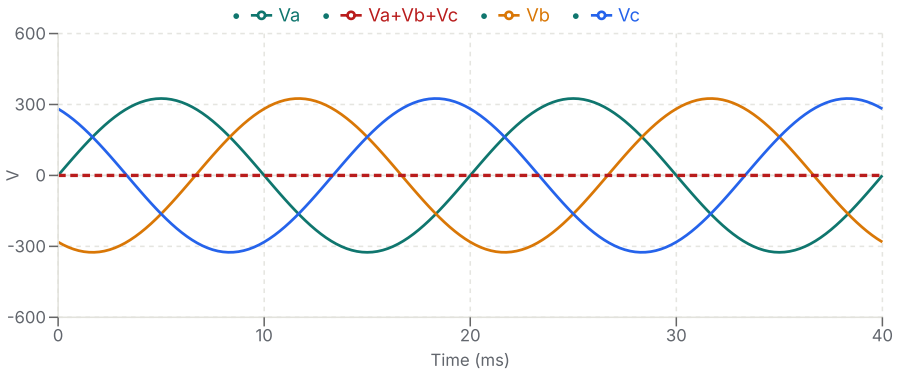
Balanced phases cancel; line voltage = √3 × phase voltage

Phase voltage (rms) 230 V

Phase A amplitude drop 0 %

Phase C angle error 0 °





What to notice: with equal amplitudes 120° apart the instantaneous sum is zero, so a balanced star load needs no neutral current. Any amplitude or angle error leaves a residual (red) that flows in the neutral/earth and causes negative-sequence heating in machines.

Phase V (rms)	Line V = √3·Vph	Actual Vab	Residual sum
230 V	398 V	398 V	0.0 % of one phase

Balanced: the three waves add to zero at every instant (no return wire needed) and line voltage is 1.73 × phase voltage. Then drag 'Phase A amplitude drop' to 20 %: the sum stops being zero, and that leftover is the unbalance (described below) that a neutral wire must carry.

Power. The one formula you will use on the job:

$$P = \sqrt{3} \times V_{\text{line}} \times I \times \text{PF}$$

Example from Track 0: a 33 kV feeder carrying 30 MW at PF 0.95 carries $30{,}000{,}000 / (1.73 \times 33{,}000 \times 0.95) \approx 553$ A.

The feeder current check in four steps (30 MW at 33 kV, PF 0.95)

1. **Start with real power** — 30 MW
2. **Divide by the power factor** — 30 / 0.95 ≈ 31.6 MVA: what the cable really carries
3. **Divide by √3 × line voltage** — 1.73 × 33 kV ≈ 57.2
4. **Current in each line** — 31.6 / 57.2 ≈ 0.553 kA ≈ 553 A

The same chain works for any feeder: MW, then MVA, then divide by 1.73 × kV to get kA. Use √3 once, with the line voltage.

Unbalance. If the three phases are loaded unequally (many single-phase AC units on one phase, a blown fuse, a fault), the currents no longer cancel. The leftover flows in the neutral, and the unequal voltages heat motors and generators and can make inverters reduce output or trip.

Numbers you will hear

- **√3 ≈ 1.73**: line voltage = 1.73 × phase voltage in star. "400 kV" is about 231 kV from each phase to earth; "132 kV" about 76 kV.

- **Current anchors** ($I = S/(\sqrt{3}V)$): about **17.5 A per MVA at 33 kV**, **1.44 A per MVA at 400 kV**. So 30 MVA on a 33 kV feeder is about 525 A.
- **Low voltage**: 400/230 V is the generic IEC example; Track 0 gives Kuwait as **415/240 V** (widely described, confirm in the MEWRE distribution standards).
- **Voltage unbalance**: planning limits are typically around **1.4 to 2 %**; many inverters derate or trip above about **2 to 3 %**; 1 % voltage unbalance can cause 6 to 10 % current unbalance in an induction motor.

Say it at MEWRE

- Are all voltages on this SLD line-to-line, and are the insulation ratings based on the phase-to-earth value?
- Did the cable sizing use MVA including the reactive requirement, with the √3 applied once?
- What voltage unbalance did the power-quality study assume at the connection point, and what can the inverters tolerate before they derate?
- Quick check: 30 MVA at 33 kV is about 525 A. Does the feeder rating match that, after site derating?

The classic trap.

Forgetting the √3, or using it twice. Writing I = P / (V × PF) with the line voltage gives a current 73 % too high; dividing a phase voltage by √3 again gives one far too low. Remember: nameplate voltage is line-to-line, and P = √3 × V_line × I × PF uses it directly.

Check yourself (answers at the end of Part 1)

Q1.

A 33 kV system: what is the voltage from one phase to earth, in kV?

Your answer: _____ kV

Q2.

A 33 kV collector feeder exports 20 MW at power factor 1. Roughly what current flows in each line, in amperes?

Your answer: _____ A

Q3.

Why can a balanced three-phase transmission line use three conductors instead of six?

A. The three balanced currents add up to zero at every instant, so no return conductor is needed

B. The earth carries the return current of all three phases in normal operation

C. Each conductor carries one-third of the voltage

D. Three-phase lines carry no reactive power

Answers: Part 1

Voltage, current and resistance: pressure, flow and the narrow pipe

Q1.

5 A

Ohm's law: $I = V/R = 240/48 = 5$ A. Halve the resistance (24 Ω) and the current doubles to 10 A.

Q2.

It increases four times

Heat is I^2R. Doubling I multiplies the heat by $2^2 = 4$. This is why extra current, even a little, matters so much for cable and transformer ratings.

Q3.

For the same power, higher voltage means much less current, so far less I²R heat lost in the lines

Power is voltage times current, so raising the voltage lets the same power travel with less current. Losses go with the square of current, so the saving is large. Transformers then step the voltage down near the customer.

Power and energy: MW is speed, MWh is distance

Q1.

200 MWh

Energy = power × time = $50 \times 4 = 200$ MWh. Batteries are described by both numbers: "50 MW / 200 MWh" or "a 4-hour battery".

Q2.

25 %

$\mathrm{CF} = 219{,}000/(100 \times 8{,}760) = 219{,}000/876{,}000 = 0.25 = 25$ %. Equivalently, 2,190 full-load hours.

Q3.

The plant's maximum output in MW (more exactly MVA), the most it carries at once

Equipment is limited by power, the rate at any instant, not by the energy over a year. Energy decides revenue; power decides equipment size.

DC and AC: one-way flow and back-and-forth

Q1.

339 V

Peak $= \sqrt{2} \times \mathrm{RMS} = 1.414 \times 240 \approx 339$ V. The wave swings between +339 V and −339 V every 20 ms.

Q2.

Transformers can step AC voltage up and down simply and efficiently

Transformers work only with a changing voltage, so AC lets the grid move between generation, transmission (132 and 400 kV) and distribution voltages cheaply. PV modules and batteries make DC and need inverters.

Q3.

1,430 MWp is the DC module capacity; 1,100 MW is the AC capacity of the inverters and grid connection

Modules make DC and are rated in MWp; inverters convert to AC and limit what reaches the grid. The DC/AC ratio here is 1,430 / 1,100 = 1.3, typical for a tracker plant. Capacity factors and PPAs must say which basis they use.

Magnetism and coils: current that resists change

Q1.

It works by induction, which needs a magnetic field that keeps changing, and only a changing current makes one

Faraday's law: only a changing magnetic field induces a voltage. Steady DC makes a steady field, so nothing appears on the other winding.

Q2.

Current lags voltage by a quarter-cycle (90°)

The coil opposes every change in current, so the current always trails behind. ELI: voltage (E) before current (I) in an inductor (L).

Q3.

It lags the voltage, so the grid must supply reactive power as well as real power

Motors need a magnetic field, so they are inductive and draw lagging current. Their power factor is typically 0.8 to 0.9, so the grid carries 10 to 25 % more current than the real power alone would need.

Capacitors: storing charge and resisting changes in voltage

Q1.

When the voltage is crossing zero, changing fastest, so current leads voltage

Charge flows fastest when the voltage changes fastest, which is at the zero crossing; at the peak the plates are full and current is zero. So current peaks a quarter-cycle before voltage. ICE: current (I) before voltage (E) in a capacitor (C).

Q2.

Its leading current cancels part of the motors' lagging current, so the supply carries less current for the same real power

This is power-factor correction: the capacitor supplies locally the reactive power the motors need, so it no longer flows through the network. Current, losses and voltage drop all fall.

Q3.

8.1 Mvar

A capacitor's reactive output goes with the square of voltage: $10 \times 0.9^2 = 10 \times 0.81 = 8.1$ Mvar. It gives less support exactly when voltage is low, which is why grid codes increasingly rely on inverters and STATCOMs.

Reactance and impedance: the AC version of resistance

Q1.

24 A

$Z = \sqrt{6^2 + 8^2} = \sqrt{100} = 10\ \Omega$, so $I = V/Z = 240/10 = 24$ A. Adding 6 + 8 = 14 Ω would give a wrong 17 A.

Q2.

The coil's reactance doubles; the capacitor's halves

$X_L = 2\pi f L$ rises in proportion to frequency; $X_C = 1/(2\pi f C)$ falls in inverse proportion. Resistance does not depend on frequency.

Q3.

The reactive power flowing through it, because the reactance dominates the voltage drop

The voltage drop is roughly $(RP + XQ)/V$. With X much larger than R, the XQ term dominates, so reactive power flow largely sets voltage on transmission lines. This is the bridge to Day 2.

Reactive power: the foam on the beer

Q1.

Absorbing reactive power (lagging)

Motors need a magnetic field, so their current lags the voltage and they absorb reactive power. ELI: voltage (E) before current (I) in an inductor (L).

Q2.

10 MVA

$S = P/\mathrm{PF} = 8/0.8 = 10$ MVA. The transformer must be sized for 10 MVA even though only 8 MW does useful work.

Q3.

To help hold the grid voltage, since reactive power flow largely sets voltage on transmission lines

Reactive power is what controls voltage on high-voltage lines. As PV replaces conventional generators, the inverters must take over part of their voltage-support job.

Three-phase: three waves taking turns

Q1.

19.05 kV

$33/\sqrt{3} = 19.05$ kV. The 33 kV on the nameplate is between two lines; each line sits 19.05 kV from earth, which is what cable and surge-arrester insulation is rated against.

Q2.

350 A

$I = P/(\sqrt{3} \times V \times \mathrm{PF}) = 20{,}000{,}000/(1.732 \times 33{,}000 \times 1) \approx 350$ A. Shortcut: 17.5 A per MVA at 33 kV, times 20.

Q3.

The three balanced currents add up to zero at every instant, so no return conductor is needed

Equal currents 120° apart sum to zero, so each phase's "return" is provided by the other two. Earth only carries current during faults or unbalance.

PART 2 OF 4

Kuwait, solar, CSP and batteries

By the end you can: Describe Kuwait's power system and who decides what, how a PV plant makes and loses MWh, how CSP and batteries fit the evening peak.

7 lessons · about 49 minutes

Lesson 9 · Part 2 · full lesson · about 8 min

Kuwait's power system in ten minutes

Why you care at MEWRE. Every memo, briefing and business case you touch at MEWRE starts from a handful of system numbers: the summer peak, what the fleet can really deliver, what it burns and what customers pay. Know these numbers and their sources and you can follow any conversation in the Undersecretary's office from week one.

The picture first

Picture a taxi company in Kuwait City that owns **21 taxis**. On the busiest afternoon of the year it needs about **17.5** on the road. Sounds comfortable.

But on that afternoon some taxis have broken down, some are in the garage for their scheduled service, and in 50 °C heat the old ones crawl. Count what is actually usable and you have about **17**. Some customers are left on the pavement.

Two more twists:

- The taxis burn fuel the owner could otherwise **sell abroad** at a good price.
- Passengers pay a fixed fare of **2 fils** for a trip that costs the company somewhere between 14 and 38 fils.

So any helper that carries passengers on the busy afternoon *without burning fuel* is worth a lot to the owner. That helper is solar.

Where the analogy breaks: electricity cannot queue like passengers. If supply falls short, frequency drops (Day 2) and the control centre must cut some areas off (**load-shedding**).

What is actually happening

Demand and the peak. Kuwait's load is dominated by **air conditioning**. The system's worst hour is a July or August afternoon at 48 to 50 °C. MEWRE reported summer peaks of 17,640 MW (2024), 17,610 MW (2025) and **17,540 MW (2026)**. New connections still add about 1 GW of load a year; demand-side measures have been holding the peak flat.

Installed vs available capacity. Installed (nameplate) capacity is what the plants are rated at: about **21 GW** (Rystad). **Available** capacity is what can actually run at the peak hour: about **17 GW**. The gap of roughly 4 GW (about one fifth of the fleet) has three causes:

- **Forced outages**: trips, like the 300 MW Az-Zour South steam turbine in June 2024.
- **Planned maintenance** that runs into the early heat (April 2025).
- **Temperature derating**: a gas turbine loses roughly 0.5 to 0.7 % of output per °C above 15 °C, so 17 to 25 % at 50 °C (generic engineering value).

Reserve margin is the spare capacity at the peak, as a share of the peak:

$$RM = (C_{avail} - P_{peak})/P_{peak}$$

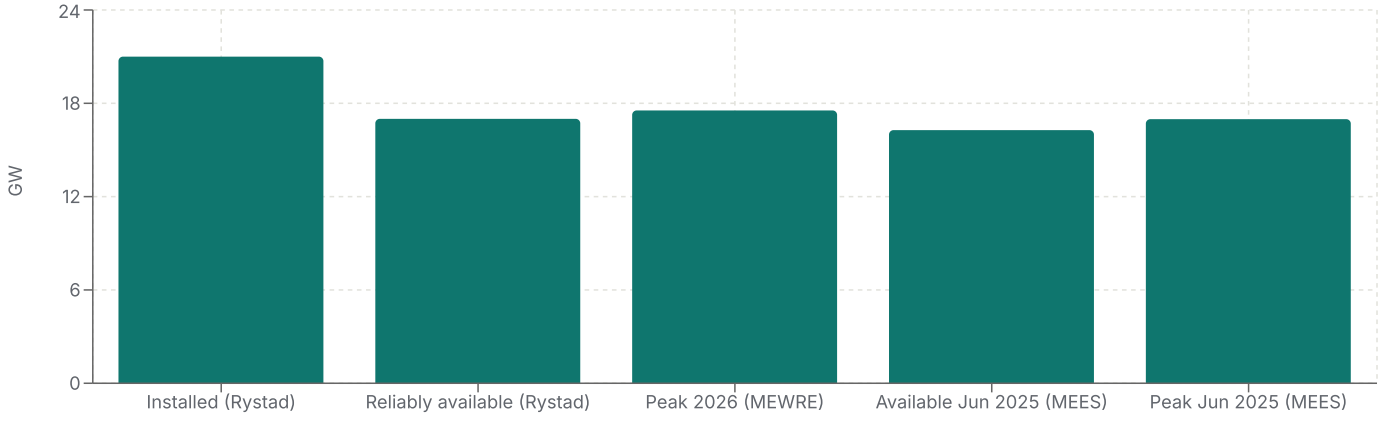
Hot-climate utilities typically plan for +10 to 20 % on *available* capacity. In June 2025 MEES reported a 16.98 GW peak against 16.275 GW available: $RM \approx -4.2\%$. A negative margin means cuts, emergency imports through the GCC interconnector (GCCIA) or appeals to save power. That is what happened in 2024 and 2025.

Why the summer margin disappears

1. **1 48 to 50 °C afternoon**
   Air conditioning pushes demand to the year's peak (17,540 MW in 2026)
2. **2 Gas turbines derate**
   The same heat costs about 0.5 to 0.7 % of output per °C above 15 °C
3. **3 Outages and maintenance**
   Forced trips and maintenance running into the heat remove more units
4. **4 Available ≈ 17 GW**
   Out of about 21 GW installed
5. **5 Margin thin or negative**
   June 2025: 16.98 GW peak vs 16.275 GW available, RM ≈ -4.2 %
6. **6 Imports, appeals, cuts**
   GCCIA imports, calls to save power or load-shedding (2024 and 2025)

Heat pushes demand up and pushes available supply down in the same hour: that double hit is why the margin vanishes.

Capacity vs peak demand (GW)



Installed capacity looks comfortable; capacity available at the peak hour does not. Sources and bases differ, which is exactly why every figure carries its label.

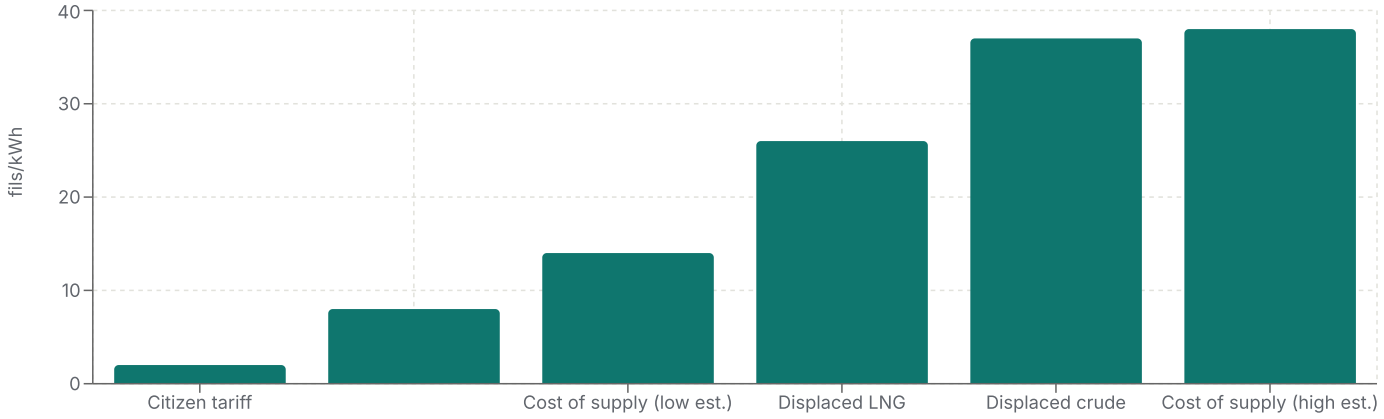
Fuel. Kuwait generates roughly **62 % from gas and 36 % from oil** (crude, fuel oil and gasoil burned directly), with about 2 % renewables. Its own gas is mostly associated gas from oil fields, so summer is topped up with **imported**

LNG (5.2 Mt in January to August 2025) and liquid fuels. The marginal summer kWh comes from the oldest, least efficient units burning the most expensive fuel.

Tariffs. Kuwaiti citizens in private housing pay **2 fils/kWh**, unchanged since 1966 (1 KWD = 1,000 fils). The cost of supply is not officially published; estimates run from **14 to 38 fils/kWh**. The state budget pays the gap.

Why every solar MWh saves money. Each solar kWh means a thermal unit burns less fuel. Valued at what that fuel could earn or would cost on world markets, that is about **37 fils/kWh** for crude in an old steam unit (at USD 70/bbl) and about **26 fils/kWh** for LNG in a combined-cycle plant (at USD 12/MMBtu). Analysts estimate solar in Kuwait at **7 to 8 fils/kWh**. The difference is the saving, and the crude not burned can be exported.

What a kWh costs, what customers pay, what solar saves (fils/kWh)



Compare solar with the fuel it displaces, not with the 2-fils tariff. Fuel values use illustrative prices (USD 70/bbl crude, USD 12/MMBtu LNG) and 1 KWD ≈ 3.25 USD.

Why it matters at MEWRE

- Ministers respond to "this plant cuts the summer shortfall by X MW and saves Y barrels", not to "Z MWp".
- PV helps the afternoon peak but gives less than nameplate at 50 °C and nothing after sunset, when cooling load is still high. Always ask how many **MWac** a project delivers at the peak hour. That is why the ministry paired the PV programme with a planned 1 to 1.5 GW / 4 to 6 GWh battery storage (BESS) programme.
- With a margin this thin, grid tie-in outages between April and October will almost certainly be refused. Plan energisation for November to March.

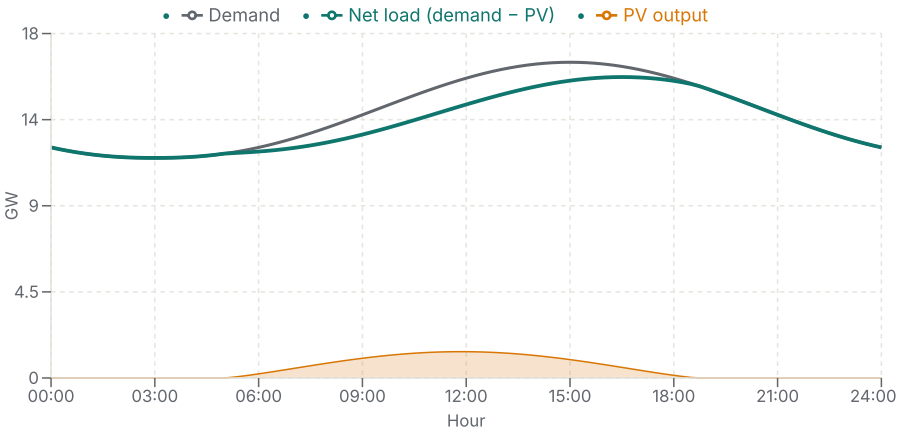
Kuwait daily load with PV — net load

Indicative shape — not measured system data

Season: Summer | Winter

PV capacity (AC): 1,600 MW

Peak load: 16.5 GW	Peak net load: 15.7 GW
Peak reduction: 0.77 GW	Min net load: 11.5 GW
Max evening ramp: 544 MW/h	PV energy share: 3.4 %



What to notice: in summer PV output lines up with the air-conditioning peak, so it shaves the peak — but the net-load peak shifts into the evening, after sunset, with a steep ramp. In winter large PV volumes push midday net load toward must-run thermal minimums (a "duck"), which is where storage and flexibility earn their value.

Indicative shape: summer peak ~16–17 GW mid-afternoon with a ~11–12 GW night minimum; winter ~6–9 GW with an evening peak. Use MEWRE dispatch data for any real study.

Indicative summer day with 1,600 MW of PV (Zones 1 and 2). PV shaves the afternoon, but load stays high after sunset, so the net peak moves toward the evening and falls by far less than 1,600 MW. Switch to winter: the peak comes after sunset and PV does not touch it at all.

🇰🇼 Numbers you will hear

- **Summer peak**: 17,540 MW in 2026 (MEWRE basis via Kuwait Times; medium-high). MEES reported a 17.71 GW monthly record in July 2025 on a different basis (medium). Always state the basis.
- **Capacity**: ~21 GW installed, ~17 GW reliably available at peak, ~14 GW gas-fired (Rystad, 2025; medium).
- **Mix**: ~62 % gas, ~36 % oil, ~2 % renewables (Ember via Inzonex; medium-low). Solar fleet about 102 MW at end-2024 (MEWRE).
- **Tariff**: 2 fils/kWh for citizens since 1966. **Cost of supply**: 14 to 38 fils/kWh, no official figure, so always show a range.
- **Displaced fuel**: ≈ 37 fils/kWh (crude) or ≈ 26 fils/kWh (LNG) vs solar ≈ 7 to 8 fils/kWh (analyst estimate; the Zone 1 tariff has not been reported).

Say it at MEWRE

- Which basis is this peak on: the ministry's seasonal figure or a MEES monthly record, net or gross?
- What does the plant deliver in MWac at 16:00 on an August day at 50 °C, not in MWp?
- Is that reserve margin computed on installed or on available capacity?
- Which fuel and heat rate are we assuming at the margin, and at what price?
- Every month this plant is late, we burn more crude or LNG: what is that worth per day?

The classic trap.

Comparing the solar tariff with the 2-fils retail tariff ("solar at 8 fils costs four times what customers pay"). The right comparison is the cost of supply or the fuel the solar kWh displaces, about 26 to 37 fils/kWh. Likewise, a reserve margin computed on 21 GW installed looks healthy; on about 17 GW available it was negative in 2025.

Check yourself (answers at the end of Part 2)

Q1.

Rystad puts Kuwait's installed generating capacity at about 21 GW. Roughly how much is **reliably available** at the summer peak hour?

A. About 21 GW

B. About 19 GW

C. About 17 GW

D. About 14 GW

Q2.

June 2025 (MEES): peak demand 16.98 GW, available capacity 16.275 GW. What was the reserve margin in **percent**? (Negative if a deficit.)

Your answer: _____ %

Q3.

A colleague says: "Solar at 8 fils/kWh is a bad deal, because citizens only pay 2 fils." What is the best reply?

A. Compare solar with the fuel it displaces (about 26 to 37 fils/kWh) or the cost of supply, not with the subsidised tariff

B. Agree: solar only makes sense once tariffs rise

C. Solar is free once built, so the tariff does not matter

D. Count KPC's 25 fils/kWh payment as the national saving instead

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Lesson 10 · Part 2 · full lesson · about 8 min

Who decides what: MEWRE, KAPP, KISR and the approval chain

Why you care at MEWRE. As a MEWRE PM you will rarely be blocked by engineering. You will be paced by a chain of institutions, each with the legal right to say "not yet". Knowing who holds each gate, and what file they need, is the fastest way to move a project.

The picture first

Think of a large company buying a very expensive machine.

- The **production department** needs the machine and writes the requirement.
- **Procurement** runs the tender and picks a supplier.
- The **board** approves the purchase.
- **Legal** checks the contract wording.
- **Internal audit** checks the money and the procedure before anyone signs.

Each of them can send the file back with comments, and each return costs a full review cycle.

Kuwait's public sector works the same way, with one difference: the "departments" are separate institutions with their own laws, and their signatories are personally accountable. Officials are rarely punished for delay but can be for an irregular approval, so incomplete files get returned.

What is actually happening

MEWRE wears four hats.

1. **Policy owner**: renewable targets and strategy.
2. **System operator and transmission owner**: dispatch, the 400/132 kV network, connecting new plants.
3. **Offtaker**: it buys the power, under a **30-year PPA** (power purchase agreement) for solar IPPs and a **25-year ECWPA** for gas-fired power and water plants (IWPPs).
4. **Direct procurer** of grid and EPC works through public tenders, for example the Shagaya to Al-Wafra 400 kV line (KD 48.6 m, March 2026).

KAPP, the Kuwait Authority for Partnership Projects (PPP Law 116/2014; Executive Regulations Decree 78/2015), runs the procurement of **PPP and IPP projects**: feasibility, RFQ, RFP, evaluation. For a solar IPP, **KAPP procures and MEWRE is the offtaker**. KAPP is attached to the Minister of Finance; on IWPPs it also becomes a shareholder (60 % in Az-Zour North 2&3).

The Higher Committee for PPP, chaired by the **Minister of Finance**, approves the project, the procurement stages and the selected investor.

CAPT, the Central Agency for Public Tenders (Public Tenders Law 49/2016), governs conventional public tenders above KD 75,000: the grid works, consultancy and O&M contracts MEWRE buys itself.

Two tracks, side by side. Every file you touch sits on one of them, and the track decides which gates apply.

	PPP / IPP track	Conventional tender track
Who runs the procurement	KAPP	MEWRE, under CAPT rules
Law	PPP Law 116/2014	Public Tenders Law 49/2016
Typical projects	Solar IPPs (Shagaya Phase III), IWPPs	Grid works (Shagaya to Al-Wafra 400 kV line), consultancy, O&M
Gatekeeper	Higher Committee for PPP (project, then investor)	CAPT (tenders above KD 75,000)
Before signature	State Audit Bureau prior review	State Audit Bureau prior review
MEWRE's hat	Offtaker (30-year PPA)	Direct buyer

The State Audit Bureau (Law 30/1964) carries out **prior review** of contracts before signature. Nothing signs without it: even the 2025 emergency GCCIA power imports needed its conditional approval.

The Fatwa and Legislation Department reviews the legal terms of contracts, PPAs included, and gives the government legal opinions. You reach it through the ministry's legal affairs department.

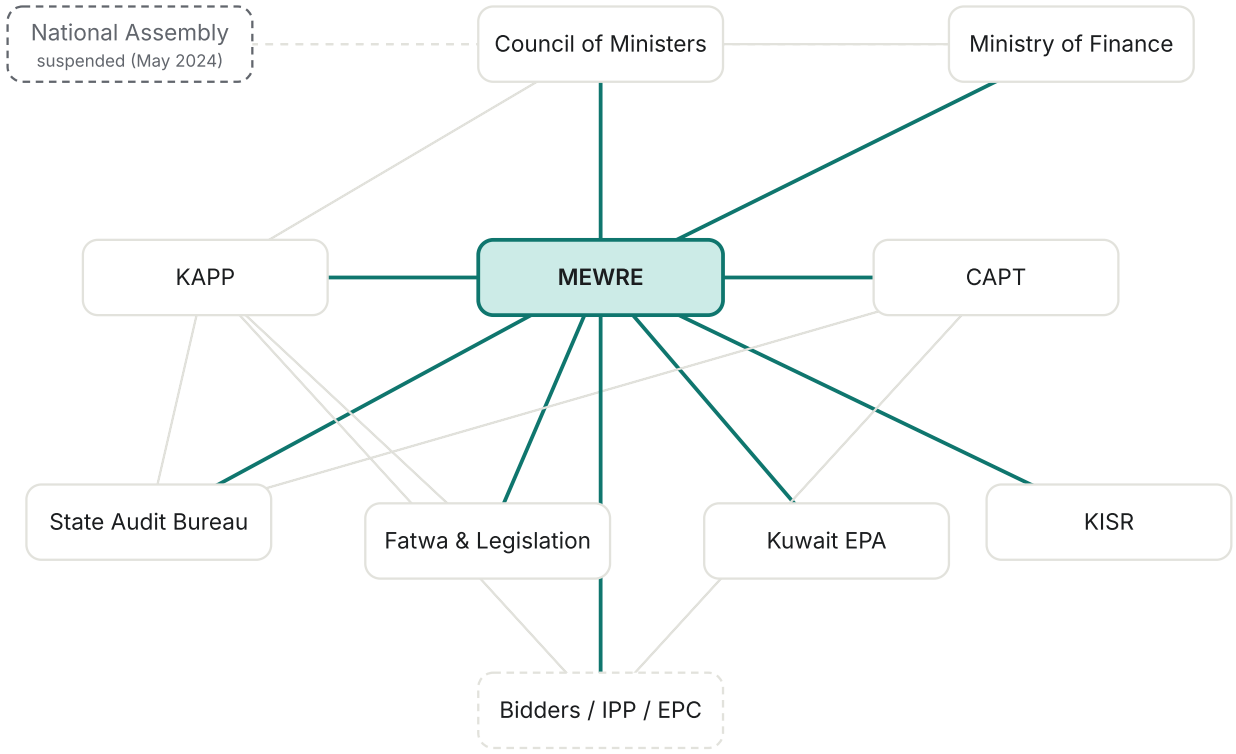
The Council of Ministers is the top executive decision maker. Since **10 May 2024** the National Assembly is suspended and laws are issued by decree-law, so the bottleneck is cabinet agendas rather than parliament.

Also on the map: the **Ministry of Finance** prepares the budget (fiscal year 1 April to 31 March) and funds ministry capital projects. **KISR**, the Kuwait Institute for Scientific Research, built Shagaya Phase I with the ministry and runs solar measurement stations at five inland sites: your source of ground-truth sunlight data. The **EPA** (Environment

Public Authority) handles environmental approvals; the **Supreme Council for Planning and Development** owns Kuwait Vision 2035.

Kuwait decision chain for power projects

Who approves what — click an institution



Ministry of Electricity, Water & Renewable Energy

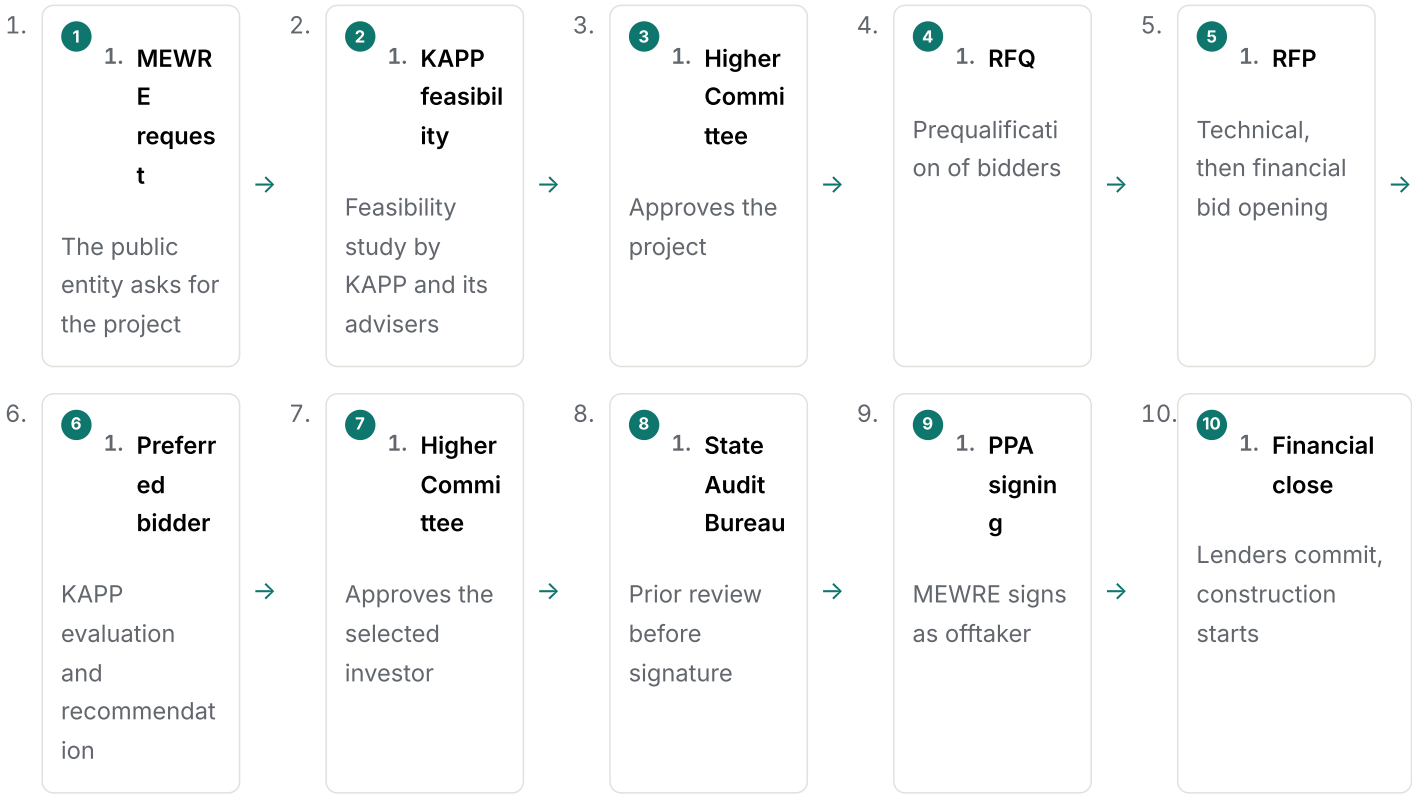
Plans, builds and operates Kuwait's power and water system; procurer/offtaker for renewable IPPs (signs the PPA and buys the energy) and owner of the transmission grid the plant connects to.

PM interaction: Your home organisation: owner's requirements, grid connection scope, internal approvals, contract administration and the interface with every other body on this map.

Simplified and general. Thresholds, committee names and procedures change — confirm current rules with MEWRE legal and procurement departments.

Click each institution for its role. Then trace a solar IPP: MEWRE requests, KAPP procures, the Higher Committee approves, the Audit Bureau reviews, MEWRE signs the PPA.

Approval chain of a KAPP solar IPP



Two Higher Committee gates (project and investor), then the Audit Bureau before signature. Synthesised from IJGlobal and MEED coverage; Fatwa's exact place in the sequence is not confirmed.

Where you sit, and the one formula that matters

You sit inside MEWRE, on the renewable-energy side (the exact unit names in the org chart are unverified: get the current chart and the delegation-of-authority matrix on day one). You report up through an Assistant Undersecretary to the Undersecretary. You do not approve contracts yourself. You **prepare files that pass the gates** and you **track every referral** (إحالة) from desk to desk in a register.

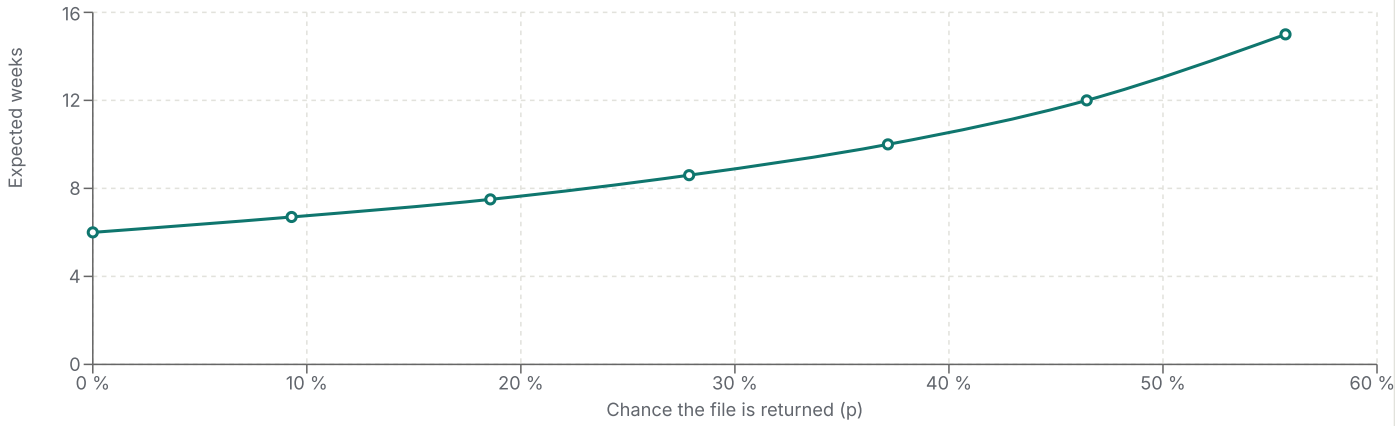
Day to day you talk to: the KAPP project team (technical and offtaker positions during procurement), MEWRE Transmission and the control centre (connection, outages), legal affairs (and through them Fatwa), the finance and budget department, the Audit Bureau liaison, CAPT for ministry tenders, and KISR for data.

Each gate takes a review time t and returns the file with probability p. On average a step then takes

$$E[T] = t/(1-p)$$

A 6-week review with a 40 % chance of return takes 10 weeks on average; at 50 % it doubles. Your biggest lever is **lowering p with complete, pre-cleared files**, not chasing the reviewer to shorten t.

Expected time at one gate with a 6-week review



E[T] = 6 / (1 - p). The curve bends upward: each extra point of return risk costs more weeks than the last, so a complete file is worth more than a faster reviewer.

🇰🇼 Numbers you will hear

- **KAPP**: PPP Law 116/2014 and Decree 78/2015. Do not confuse it with FDI Law 116/2013 (KDIPA): same number, different year.
- **CAPT**: tenders above KD 75,000 (Law 49/2016). Law 94/2026 (gazetted 27 Sep 2026, effective about three months later) allows direct contracting up to KD 1 m per financial year without CAPT approval (single law-firm source: verify).
- **Audit Bureau** prior review: reported threshold KD 100,000 and above (low-to-medium confidence; may have been amended).
- **Contract terms**: PPA 30 years (solar IPP); ECWPA 25 years (IWPP). Fiscal year 1 April to 31 March.
- **Benchmark**: Zone 1 went from RFQ (January 2024) to a reported preferred bidder (about August 2026) in roughly 31 months, with the Higher Committee, Audit Bureau and PPA still to come.

Say it at MEWRE

- Which approvals are pending on this file today, with whom, since when, and what observations were raised?
- Is this a KAPP PPP track or a CAPT tender track? That decides which gates apply.
- What did the Audit Bureau observe on the last similar contract, and is every point closed in our file?
- Who on MEWRE's side has the authority to agree the offtaker positions in the PPA mark-up?
- A press report of a 'selected' bidder is not a Higher Committee approval or a signed PPA: what is the formal status?

The classic trap.

Thinking KAPP speaks for MEWRE, or that CAPT runs IPP tenders. KAPP procures PPP projects under the PPP Law, CAPT governs conventional ministry tenders, and MEWRE as offtaker must state its own positions in writing. And never go around the chain to the Audit Bureau or Fatwa directly: go through the ministry liaison and legal affairs.

Check yourself (answers at the end of Part 2)

Q1.

For a Shagaya Phase III solar IPP, which statement is correct?

A. KAPP runs the procurement and MEWRE is the offtaker under a 30-year PPA

B. MEWRE runs the RFP and KAPP signs the PPA

C. CAPT runs the procurement because the value exceeds KD 75,000

D. The Council of Ministers selects the developer directly

Q2.

Audit Bureau prior review of your contract normally takes 8 weeks, and files like yours are returned for rework half the time (p = 0.5). Using $E[T] = t/(1-p)$, what is the expected duration in weeks?

Your answer: _____ weeks

Q3.

The Audit Bureau returns your Owner's Engineer consultancy contract with five observations. Three concern documents you had sent separately by email. What do you do?

A. Call the Audit Bureau examiner directly and explain the documents were emailed.

B. Answer all five observations point by point through the ministry liaison, put the missing documents in the official file, and add them to your pre-submission checklist.

C. Ask the Undersecretary to pressure the Bureau.

D. Resubmit only the three missing documents and argue the other two observations later.

Lesson 11 · Part 2 · full lesson · about 7 min

How a solar cell makes electricity

Why you care at MEWRE. Bidders will quote module wattages, temperature coefficients and bifacial gains at you from the first tender meeting. In Kuwait, how a module behaves at 70 °C matters more than its rating at 25 °C, and one curve shows you why.

The picture first

A solar cell is a pump driven by light. Think of a garden hose connected to that pump.

- **Nozzle wide open**: lots of water flows, but there is no pressure. That is **short circuit**: maximum current (I_{sc}), zero voltage.
- **Thumb fully over the end**: full pressure, nothing flows. That is **open circuit**: maximum voltage (V_{oc}), zero current.
- Useful work needs both flow and pressure. Somewhere in between is the setting that gives the most power: the **maximum power point**.

More sunlight drives the pump harder: more flow (current). Heat makes the pump leakier: the pressure (voltage) it can hold drops.

Where the analogy breaks: in a real cell the current stays almost constant as voltage rises, then falls off steeply near V_{oc} (the "knee" of the curve). The maximum power point sits just at that knee.

What is actually happening

Photons free electrons. A silicon cell is a large, thin diode. Light arrives as packets of energy (**photons**). A photon with enough energy (more than silicon's **band gap** of 1.12 eV, which means wavelengths below about 1,100 nm) knocks an electron loose. The cell is made of two layers, **p-type** and **n-type** silicon. Where they meet, the **p-n junction**, a built-in electric field sweeps the freed electrons to one side. Connect a load and they flow round the circuit: the **photocurrent**. Photon energy above the band gap ends up as heat, one reason commercial cells convert only about 23 to 26 % of the light (modules about 22 to 23 %).

The output is DC. Current flows one way only. A string of modules gives up to 1,500 V DC, and the **inverter** turns it into grid AC (next explainer).

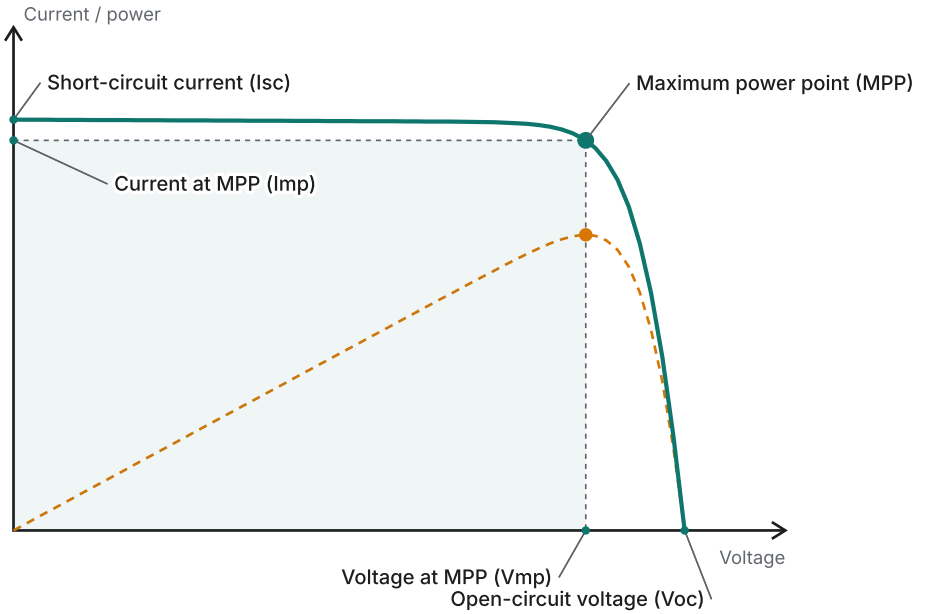
From a photon to grid power

1. **1 Photon arrives** — Needs more than 1.12 eV (wavelength below about 1,100 nm)
2. **2 Electron knocked loose** — Any energy above the band gap becomes heat
3. **3 p-n junction field** — Sweeps the freed electrons to one side
4. **4 Load connected** — Electrons flow round the circuit: the DC photocurrent
5. **5 Strings in series** — Voltages add, up to 1,500 V DC
6. **6 Inverter** — Turns DC into three-phase grid AC

Light in, one-way current out. Everything before the inverter is DC.

The I-V curve shows every combination of current and voltage the module can deliver:

- I_{sc} is proportional to sunlight: half the irradiance, half the current.
- V_{oc} barely changes with sunlight but **falls as the cell heats up**.
- At the **maximum power point** (MPP), V_{mp} is about 80 to 84 % of V_{oc} and I_{mp} about 94 to 96 % of I_{sc}. The inverter's **MPPT** (maximum power point tracker) hunts for this point all day.
- **Watts-peak (Wp)** on the datasheet are measured at **STC**: 1,000 W/m², 25 °C cell temperature and a standard spectrum (AM1.5). The field is rarely like that.



The hose picture on one curve: Isc is the nozzle wide open, Voc the thumb fully over the end, and the MPP sits at the knee, where current times voltage is largest.

Heat is the enemy. Every °C of cell temperature above 25 °C costs power, set by the **temperature coefficient** γ (in %/°C):

$$P = P_{STC} \times \frac{G}{1000} \times [1 + \gamma(T_c - 25)]$$

On a Kuwaiti summer afternoon at 50 °C ambient the cell runs at about **70 to 75 °C**. A TOPCon module (γ = -0.30 %/°C) at 1,000 W/m² and 72 °C gives $1 - 0.0030 \times 47 \approx 0.86$, so about **86 % of its rating** even in full sun.

PV module I-V and P-V curves

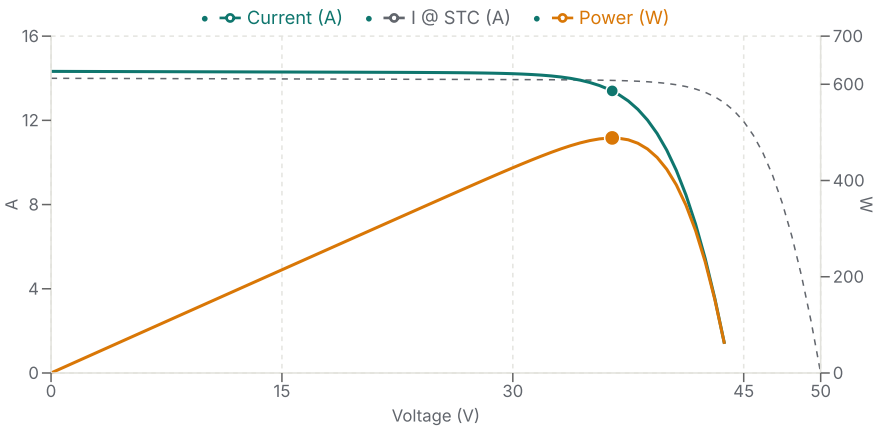
Single-diode model of a generic ~565 W module (144 half-cut cells)

Irradiance G 1,000 W/m²

Cell temperature 72 °C

STC | Kuwait winter noon | Kuwait summer noon | Dust haze

Pmp 489 W	vs STC −13.8 %
Voc 44.1 V	Isc 14.3 A
Vmp 36.5 V	Imp 13.4 A
Fill factor 0.773	Efficiency 18.9 % 2.58 m² module



What to notice: irradiance mostly scales current; heat mostly cuts voltage. At a 75 °C summer cell temperature the module loses roughly 15 % of its STC power — which is why string voltage design uses the coldest morning (max Voc) and yield uses the hottest afternoon.

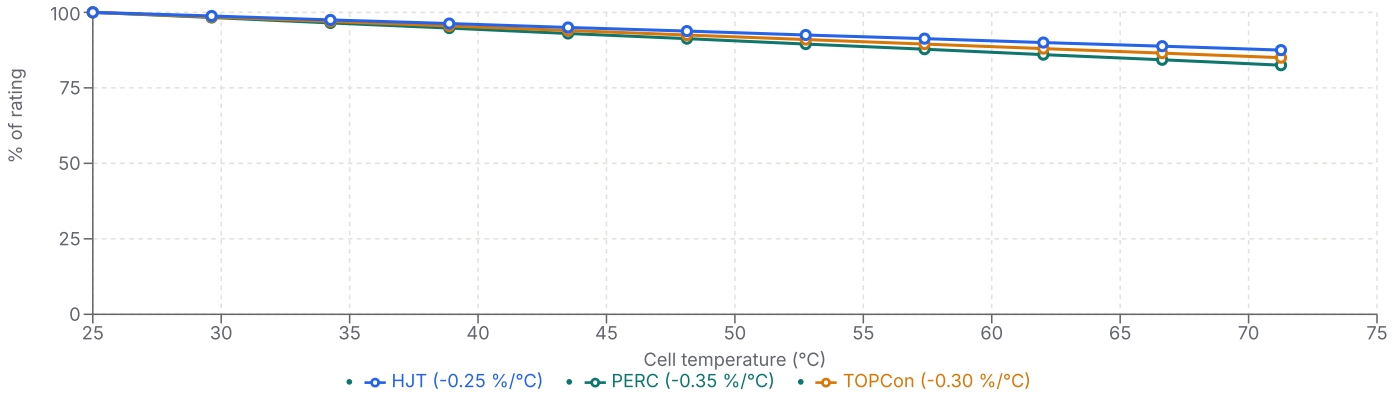
Set to a Kuwaiti summer afternoon (1,000 W/m², cell 72 °C). Slide the temperature back to 25 °C: voltage and power recover, current barely changes. Then drop irradiance to 500 W/m²: current halves, voltage hardly moves.

Module types.

- **PERC** (p-type): the older workhorse, γ about -0.35 %/°C.
- **TOPCon** (n-type): now mainstream, γ about -0.30 %/°C.
- **HJT** (heterojunction, n-type): best in heat, γ about -0.25 %/°C.

Better passivation of the cell surfaces gives higher voltage and a smaller heat loss, which is why n-type modules dominate Gulf tenders.

Output vs cell temperature at 1,000 W/m² (% of STC rating)



The lines fan out as the cell heats up. At the 70 to 75 °C of a Kuwaiti summer afternoon, HJT keeps about 4 to 5 points more output than PERC, every sunny day.

Bifacial modules have a glass rear and also collect light reflected from the ground. Over light desert sand (**albedo**, the share of light reflected, about 0.25 to 0.4) on trackers, they produce about **5 to 9 %** more energy. **Bifaciality** (rear efficiency relative to front) is about 0.70 for PERC, 0.80 for TOPCon and 0.85 to 0.90 for HJT.

Numbers you will hear

- **STC**: 1,000 W/m², 25 °C cell, AM1.5. Utility modules ≈ 580 to 720 Wp, ≈ 50 V each, efficiency ≈ 22 to 23 %.
- **Temperature coefficient of power**: PERC ≈ -0.35, TOPCon ≈ -0.30, HJT ≈ -0.25 %/°C.
- **Kuwait summer**: cell ≈ 70 to 75 °C at 50 °C ambient, so a TOPCon module delivers ≈ 82 to 87 % of nameplate in full sun.
- **Fill factor** $P_{mp}/(V_{oc}I_{sc})$ ≈ 0.79: how "square" the I-V curve is.
- **Bifacial gain** ≈ 5 to 9 % on trackers over sand (albedo 0.25 to 0.4).

Say it at MEWRE

- What temperature coefficient and which cell-temperature model did you assume at 50 °C ambient?
- Is the module PERC, TOPCon or HJT, and is the supplied bill of materials the one that was reliability-tested?
- What bifaciality, albedo and rear-shading loss sit behind your bifacial gain? Was the albedo measured on site?
- What does the module deliver at 1,000 W/m² and 70 °C, not at STC?

The classic trap.

Assuming a 600 Wp module gives 600 W in Kuwait's full sun. Wp is a laboratory rating at 25 °C; with the cell at 70 to 75 °C it delivers roughly 85 % of that. And remember the directions: more sun means more current (not much more voltage); more heat means less voltage.

Check yourself (answers at the end of Part 2)

Q1.

A 600 Wp TOPCon module (γ = -0.30 %/°C) is in full sun (1,000 W/m²) with a cell temperature of 70 °C. Roughly what power (W) does it produce?

Your answer: _____ W

Q2.

Irradiance falls from 1,000 to 500 W/m² while the cell temperature stays the same. What happens to the I-V curve?

A. Current roughly halves; voltage drops only slightly

B. Voltage halves; current stays the same

C. Both current and voltage halve

D. Nothing changes until irradiance falls below 200 W/m²

Q3.

A bidder claims a **15 % bifacial gain** for a tracker plant on Kuwaiti sand using TOPCon modules. What do you do?

A. Accept it: bifacial modules are proven technology.

B. Ask for the bifaciality, albedo (measured or assumed), rear-shading and rear-mismatch assumptions, and compare the result with the typical 5 to 9 % for trackers over sand.

C. Reject the bid for overstating its yield.

D. Ignore bifacial gain for every bidder in the evaluation.

Lesson 12 · Part 2 · full lesson · about 8 min

A utility PV plant from module to grid

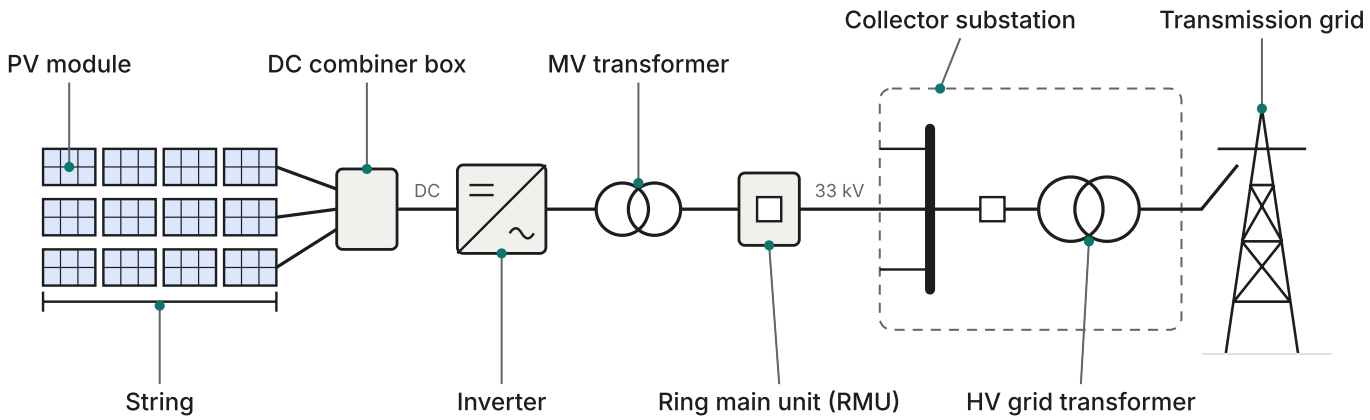
Why you care at MEWRE. When you open a Zone 1-type design submittal you will see single-line diagrams, inverter blocks, DC/AC ratios and tracker layouts. This is the map: what each box does, which ratings matter and where a PM asks questions.

The picture first

Think of a river system. Thousands of small springs feed little streams; streams join into creeks, creeks into a river, and the river reaches the sea through a wide estuary.

- **Modules** are the springs. Wired one after another in **series** they form a **string**: like batteries stacked in a torch, the voltages add.
- Strings wired side by side in **parallel** add their currents, in a combiner box or directly at an inverter.
- The **inverter** is a currency exchange counter: DC comes in, grid-quality AC goes out.
- **Transformers** raise the voltage step by step (33 kV, then 132 or 400 kV) so the power can travel far with little loss: the river widening into the estuary.

One more idea, the most debated one: designers deliberately install **more DC (modules) than AC (inverters)**. Picture an exchange counter sized for a normal busy hour rather than the single busiest minute of the year. At the very peak a few customers are turned away (**clipping**), but the counter is busy all day and you paid for a smaller one.



The river on one drawing: modules (springs) join into strings, strings meet at combiners, the inverter changes DC into AC, then the MV and HV transformers widen the flow toward the grid.

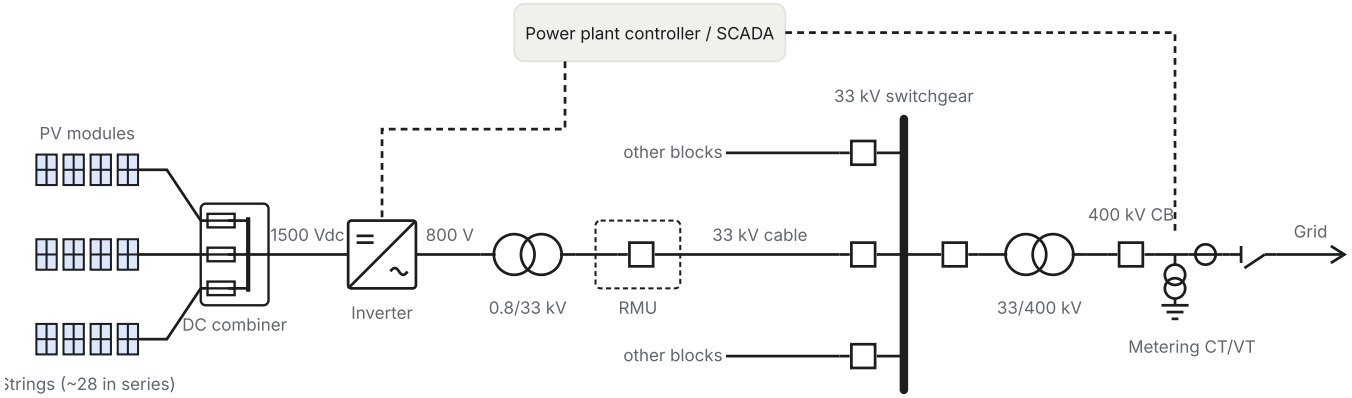
What is actually happening

The power path.

1. **Modules**: about 580 to 720 Wp and about 50 V each.
2. **Strings** of 27 to 30 modules in series, up to **1,500 V DC**, about 15 to 20 kWp each. The coldest morning sets the maximum length (voltage rises in the cold); the hottest afternoon sets the minimum (voltage must stay inside the inverter's MPPT window).
3. **Combiner boxes** parallel the strings with fuses (central design), or strings land directly on **string inverters**.
4. **Inverters** convert DC to three-phase AC.
   - **Central**: 3 to 5 MVA per unit on a skid. Lowest cost per watt at GW scale, but one fault takes 3 to 5 MW down.
   - **String**: 250 to 350 kW each, spread across the field. More MPPT inputs (less mismatch); a fault costs about 0.3 MW and the unit is swapped in hours. Both are bankable. In Kuwait the choice turns on heat derating (full power typically up to 35 to 45 °C ambient, then down to about 80 to 90 % at 50 °C, OEM-specific), dust filters and in-country service.
5. **MV skid**: inverter(s) plus a step-up transformer to **33 kV** plus a ring main unit (RMU, the MV switchgear), about 4 to 7 MVA per block.
6. **33 kV collector cables**, buried, loop back to the **collector substation**.
7. The **HV transformer** steps up to 132 or 400 kV at the **point of interconnection**. Zone 1's scope includes a **400 kV substation**; the Shagaya to Al-Wafra 400 kV line is a separate MEWRE contract, a classic interface risk.

Utility PV plant — single-line diagram

Click a component for its role, typical rating in a 1 GW Kuwait plant and PM checks



What to notice: power flows left to right through three voltage levels — 1500 V DC, ~800 V AC and 33 kV — before the grid transformer lifts it to transmission voltage. Each step is a contract interface and a commissioning hold point.

Select a component on the diagram (click or press Enter) to see its role, typical rating and what a PM should check.

Click each box from module to grid: role, typical rating and the PM checks. Notice where DC becomes AC and where the voltage steps up.

Central or string, at a glance.

	Central inverter	String inverter
Unit size	3 to 5 MVA on a skid	250 to 350 kW, spread across the field
Cost per watt at GW scale	Lowest	Higher
Output lost when one unit fails	3 to 5 MW	About 0.3 MW
After a fault	Repaired in place	Swapped in hours
MPPT inputs	Fewer	More (less mismatch)
Kuwait questions (both)	Heat derating, dust filters, in-country service	Heat derating, dust filters, in-country service

DC/AC ratio.

$$\mathrm{DC/AC} = P_{DC,STC} / P_{AC,rated}$$

that is, module nameplate (MWp) divided by inverter AC rating (MWac). Typical values are **1.15 to 1.40**. Real DC output rarely reaches nameplate (heat, dust and wiring keep it to about 80 to 90 % at best), so a 1:1 inverter would rarely run at full output. At 1.30 the clipped energy is about 3 % on a clear day and 0.5 to 3 % over a year. Since the grid connection is contracted in MWac, a little clipping is the price of using that connection for more hours.

DC/AC ratio and inverter clipping

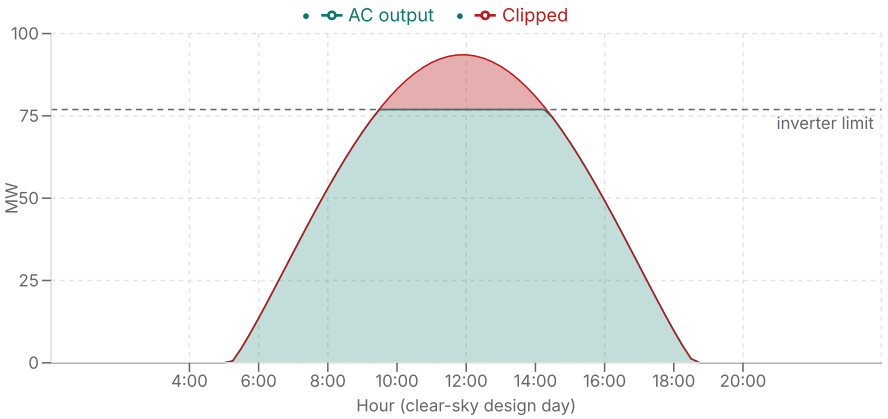
100 MWp DC array; inverter AC rating = DC ÷ ratio

DC/AC ratio 1.30

Inverter + MV cost 60 USD/kWac

Energy value 17 USD/MWh

AC capacity **76.9** MWac	Clipping (clear day) **7.0 %**
Clipping (annual e... **3.7 %**	Clipped energy **8,590** MWh/yr
Lost revenue **146** kUSD/yr	Inverter capex sav... **1.38** USD m vs ratio 1.00



What to notice: clipping only shaves the top of clear, cool middays, so a ratio of 1.2–1.35 loses little energy while cutting inverter, transformer and grid-connection capacity. Beyond ~1.4 the clipped energy grows quickly.

Annual estimate uses 12 representative monthly days scaled from indicative Kuwait GHI; a full 8760-h simulation (PVsyst) is needed for design.

Clear-sky day at DC/AC 1.3: the flat top is clipped energy. Push the ratio to 1.5 and back to 1.1 to see the trade between clipped energy and inverter cost.

Trackers vs fixed tilt. A **single-axis tracker** turns rows from east to west (about ±52° to ±60°) on a north-south axis. At 29°N, with the summer sun rising north of east, it gains about **15 to 25 %** more energy than fixed tilt, for a few US cents per Wp more. At low sun it **backtracks** (rotates back toward flat) so rows do not shade each other, and in high wind it **stows**. The ground coverage ratio (GCR, module width divided by row spacing) is typically 0.28 to 0.40.

Sun path, tracker angle and plane-of-array irradiance

Clear-sky model; local time UTC+3, longitude 47.7°E

Latitude 29.3 °

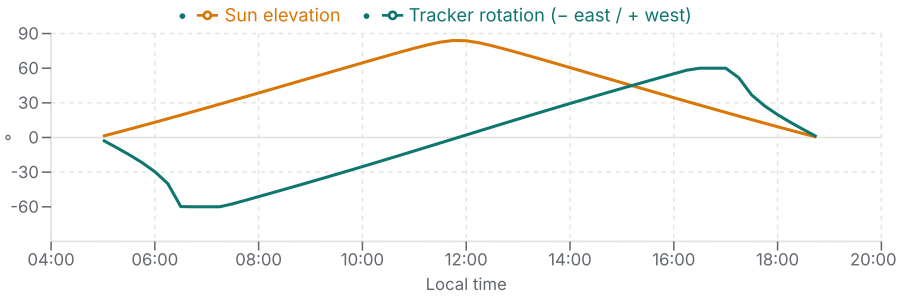
Day of year 172 (21 Jun)

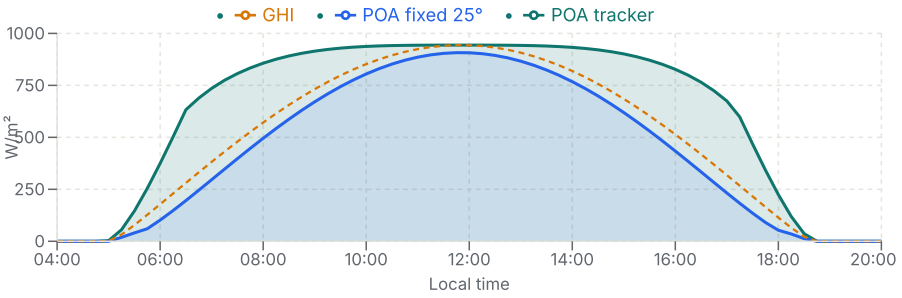
Fixed-tilt angle (south) 25 °

☑ Tracker backtracking (avoid row shading)

Ground coverage ratio 0.35

Day length **13.9** h	Noon elevation **84.1** °
Clear-sky GHI **7.94** kWh/m²	POA fixed tilt **7.14** kWh/m²
POA tracker **10.2** kWh/m²	Tracker gain **43.1** %





What to notice: the tracker's gain comes from mornings and afternoons (the broad shoulders), and is largest in summer when the sun rises north of east. In winter a well-tilted fixed array can match a horizontal tracker. Backtracking flattens the tracker at low sun to stop rows shading each other.

21 June at Shagaya: compare the tracker and fixed-tilt irradiance curves. The tracker's extra energy sits in the morning and late afternoon, when the sun is north of east or west. A clear June day shows a gain well above the annual 15 to 25 %; move to December (day 355) and the gain disappears, because a south-facing fixed tilt suits the low winter sun. The year averages the two.

✳ Numbers you will hear

- **MWp vs MWac**: MWp is DC module nameplate; MWac is inverter or grid export capacity. 1,000 MWac at DC/AC 1.3 means 1,300 MWp. Quote capacity at the peak hour in MWac.
- **Strings**: 27 to 30 modules at 1,500 V DC. **Inverters**: string 250 to 350 kW, central 3 to 5 MVA; MV blocks 4 to 7 MVA at 33 kV.
- **DC/AC** 1.15 to 1.40; annual clipping 0.5 to 3 %.
- **Trackers**: +15 to 25 % energy vs fixed tilt; GCR 0.28 to 0.40.
- **Inverter heat derating**: full power to 35 to 45 °C ambient, about 80 to 90 % at 50 °C (check the OEM curve).

Say it at MEWRE

- What is the plant's guaranteed export in MWac at the point of connection at 50 °C ambient?
- What DC/AC ratio did you choose, what annual clipping does the model show, and why is that the optimum?
- Which inverter derating curve did you use at 50 and 55 °C, and is it in the yield model?
- Central or string inverters: what spares and OEM service are in Kuwait, and what availability is guaranteed?
- At what wind speed do the trackers stow, and what do they do if they lose communications?

The classic trap.

Treating MWp and MWac as the same thing. A 1,300 MWp array on 1,000 MWac of inverters can never export more than about 1,000 MW. Clipping is a deliberate design choice, not a defect, as long as it was modelled and optimised.

Check yourself (answers at the end of Part 2)

Q1.

A 500 MWac plant is designed with a DC/AC ratio of 1.3. How much module capacity (MWp) is installed?

Your answer: _____ MWp

Q2.

Which is a real advantage of string inverters over central inverters?

A. A single failure takes only about 0.3 MW offline and the unit can be swapped in hours

B. They always have the lowest cost per watt at GW scale

C. They are not affected by dust or heat

D. They remove the need for MV transformers

Q3.

Why do single-axis trackers gain about 15 to 25 % energy over fixed tilt in Kuwait?

A. In summer the sun rises north of east and sets north of west; a tracker turning east to west faces the morning and evening sun that a fixed south-facing array wastes

B. They point the modules north in summer

C. They cool the cells by about 25 °C

D. They stop dust settling on the modules

Lesson 13 · Part 2 · full lesson · about 7 min

Where the MWh go: losses, PR and P50/P90

Why you care at MEWRE. The yield report sets the tariff bidders offer, how much debt the plant can carry and the energy MEWRE plans around for 30 years. You do not need to run PVsyst, but you must be able to read its loss diagram and say why a number looks too good.

The picture first

Think of your salary.

- Your **gross salary** is the sunlight landing on the site.
- Each line on the payslip takes a little: social security, a loan instalment, insurance. Each is small; together they matter.
- **Take-home pay** is the energy that reaches the grid.

The **performance ratio** is take-home pay divided by what you would have received with no deductions. It tells you how well the plant turns the sunlight it actually got into energy, whatever the weather that year.

Now the bonus. You expect roughly the usual amount, but it varies. **P50** is the figure you have a 50:50 chance of beating. **P90** is the figure you are 90 % sure to beat. A bank lending you money looks at P90, not P50.

Where the analogy breaks: payslip deductions are subtracted from the gross; plant losses multiply, each a percentage of what is left. At these sizes the difference is small, but that is how the yield model works.

What is actually happening

A yield model (usually **PVsyst**) runs hour by hour through a typical year and draws a **loss waterfall** from sunlight to grid energy. Typical values for a tracker plant with bifacial TOPCon modules in a Kuwait-like climate (illustrative):

- **Gains first**: tracking raises the irradiation on the modules by about 22 to 28 % over GHI; the bifacial rear adds about 5 to 8 %.
- **Temperature: about 7 to 10 %**, the largest single loss in Kuwait.
- **Soiling** (dust on the glass): about 2 to 5 % with a good cleaning regime. It rises fast if cleaning slips: Kuwaiti test studies saw 8.7 to 15.1 % with only monthly cleaning.
- **Angle of incidence (IAM)**: 1.5 to 2.5 %. **Shading and backtracking**: 0.5 to 1.5 %.
- **Mismatch**: 0.7 to 1.5 %. Modules in a string are never identical, and the weakest drags the rest.
- **DC cables** 1 to 1.5 %; **inverter efficiency** 1.2 to 2 %; **clipping** 0.5 to 3 %; **AC cables and transformers** 1 to 2 %; **unavailability** 0.5 to 1.5 %.

Energy loss waterfall

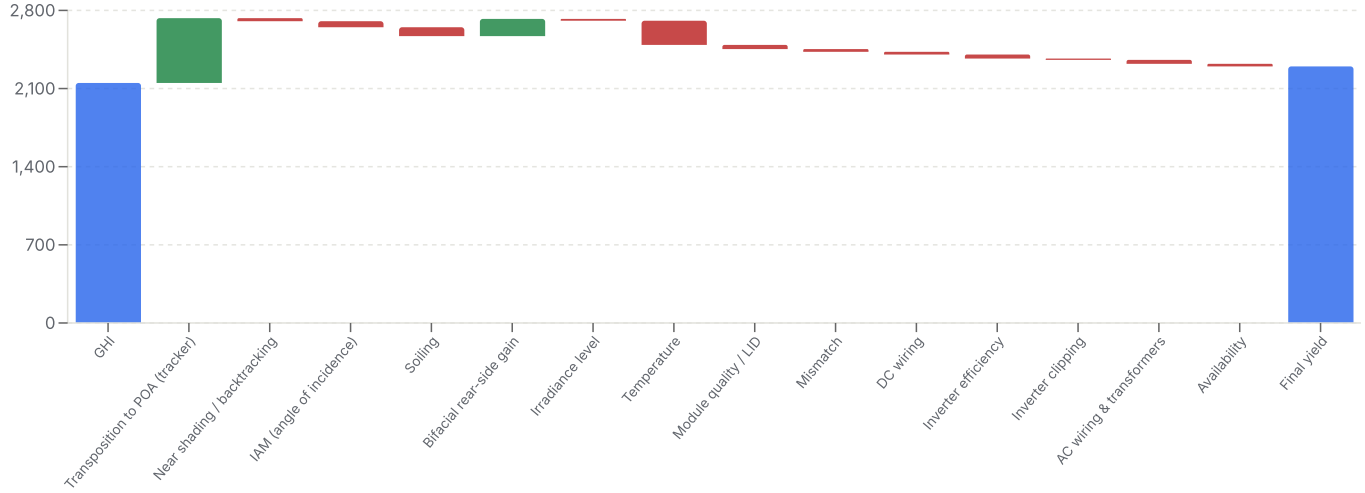
PVsyst-style chain from horizontal irradiation to grid-injected energy (multiplicative losses), in kWh/kWp

Annual GHI	2150	kWh/m²
DC capacity	1250	MWp

Transposition to POA (tracker)	-27	% gain
Near shading / backtracking	1	% loss
IAM (angle of incidence)	2	% loss
Soiling	3	% loss
Bifacial rear-side gain	-6	% gain
Irradiance level	0.6	% loss
Temperature	8	% loss
Module quality / LID	1.5	% loss
Mismatch	1	% loss
DC wiring	1	% loss
Inverter efficiency	1.5	% loss

Negative values are gains.

POA irradiation	Specific yield	Performance ratio	Annual energy
2,731 kWh/m²	**2,298** kWh/kWp	**84.2 %** vs front POA	**2,873** GWh



What to notice: losses multiply rather than add, so the total loss is less than the sum of the percentages. In Kuwait the temperature and soiling bars are the big controllable ones — module choice, row spacing for airflow and cleaning strategy all show up here.

Default values are illustrative for a single-axis-tracked bifacial plant in Kuwait. A bankable estimate comes from an independent energy-yield assessment with site data.

Defaults are a Kuwait-like tracker plant. Set GHI to 2,050 kWh/m², then raise temperature to 10 % and soiling to 5 %: watch the PR and specific yield fall.

Soiling & cleaning optimiser

Linear dust build-up, reset at each full-plant cleaning

Soiling rate 0.3 %/day

Soiling loss cap 15 %

Plant size 1,000 MWp

Specific yield (clean) 2,300 kWh/kWp

Energy value 15 USD/MWh

Cost per cleaning cycle 0.3 USD/kWp

Optimal interval **46** days	Cleanings / year **7.93**
Average soiling loss **6.9 %**	Lost energy **159** GWh/yr
Lost revenue **2.38** USD m/yr	Total annual cost **4.76** USD m/yr



What to notice: the optimum sits where the marginal cost of one more cleaning equals the revenue it recovers (roughly T* ≈ √(2·365·C / (r·E·price))). Kuwait dust events can multiply the soiling rate for days — real O&M schedules add event-driven cleans on top of the base cycle.

Soiling on its own: the sawtooth is dust building up day by day and dropping to zero at each cleaning. Raise the soiling rate (a shamal week) and see the optimal cleaning interval shorten and the lost energy grow.

Performance ratio (PR):

$$PR = Y_f / Y_r$$

The **final yield** Y_f is the kWh exported per kWp installed. The **reference yield** Y_r is the irradiation on the module plane in kWh/m² divided by 1 kW/m²: the number of hours of "full STC sun". Kuwait tracker plants: PR about **0.78 to 0.84** on a front-side basis. Bifacial gain inflates PR, so always check which definition is used.

Specific yield (kWh per kWp per year) is the number bidders compete on. Plausible for Kuwait trackers: about **2,000 to 2,300 kWh/kWp**. Claims of 2,300 to 2,500 are optimistic and unverified for Shagaya.

P50 and P90. Every input is uncertain: the long-term sunlight estimate (4 to 6 % without site data, 2.5 to 3.5 % with 12 months of site-adapted data), the model, soiling, availability and year-to-year weather (2.5 to 4 %). Combining the mid-range values of Track 0's table gives a total of about 6 % with good site data and 7 to 8 % without. Assuming a normal distribution, **P90 sits 1.28 standard deviations below P50**: at 6 % uncertainty, about 7.7 % below.

Energy-yield exceedance (P50 / P90)

Normal distribution: $P_x = P50 \cdot (1 - z_x \cdot \sigma)$; independent uncertainties combined in quadrature

P50 annual energy 2,400 GWh

Build σ from components | Total σ

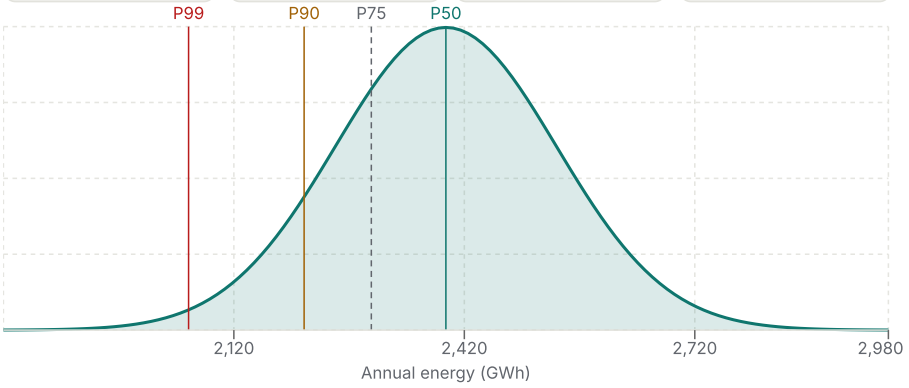
Total uncertainty σ 6 %

Combined σ 6.0 %

P75 2,303 GWh

P90 2,215 GWh
−7.7 % vs P50

P99 2,065 GWh



What to notice: P90 is the energy exceeded in 90 % of cases — lenders size debt on it (often the 1-year P90 for DSCR), while equity values the project on P50. Reducing resource uncertainty with on-site measurement directly raises P90 and so debt capacity.

An illustrative plant with P50 = 2,400 GWh/yr. Move σ from 4 % to 8 % and watch P90 drift away from P50: that gap is lost debt capacity.

Why lenders care. Debt is sized so that cash flow in a **P90 (or P99) year** still covers debt service with a margin (the DSCR covenant, for example 1.20 to 1.30×). Lower uncertainty means a higher P90, more debt, less equity and a **lower tariff for MEWRE**. That is why a ground measurement station installed a year before the tender pays for itself.

How a ground station lowers MEWRE's tariff

1. **1 Site data before the tender**
   12 months of site-adapted measurements
2. → **2 Resource uncertainty falls**
   From 4 to 6 % to 2.5 to 3.5 %
3. → **3 P90 moves up toward P50**
   Total uncertainty about 6 % instead of 7 to 8 %
4. → **4 More debt**
   Same DSCR covenant (1.20 to 1.30×) on a higher P90 cash flow
5. → **5 Less equity**
   Cheaper money funds more of the plant
6. → **6 Lower tariff**
   Bidders can offer MEWRE a lower price

Each box causes the next. The station is cheap; the tariff it lowers runs for 30 years.

✳ **Numbers you will hear**

- **Losses**: temperature 7 to 10 % (largest); soiling 2 to 5 % (insist on site data); inverter 1.2 to 2 %; clipping 0.5 to 3 %.
- **PR**, Kuwait trackers ≈ 0.78 to 0.84 (front-side). **Specific yield** ≈ 2,000 to 2,300 kWh/kWp plausible; 2,300 to 2,500 optimistic (unverified).
- **Exceedance**: $P_{90} = P_{50}(1 - 1.282\,\sigma)$; P99 uses 2.326. 1-year P90 for liquidity reserves, 10-year P90 for debt sizing.
- **DSCR** covenants ≈ 1.20 to 1.30× on P90 or P99 cash flow.
- **Sanity check**: specific yield ≈ GHI × (POA/GHI) × PR, e.g. 2,100 × 1.25 × 0.82 ≈ 2,150 kWh/kWp.

Say it at MEWRE

- Which PR definition is this: front-side or including rear irradiance, temperature-corrected or not?
- Which soiling profile did you assume month by month, and what cleaning frequency and water source does it imply?
- Show me the uncertainty table: which components, and is the year-to-year term for 1 year or 10 years?
- Your specific yield is 2,450 kWh/kWp: which GHI, POA/GHI ratio and PR get you there?
- On which P-value and DSCR did the lenders size the debt?

The classic trap.

Thinking P90 is a pessimistic scenario someone picked. It is a statistical level computed from the uncertainty: a 90 % chance that the year's energy beats it. And a high PR is not automatically good news: bifacial gain or a generous irradiance basis can inflate it.

Check yourself (answers at the end of Part 2)

Q1.

A plant's P50 is 1,000 GWh/yr with a combined uncertainty of 7 % (1σ). Using z = 1.282, what is the P90 in GWh/yr?

Your answer: _____ GWh

Q2.

In a PV yield report for a Kuwaiti site, which single loss is usually the largest?

A. Temperature (about 7 to 10 %)

B. Inverter efficiency

C. DC cable losses

D. Module mismatch

Q3.

For the same Shagaya site and tracker design, bid A claims 2,480 kWh/kWp and bid B claims 2,150 kWh/kWp. What is your move?

A. Rank A higher: more energy per kWp means a better plant.

B. Ask both for their yield reports and check GHI, POA/GHI, PR, soiling and the uncertainty table against an independent check; flag A as optimistic until justified.

C. Average the two figures for both bids.

D. Disqualify bid A immediately.

Lesson 14 · Part 2 · quick lesson · about 4 min

Concentrated solar power: mirrors, heat and a steam turbine

Why you care at MEWRE. Shagaya hosts Kuwait's only utility CSP plant: a 50 MW parabolic trough with about 10 hours of molten-salt storage, whose performance since 2019 is not public. Sooner or later someone will ask you whether it is worth keeping, or a developer will pitch CSP for the evening peak. You need to know how it works to ask the right questions.

Quick lesson: the essentials only. The full version, with more drawings and worked numbers, is on the website under Trial week.

The picture first

Think of three things in a kitchen: a **magnifying glass**, a **kettle** and a **thermos flask**.

- The magnifying glass focuses sunlight onto a hot spot. It only works with **direct** sun: on a hazy day there is plenty of light around you, but no sharp hot spot.
- The heat boils water, and the steam spins a turbine, like a kettle driving a steam engine.
- The thermos keeps heat for later. Heat is cheap to store; electricity is not.

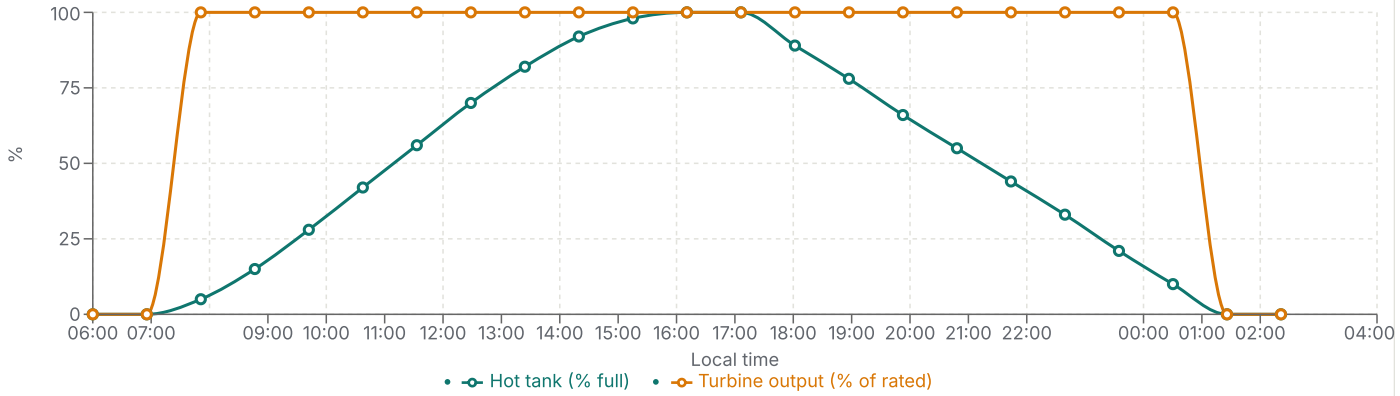
That is CSP: mirrors make heat, heat makes steam, steam makes electricity, and a hot "thermos" of molten salt lets it keep generating after sunset. Where the picture breaks: in a trough plant the mirrors do not boil water directly. They heat an oil loop, and the oil heats the salt and the steam through heat exchangers.

What is actually happening

Two main designs concentrate the sun:

- **Parabolic trough** (the workhorse, about 75 % of installed CSP worldwide). Long curved mirrors in north-south rows track the sun east to west and focus it onto a **receiver tube** (a black steel pipe inside a vacuum glass tube). A **heat transfer fluid (HTF)**, usually synthetic oil, is heated from about 293 °C to 393 °C. Decades of bankable operating history.
- **Solar tower**. A field of flat, individually tracking mirrors (**heliostats**) focuses onto a receiver at the top of a tower, heating **molten salt** to about 565 °C. Hotter means a more efficient turbine and about three times more energy stored per tonne of salt, but fewer reference plants, and the hot tank has been the weak point (Noor III in Morocco and Crescent Dunes in the US both had long storage-related outages).

A trough plant's day with storage (illustrative summer day)



The mirror field runs the turbine and fills the hot tank during the day; after sunset the tank drains and the turbine keeps going at full load until about 03:00 (1,200 MWh_th × 0.37 ÷ 50 MW ≈ 9 h). Shape only, not Shagaya data.

Numbers you will hear

- **Shagaya CSP**: 50 MW trough, about 10 h molten-salt storage, quoted as 1,200 MWh_th (Worley), dry cooled; full operation from about February 2019 (sources differ). Source: IEA policy database, Reuters Events.
- **Storage check**: $1{,}200 \times 0.37 / 50 \approx 9$ h at full load, close to the quoted 10 h. Always ask whether storage hours are gross or net, at full or part load.
- **Trough** oil 293 to 393 °C, turbine cycle about 37 to 38 %; **tower** salt 290 to 565 °C, about 41 to 44 %.
- **Capacity factor**: 25 to 28 % without storage, 45 to 55 % with 10 h storage.
- **Later CSP**: the planned 200 MW / 5 h Phase 2 CSP plant was never procured; Shagaya Phase III zones are PV.

Say it at MEWRE

- Is the yield based on ground-measured DNI with a P90, or on satellite data only?
- When you say 10 hours of storage, is that gross or net, at full load, and from usable or total salt?
- What is the turbine output at 48 °C ambient with dry cooling, compared with the design point?
- Since the design-build-operate period ended, who operates Shagaya CSP, and can we get monthly generation and measured DNI since 2019?
- What is the freeze-protection procedure if heat tracing fails?

The classic trap.

Thinking CSP is 'PV with mirrors' that works on any sunny day. CSP turns sunlight into heat and then into electricity through a steam turbine, and it uses only the direct beam, so dust and haze that barely bother PV can cut CSP output sharply. Its real advantage is not efficiency but cheap heat storage that lets it generate after sunset.

Check yourself (answers at the end of Part 2)

Q1.

Why does a CSP plant lose much more output than a PV plant on a hazy, dusty day?

A. Mirrors can only focus the direct beam (DNI), and haze scatters the beam into diffuse light

B. Dust blocks the steam turbine's air intake

C. Molten salt freezes when the sky is hazy

D. CSP mirrors are less reflective than PV modules

Q2.

A trough plant's storage holds 1,200 MWh of heat. The turbine converts heat to electricity at 37 % and is rated 50 MW. How many hours can it run at full load from storage alone?

Your answer: _____ h

Q3.

A colleague's briefing note says "Shagaya CSP produces 225,000 MWh a year". What do you do?

A. Leave it: the figure appears in published sources

B. Flag that it is a reported target of unclear scope (51 % CF if CSP only, 37 % if the whole 70 MW), and that actual output since 2019 is not public; ask KISR for the operating data

C. Replace it with 50 MW × 8,760 h = 438,000 MWh

D. Delete all Shagaya numbers from the note

Lesson 15 · Part 2 · full lesson · about 7 min

Batteries: MW is the tap, MWh is the tank

Why you care at MEWRE. MEWRE has announced a large battery programme for the evening peak (reported as 1 to 1.5 GW / 4 to 6 GWh, to be verified). Every bid, test and guarantee for it will be written in MW, MWh, hours, efficiency and depth of discharge, measured at some boundary. Mixing those up costs real money.

The picture first

A water tank with a tap.

- The **tank size** is the energy, in **MWh**: how much is stored.
- The **tap size** is the power, in **MW**: how fast it can flow in or out.
- **Duration** is how long a full tank lasts with the tap fully open: 2,000 MWh through a 500 MW tap lasts **4 hours**.

You never drain the tank to the last drop or fill it to the brim (that is the **depth of discharge**), and some water spills on the way in and on the way out (that is the **round-trip efficiency**). Where the picture breaks: a battery tank slowly shrinks with age and heat. That is the next explainer.

What is actually happening

A **battery energy storage system (BESS)** has two independent sizes:

- **Power (MW)**, set by the **PCS** (power conversion system: bidirectional inverters that turn the battery's DC into grid AC and back) and by how much current the cells can take.

- **Energy (MWh)**, set by how many cells you buy.

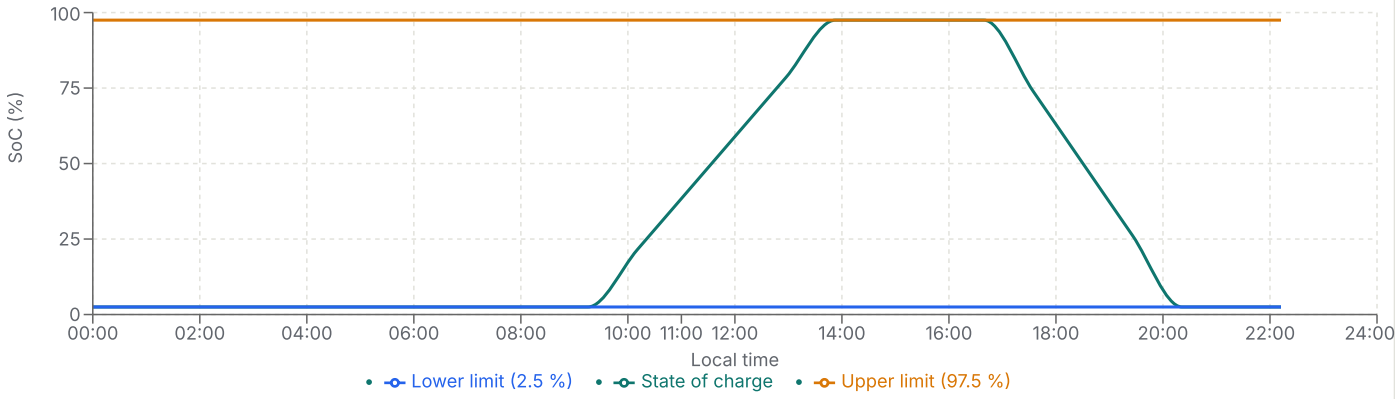
Duration and C-rate say the same thing two ways:

$$\text{duration (h)} = \frac{\text{energy (MWh)}}{\text{power (MW)}}, \qquad \text{C-rate} = \frac{1}{\text{duration}}$$

A 4-hour system runs at **0.25C**, a 2-hour system at 0.5C, a 1-hour system at 1C. Lower C-rate means less heat and slower aging, but more cells per MW.

State of charge (SoC) is how full the battery is. The battery management system keeps it inside a window, for example 2.5 to 97.5 %; the width of that window (here 95 %) is the **depth of discharge (DoD)**. Usable energy = nameplate × DoD × state of health.

State of charge of a 4-hour battery over a summer day (illustrative)

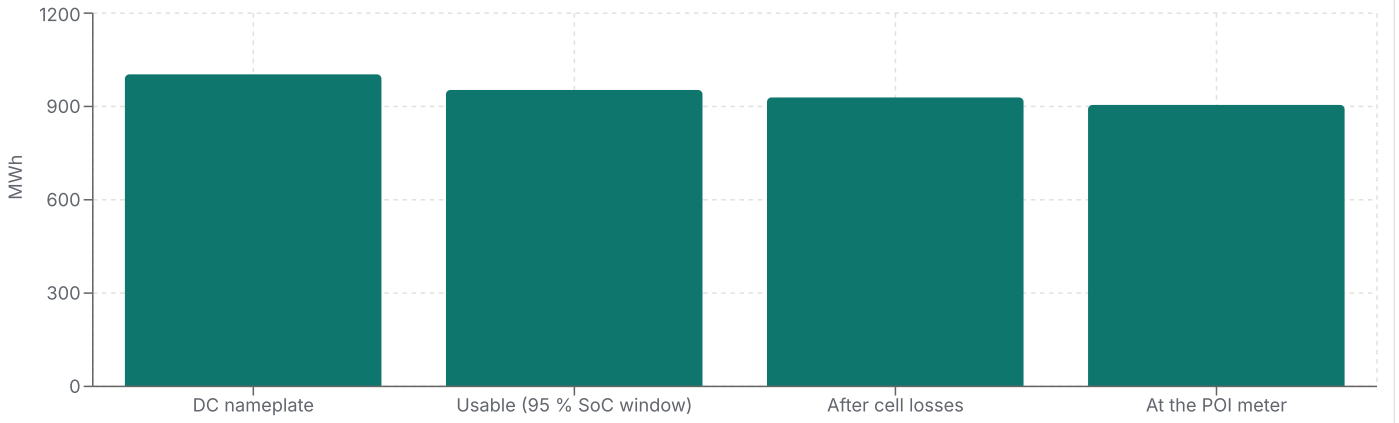


Charge from PV between 10:00 and 15:00, sit full through the afternoon, empty the tank between 18:00 and 22:00. The SoC never leaves the window: the gap between the two limit lines is the DoD (95 %).

Round-trip efficiency (RTE) = energy out ÷ energy in. The cells alone lose only 4 to 6 %, but every step to the grid loses a little more, and in a 50 °C summer the air conditioning (**auxiliaries**) runs all day.

The boundary matters most. Energy flows cells → PCS → transformers → **point of interconnection (POI)** revenue meter. A plant sold as "1,000 MWh" (DC nameplate) can deliver about 905 MWh at the POI. MEWRE only cares about the POI.

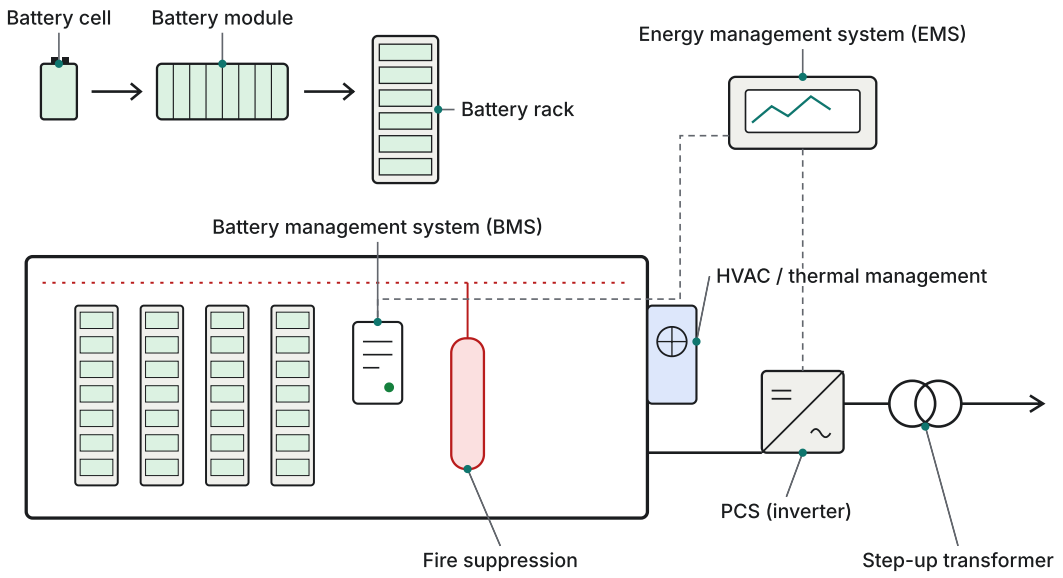
A battery sold as "1,000 MWh": what reaches the meter (Track 0 worked example, MWh)



Each step loses a little and the bars step down: about 98 MWh gone before the meter, plus HVAC auxiliaries drawn from the grid on top. Only the last bar is what MEWRE buys.

The three brains:

- **BMS** (battery management system): watches the voltage, current and temperature of every cell, estimates SoC and health, balances cells and opens contactors on a fault.
- **PCS**: does the conversion and the grid-code behaviour (reactive power, fault ride-through).
- **EMS** (energy management system): decides when to charge and discharge, following the control centre's dispatch.



From cell to grid: cells, modules and racks (watched by the BMS), then the PCS and transformer, with HVAC, fire suppression and the EMS around them. Every step between the cell and the POI loses a little energy.

Chemistry: LFP vs NMC. Both are lithium-ion.

- **LFP** (lithium iron phosphate): 3.2 V per cell, 6,000 to 10,000+ cycles, goes into thermal runaway later and releases no oxygen. Heavier, which does not matter for a stationary plant. The utility standard.
- **NMC** (nickel manganese cobalt): more energy per kg (220 to 280 vs 160 to 190 Wh/kg, good for cars), but 2,000 to 5,000 cycles and earlier runaway with oxygen release. Avoid for a Kuwaiti grid battery.

Today's standard is a 314 Ah LFP cell of about 1 kWh; a 20-ft container holds about 5 MWh.

	LFP	NMC
Cycle life	6,000 to 10,000+	2,000 to 5,000
Energy per kg	160 to 190 Wh/kg	220 to 280 Wh/kg
Thermal runaway	Later, no oxygen released	Earlier, with oxygen release
Best use	Stationary grid storage	Cars, where weight matters

BESS sizing, degradation & augmentation

Requirement = MW × hours delivered at the point of connection, every year

Power 500 MW

Duration 4 h

Depth of discharge 90 %

Round-trip efficiency 87 %

Cycles per year 365

Calendar fade 1 %/yr

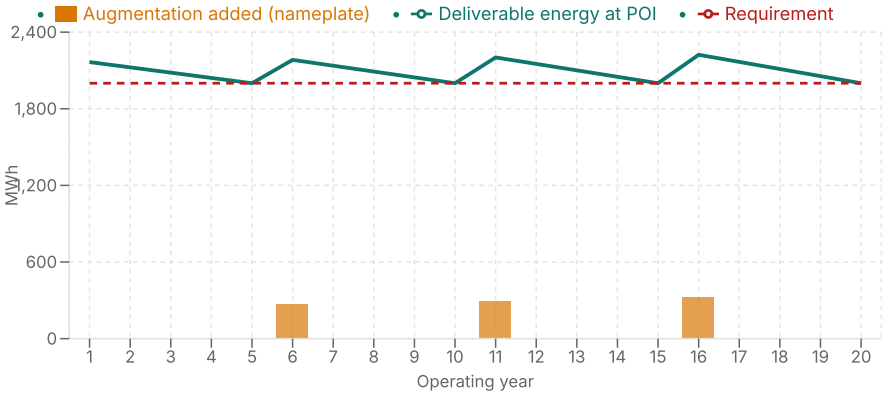
Cycle fade 2.5 % per 1000 cycles

Project life 20 years

Strategy Periodic augmentatic

Augmentation interval 5 years

Initial nameplate	C-rate	Total augmentation	Years below requi...
2,580 MWh	0.19 1/h	887 MWh	0



What to notice: nameplate must exceed MW × h because of DoD and round-trip losses, and capacity fades every year. Overbuilding buys all the fade up front; periodic augmentation defers capex to when cells are cheaper but needs space, compatible equipment and contract provisions planned from day 1.

Lifetime energy delivered	Total installed (incl. augmentation)
14,600 GWh	3,470 MWh

500 MW for 4 hours, the size of the reported first phase. Change DoD and RTE and watch the usable energy move.

✳ Numbers you will hear

- **0.25C (4 h)** is typical for evening energy shifting; Gulf procurements and MEWRE's reported plan sit there.
- **RTE at the POI**: about 90 % excluding auxiliaries, about 85 % including auxiliaries in a 50 °C summer. Cells alone: 94 to 96 %.
- **Usable energy** is usually guaranteed at about 90 to 95 % DoD.
- **LFP warranties**: about 6,000 to 8,000 equivalent full cycles or 15 to 20 years, whichever comes first.
- **Kuwait (verify before use)**: 1 to 1.5 GW / 4 to 6 GWh announced November 2025; sites at Al-Mutlaa and Jaber Al-Ahmad (1.5 GW) reported June 2026; first 500 MW targeted for summer 2027. Counterparty, procurement route and chemistry are unverified.

Say it at MEWRE

- Is that MWh at the DC nameplate or at the POI revenue meter, and at what ambient temperature?
- Does the round-trip efficiency include auxiliaries such as the HVAC?
- What SoC window does the usable energy assume?
- Which cell, from which factory, and is a change of cell treated as a variation?
- Is the PCS rated to deliver full MW and the required reactive power at 50 °C?

The classic trap.

Reading '500 MW / 2,000 MWh' as a single size, or assuming the MWh is what reaches the grid. MW and MWh are two separate ratings, and the MWh shrinks between the cells and the POI meter through the SoC window, conversion losses and auxiliaries: a plant sold as 1,000 MWh can deliver about 905 MWh at the POI. Pin every number to the POI, at 50 °C, with auxiliaries defined.

Check yourself (answers at the end of Part 2)

Q1.

A BESS is rated 300 MW / 1,200 MWh at the POI. What is its C-rate?

Your answer: _____ C

Q2.

A battery has 1,000 MWh of DC nameplate energy, a 95 % SoC window (DoD) and a state of health of 0.97 after its first year. How much usable DC energy does it have?

Your answer: _____ MWh

Q3.

Why do utility batteries in a hot climate like Kuwait's almost always use LFP rather than NMC?

A. LFP lasts more cycles and goes into thermal runaway later, without releasing oxygen

B. LFP stores more energy per kilogram

C. NMC cannot be connected to the grid

D. LFP needs no cooling

Answers: Part 2

Kuwait's power system in ten minutes

Q1.

About 17 GW

About **17 GW**. Forced outages, planned maintenance and high-temperature derating remove about 4 GW at the peak. 14 GW is the gas-fired part of the fleet, not the available capacity.

Q2.

-4.15 %

$RM = (16.275 - 16.98)/16.98 = -0.705/16.98 \approx -4.15\%$. About 705 MW had to be imported, saved or shed.

Q3.

Compare solar with the fuel it displaces (about 26 to 37 fils/kWh) or the cost of supply, not with the subsidised tariff

The 2-fils tariff is a subsidised price, not a cost. The economic value of a solar kWh is the fuel not burned, valued at its export or import price. KPC's 25 fils is a transfer between two public bodies, not a national saving.

Who decides what: MEWRE, KAPP, KISR and the approval chain

Q1.

KAPP runs the procurement and MEWRE is the offtaker under a 30-year PPA

KAPP procures under PPP Law 116/2014, the Higher Committee for PPP approves, and MEWRE signs the 30-year PPA as offtaker. CAPT governs conventional public tenders, not PPP procurement.

Q2.

16 weeks

$E[T] = 8/(1 - 0.5) = 16$ weeks. A 50 % return probability doubles the step. Cutting p to 0.2 with a complete file brings it down to 10 weeks.

Q3.

Answer all five observations point by point through the ministry liaison, put the missing documents in the official file, and add them to your pre-submission checklist.

Every return is a full review cycle. Respond formally and completely through the proper channel, then fix the process that caused the return.

How a solar cell makes electricity

Q1.

519 W

$P = 600 \times 1 \times [1 - 0.0030 \times (70 - 25)] = 600 \times 0.865 = 519$ W. Heat alone takes about 13.5 % off the nameplate.

Q2.

Current roughly halves; voltage drops only slightly

I_{sc} is proportional to irradiance, while V_{oc} grows only logarithmically with it. Power therefore falls roughly in proportion to irradiance.

Q3.

Ask for the bifaciality, albedo (measured or assumed), rear-shading and rear-mismatch assumptions, and compare the result with the typical 5 to 9 % for trackers over sand.

Bifacial gain over desert sand on trackers is typically 5 to 9 % (TOPCon bifaciality about 0.80, albedo 0.25 to 0.4). Larger claims need justified inputs, ideally measured albedo.

A utility PV plant from module to grid

Q1.

650 MWp

$P_{DC} = 1.3 \times 500 = 650$ MWp. The extra 150 MWp raises output in the mornings, evenings and hazy hours; at midday on clear days the inverters clip at 500 MWac.

Q2.

A single failure takes only about 0.3 MW offline and the unit can be swapped in hours

String inverters (250 to 350 kW) limit the impact of a fault and are replaced rather than repaired. Central inverters are usually cheapest per watt at GW scale, but one fault costs 3 to 5 MW. Both need MV transformers and both suffer heat derating.

Q3.

In summer the sun rises north of east and sets north of west; a tracker turning east to west faces the morning and evening sun that a fixed south-facing array wastes

At 29°N the long summer days and north-of-east sunrise favour tracking. The extra energy arrives in the morning and late afternoon, which also fits Kuwait's long afternoon load.

Where the MWh go: losses, PR and P50/P90

Q1.

910.3 GWh

$P_{90} = 1{,}000 \times (1 - 1.282 \times 0.07) = 1{,}000 \times 0.9103 = 910.3$ GWh/yr. Cutting the uncertainty to 5 % would raise P90 to about 936 GWh/yr, and with it the debt the plant can carry.

Q2.

Temperature (about 7 to 10 %)

With cells at 70 to 75 °C on summer afternoons, the temperature loss is about 7 to 10 % of annual energy. Inverter, DC cable and mismatch losses are each around 1 to 2 %.

Q3.

Ask both for their yield reports and check GHI, POA/GHI, PR, soiling and the uncertainty table against an independent check; flag A as optimistic until justified.

Check the claim with specific yield ≈ GHI × (POA/GHI) × PR. Reaching 2,480 needs GHI near 2,300 kWh/m² or an unusually high PR, neither verified for Shagaya.

Concentrated solar power: mirrors, heat and a steam turbine

Q1.

Mirrors can only focus the direct beam (DNI), and haze scatters the beam into diffuse light

A mirror can only redirect rays arriving from a known direction, the sun's disc. Haze turns direct light into diffuse light, which PV can still use but CSP cannot. Because receiver losses stay roughly fixed, a 10 % DNI shortfall costs roughly 12 to 15 % of the electricity.

Q2.

8.88 h

Electricity from storage $= 1{,}200 \times 0.37 = 444$ MWh. At 50 MW: $444/50 \approx 8.9$ h. That is close to, but not the same as, the "about 10 h" usually quoted for Shagaya, which is why you ask how storage hours are defined.

Q3.

Flag that it is a reported target of unclear scope (51 % CF if CSP only, 37 % if the whole 70 MW), and that actual output since 2019 is not public; ask KISR for the operating data

$225{,}000/(50 \times 8{,}760) = 51$ % and $225{,}000/(70 \times 8{,}760) = 37$ %. A 51 % CF for a 10 h trough in a dusty site with dry cooling would be world-class, so the 70 MW reading is more plausible. Either way it is a reported target, not measured output.

Batteries: MW is the tap, MWh is the tank

Q1.

0.25 C

C-rate $= P/E = 300/1{,}200 = 0.25$C, i.e. a 4-hour system: the typical evening energy-shifting design.

Q2.

921.5 MWh

Usable $= 1{,}000 \times 0.95 \times 0.97 = 921.5$ MWh, before any conversion losses on the way to the POI.

Q3.

LFP lasts more cycles and goes into thermal runaway later, without releasing oxygen

For a stationary plant, weight hardly matters, while cycle life and thermal stability do. LFP gives 6,000 to 10,000+ cycles and a later, oxygen-free runaway. It still needs good cooling: aging roughly doubles with every 10 °C.

PART 3 OF 4

Grid connection and project delivery

By the end you can: Follow a grid-connection discussion and walk a project from feasibility to handover, reading a schedule, an earned-value report and a risk register.

8 lessons · about 42 minutes

Lesson 16 · Part 3 · quick lesson · about 5 min

Transformers and voltage levels: why 400 kV

Why you care at MEWRE. A Shagaya-scale PV plant's output goes through two sets of transformers before it reaches the 400 kV grid, and the main transformer is one of the longest-lead, most expensive items on the project. Knowing why voltages step up, and how a transformer is rated for a 50 °C summer, lets you follow the grid-connection and procurement discussions from day one.

Quick lesson: the essentials only. The full version, with more drawings and worked numbers, is on the website under Trial week.

The picture first

Think of moving water through a pipe. To move a lot of water through a thin pipe you need high pressure; at low pressure you need a fat pipe, and friction wastes a lot of the effort. Voltage is the pressure, current is the flow rate, and losses in the line are the friction.

Friction in a wire grows with the **square** of the current. So the trick for long distances is: raise the pressure (voltage), shrink the flow (current) for the same power, and the losses collapse. A **transformer** is the device that trades voltage for current, up at the power station and down near the customer.

Where the analogy breaks: a transformer has no moving parts and does not "pump". It only works with AC, because it relies on a changing magnetic field.

What is actually happening

A transformer is two **coils (windings)** wound on the same iron **core**. AC in the first coil makes a magnetic field that keeps changing direction (Day 1's magnetism). The core carries that changing field through the second coil, and a changing field induces a voltage in it.

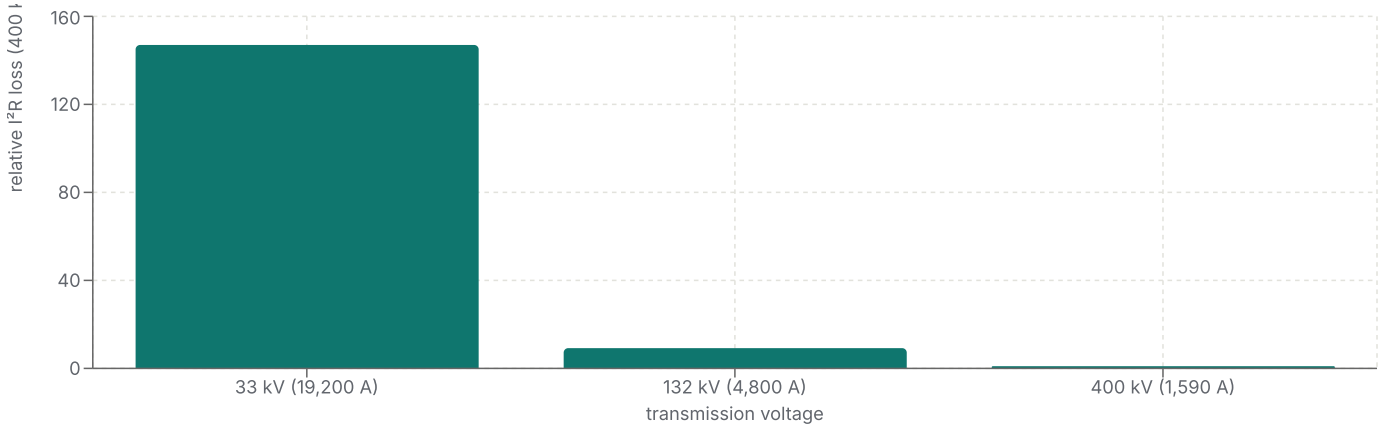
- The voltage on each side is set by the **turns ratio**: a coil with 12 times more turns has 12 times the voltage. A 400/33 kV transformer has a ratio of about 12.1.
- Power in ≈ power out (large units are 99.5 to 99.8 % efficient), so **current goes the other way**: 12 times the voltage means one-twelfth of the current.

Why 400 kV. For the same power, current falls in proportion to voltage and losses fall with the current squared. Moving 1,100 MW at power factor 1:

Voltage	Current	Relative I^2R loss (same wire)
33 kV	about 19,200 A	about 147 ×
132 kV	about 4,800 A	about 9 ×
400 kV	about 1,590 A	1 ×

The price is bigger towers, wider clearances and costlier substations, worth it for bulk power over long distances.

Line losses for 1,100 MW on the same wire, relative to 400 kV



The 400 kV bar is barely visible next to 33 kV: about 12 times less current means about 147 times less heat in the wire. That is the whole case for stepping up before leaving site.

🇰🇼 Numbers you will hear

- **Voltage levels**: **400 kV** for bulk transmission and new renewable connections is **confirmed** in Track 0 (Shagaya to Al-Wafra 400 kV line; the Shagaya Phase III Zone 1 IPP scope includes a 400 kV substation). **132 kV** (sub-transmission) and **33 / 11 kV** (distribution, PV collector systems at 33 kV) are commonly cited but flagged **unverified** in Track 0; an older intermediate level (300 or 275 kV) is also unverified.
- **Ratios**: 400/33 kV main transformers, 0.8/33 kV inverter transformers; impedance about 12 to 16 % for 400/33 kV and 132/33 kV units.
- **Ambient basis**: IEC default 40 / 30 / 20 °C (max / hottest month / year); Kuwait specifications should state the site values (for example max 55 °C).
- **Lead time**: large power transformers have taken **2 to 4 years** in the 2023 to 2026 market.

Say it at MEWRE

- On what ambient temperature is this transformer's MVA rating guaranteed: IEC normal conditions or our site values?
- What is the ONAN rating, and can we carry the critical load at 50 °C with one fan group out of service?
- Are the transformer losses guaranteed, with penalties that match how we evaluated the bids?
- What is the delivery lead time, and has the transport route to site been surveyed for a unit of this weight?

The classic trap.

Thinking a transformer's nameplate MVA is what it can carry on a Kuwaiti summer afternoon. The nameplate is usually valid for IEC normal ambient (40 °C maximum); at 50 °C plus, a standard unit must be derated or it ages fast. The rating that matters is the one at site conditions, written into the specification.

Check yourself (answers at the end of Part 3)

Q1.

The same power is sent over the same conductor at 400 kV instead of 132 kV. By roughly what factor do the I^2R losses fall?

Your answer: _____ times

Q2.

A contractor offers a "standard" 400/33 kV transformer rated to IEC normal service conditions. Why is that a problem in Kuwait?

A. The IEC rating assumes at most 40 °C ambient; Kuwait exceeds it, so the unit will run hot and age faster unless derated or designed for site ambient

B. IEC transformers cannot operate at 50 Hz

C. IEC ratings are in MW, but Kuwait uses MVA

D. Standard transformers cannot step up from 33 kV to 400 kV

Q3.

A 1,100 MW PV plant exports at power factor 1 at 400 kV. What is the line current, in amperes?

Your answer: _____ A

Lesson 17 · Part 3 · full lesson · about 7 min

Connecting a plant: the grid code and interconnection studies

Why you care at MEWRE. A plant earns nothing until it is connected, and the connection is where schedules slip: studies, models, witnessed tests and the line itself. Kuwait has no published renewable grid code that Track 0 could verify, so knowing what a grid code normally asks lets you ask MEWRE Transmission the right questions.

The picture first

A plant joining the grid is like a new driver joining a motorway. You need a licence and you must follow the rules:

- Drive within the limits (**operating ranges** of frequency and voltage).
- Do not slam on the brakes when something happens ahead (**fault ride-through**: stay connected through a short disturbance).
- Help keep traffic flowing (**frequency response**) and give way when asked (**reactive power** for voltage).
- No sudden swerves (**ramp rates**), and keep your exhaust clean (**power quality**).

Before you are allowed on, the highway authority checks that the junction can take your traffic (**interconnection studies**). When the road is jammed, you are held on the on-ramp (**curtailment**).

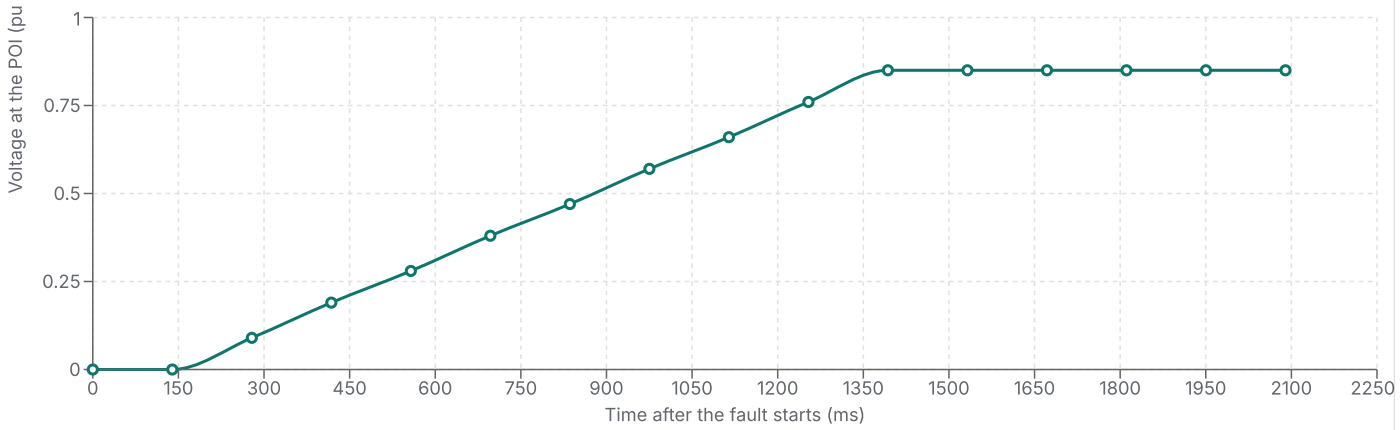
What is actually happening

A **grid code** is the technical contract between every connected plant and the system operator. Inverters do not naturally behave like spinning generators, so the code makes them do deliberately what generators do by physics. Typical requirements (values from international comparators such as ENTSO-E RfG and IEEE 2800):

- **Operating ranges**: stay connected, for example 49.0 to 51.0 Hz continuously and 0.90 to 1.05 pu voltage at 400 kV, with wider bands for limited times. A plant that trips during an excursion turns a disturbance into a blackout.

- **Fault ride-through (FRT)**: a transmission fault drops voltage over a wide area for about 100 ms. The plant must stay connected, even down to 0 pu for about 150 ms, and inject **reactive current** to prop up voltage. Mass PV tripping has lost gigawatts in real events.

Fault ride-through: stay connected above this line (typical profile)



A typical international profile, not a Kuwaiti requirement: 0 pu for 150 ms, then a ramp to about 0.85 pu by 1.5 s. While the POI voltage stays above the line the plant must stay on and push reactive current; only below it may it trip.

- **Reactive power**: typically 0.95 power factor leading and lagging at the **POI**, about ±0.33 × Pmax, across 0.90 to 1.10 pu voltage, and in Kuwait it must hold at 50 °C.
- **Active power and ramp rates**: follow the control centre's set-point; ramp no faster than about 10 % of Pmax per minute. PV can hold back upward ramps, but a cloud or dust front pulling output down needs storage.
- **Frequency response**: cut output when frequency is high (from about 50.2 Hz, with a **droop** of 2 to 12 %); raise output when it is low, which needs headroom or a battery.
- **Power quality**: limits on harmonics, flicker and unbalance (IEEE 519: total harmonic distortion at most 1.5 % above 161 kV).
- **Models and tests**: validated simulation models and witnessed on-site tests.

Frequency droop response

1,000 MW PV plant, 900 MW available irradiance-limited; 50 Hz system

Droop 4 %

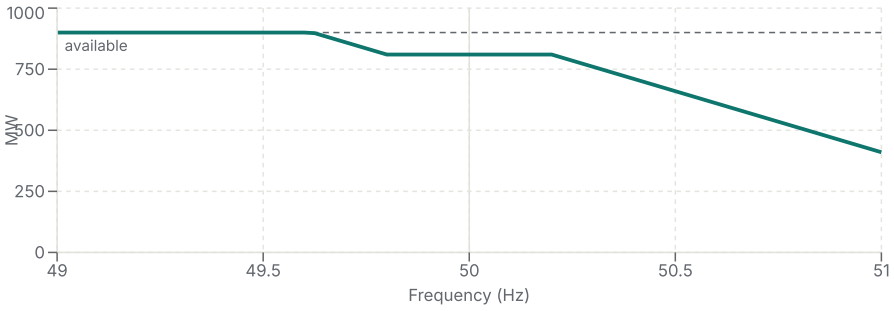
Deadband 200 mHz

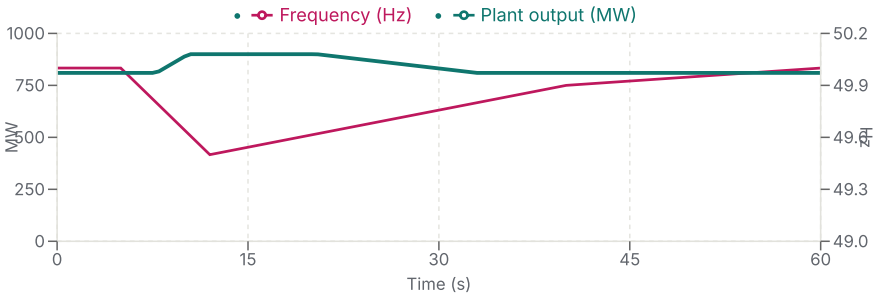
Headroom (curtailment reserve) 10 %

Event frequency nadir 49.50 Hz

Pre-event output	Output at nadir
810 MW	900 MW

Response ΔP	Wanted by droop
90 MW	150 MW limited by headroom





What to notice: droop % is the frequency change (as a share of 50 Hz) that would swing the plant through 100 % of rated power — 4 % droop means 2 Hz for full range. A PV plant can only respond upward if it is already curtailed (headroom); over-frequency response is always available by reducing output.

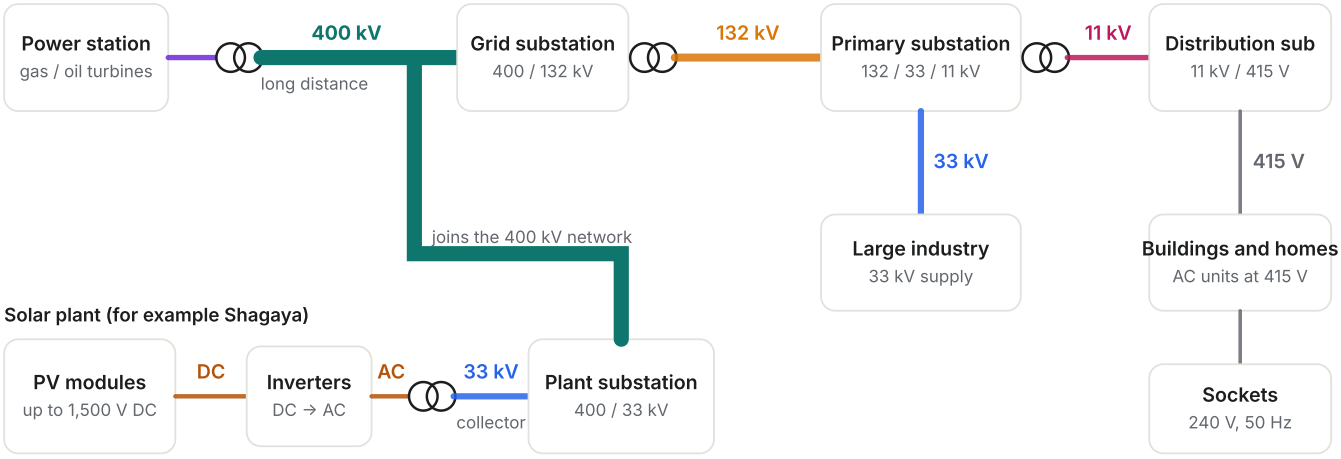
Lower the grid frequency and watch the plant's output move along the droop line once outside the deadband. A PV plant can only add power if it is running with headroom; a battery can do it either way.

Interconnection studies prove the network can take the plant:

- **Load flow**: voltages and line loadings in normal and single-outage (N-1) cases. In Kuwait, **winter midday** (low load, strong sun) is as critical as the summer peak, plus the evening with batteries discharging.
- **Short circuit**: fault levels against switchgear ratings.
- **Stability (RMS) and EMT**: does the plant ride through faults; are its controls stable on a weak grid?
- **Harmonics** and **protection coordination**.

The grid by voltage level: from power station to socket

Click any line or box. Thicker line = higher voltage. Power steps up to travel far, then steps down to be used.



Transmission (EHV): 400 kV Kuwait: confirmed

The backbone: carries bulk power over long distances between power stations and grid substations.

Confirmed for Kuwait (for example the Shagaya to Al-Wafra 400 kV line). Older EHV circuits are quoted as 300 or 275 kV; sources differ.

Power to deliver 400 MW

Current at 400 kV
577 A

Heat loss vs 400 kV, same wire
1×

Same power, different voltage: current is power divided by voltage, and the heat lost in a wire grows with current squared. That is why power leaves the station at 400 kV (100 MW needs about 144 A) and is only brought down to 240 V at the last few metres (the same 100 MW at 240 V would need over 400,000 A). Each step down happens in a substation through a transformer, the two-circle symbol. A solar plant does the reverse: low DC voltage at the modules, stepped up twice to reach the 400 kV network. Only 400 kV is confirmed for Kuwait in our sources; the other levels are commonly cited and worth confirming with MEWRE.

The network the studies test: a PV plant collects at 33 kV, steps up and joins the 400 kV backbone. Click each level and compare the current that 400 MW needs at 400 kV and at 33 kV: this is why big plants connect high.

Curtailment risk: when the grid cannot take the power, the control centre caps output. Causes: a grid constraint, system security, oversupply, grid unavailability or a late grid connection. In typical PPAs these are compensated as **deemed energy** (what the plant could have produced), sometimes above an annual allowance; curtailment caused by the plant itself is not.

Why output was capped	Typical PPA treatment
Grid constraint	Deemed energy, paid (sometimes only above an annual allowance)
System security	Deemed energy, paid
Oversupply	Deemed energy, paid
Grid unavailability	Deemed energy, paid
Late grid connection	Deemed energy, paid
The plant's own fault	Not paid

From connection request to compliance

1. **1 Connection requirements**
   Issued by MEWRE Transmission; for a KAPP IPP, in the RFP and PPA schedules
2. **2 Interconnection studies**
   Developer's consultant runs them; MEWRE reviews
3. **3 Validated models**
   RMS and EMT models from the inverter maker, checked against type tests
4. **4 Unit certificates**
   Inverter and controller type tests, including fault ride-through
5. **5 Compliance plan**
   Test list, procedures, acceptance criteria, witnesses
6. **6 Witnessed site tests**
   Reactive capability, voltage steps, set-point and ramp, frequency response, power quality
7. **7 Compliance and monitoring**
   Certificate, then re-tests after any firmware change

Models and studies run in parallel with design. A missing EMT model or a late study can hold the whole energisation date.

❔ **Kuwait: what is not public**

No published Kuwaiti grid code for renewable or inverter-based plants was found. Requirements are likely written into RFP technical schedules, the PPA and connection agreements with MEWRE's Transmission sector. Kuwait's solar IPPs use a 30-year PPA with MEWRE (confirmed), but the curtailment allowance and deemed-energy method are not public. Obtain MEWRE's current requirements before relying on any value.

✳ **Numbers you will hear**

- **Reactive power**: 0.95 PF at the POI means $Q = P\tan(\arccos 0.95) = 0.329\,P$, so 100 MW needs about ±33 Mvar.
- **Fault ride-through**: about 0 pu retained voltage for about 150 ms; protection clears 400 kV faults in about 80 to 120 ms.
- **Ramp**: at most about 10 % of Pmax per minute; over-frequency response from about 50.2 Hz with 2 to 12 % droop.
- **Droop example**: a 500 MW battery with 4 % droop and a ±0.2 Hz deadband gives 50 MW at 49.6 Hz (100 MW without the deadband).
- **Harmonics**: THD at most 1.5 % on buses above 161 kV (IEEE 519).

Say it at MEWRE

- Which grid code or comparator applies, and where exactly are the requirements written: RFP schedules, PPA or connection agreement?
- Is the reactive capability stated at the POI, at 50 °C, across 0.90 to 1.10 pu?
- Do the studies cover winter midday, the evening battery case and the neighbouring plants together?
- Are the RMS and EMT models validated by the inverter maker, and who re-tests after a firmware update?
- Which curtailment is compensable, and how is the available energy estimated and logged?

The classic trap.

Thinking a grid connection is just a cable to the substation. The grid code makes the plant behave like a good citizen during faults and frequency events, and proving it takes studies, validated models and witnessed tests that often sit on the critical path. Second trap: assuming Kuwait has a published renewable grid code. Track 0 found none, so the requirements must come from MEWRE Transmission and the contract schedules.

Check yourself (answers at the end of Part 3)

Q1.

What does "fault ride-through" require of a PV or battery plant?

A. Stay connected through a short voltage dip caused by a grid fault, and inject reactive current to support the voltage

B. Disconnect within 100 ms of any fault to protect the inverters

C. Restart automatically after a blackout

D. Ride through dust storms without losing output

Q2.

A 400 MW plant must provide reactive power at 0.95 power factor (lagging) at the POI at full output. How many Mvar is that?

Your answer: _____ Mvar

Q3.

The developer's load-flow study for a large Shagaya PV plant covers only the summer peak. What do you ask for?

A. Nothing: summer peak is the worst case in Kuwait

B. Add winter midday and the evening battery-discharge case, with N-1 outages and neighbouring plants included

C. Ask for a harmonic study instead

D. Accept and plan to curtail the plant if problems appear

Lesson 18 · Part 3 · full lesson · about 8 min

Weak grids and inverters: SCR, current clamping and EMT studies

Why you care at MEWRE. Grid engineers at MEWRE will ask whether a solar or battery plant is stable where it connects. The answer depends on how strong the grid is there (its short-circuit ratio), on what inverters do in a fault (they clamp their current), and on whether the study used detailed EMT models. These three ideas decide which studies you demand and which risks you flag.

The picture first

Imagine pushing a small boat away from a **big ship**. The ship does not move; your push goes exactly where you aim it. Now push the same boat away from **another small boat**: both drift, and if you keep pushing in rhythm the two can start rocking each other.

- The **grid** is the ship or the small boat. A grid with lots of large spinning generators and short, strong connections is the big ship: **strong**. A remote node at the end of a long line with few generators nearby is the small boat: **weak**.
- The **inverter** of a PV or battery plant is the person pushing. On a strong grid its push changes little; on a weak grid its own push moves the voltage it is trying to follow, and its controls can start to oscillate.

Where the picture breaks: the "rocking" is not mechanical but in the inverter's control loops, which react in milliseconds.

What is actually happening

Grid strength is measured by the fault level. If you short-circuited the grid at the plant's connection point, how much current would flow? A strong point delivers a huge current (many thousands of MVA); a weak point much less. Fault level is set by the generators and lines behind that point.

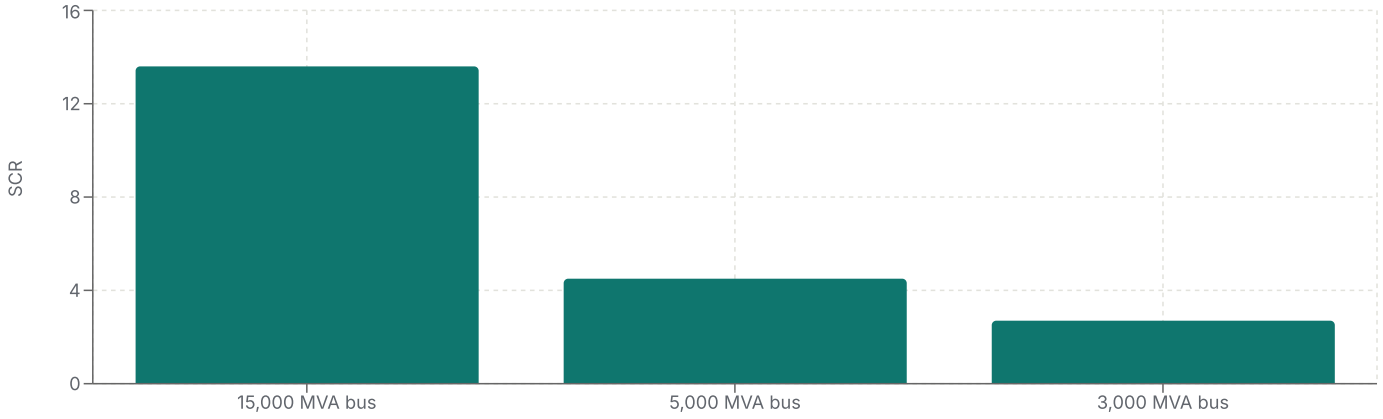
The short-circuit ratio (SCR) compares that strength with the plant's size:

$$\mathrm{SCR} = \frac{S_{sc}}{P_{plant}}$$

where S_{sc} is the fault level at the connection point (MVA) and P_{plant} the plant rating (MW). Several plants on the same weak node share it, so engineers also use the **weighted SCR (WSCR)**: fault level divided by the **sum** of all nearby inverter plants.

- **SCR above about 5:** strong. Standard studies are usually enough.
- **SCR about 3 to 5:** moderate. Detailed EMT studies recommended.
- **SCR below about 3: weak** for today's standard inverters; risk of oscillations and voltage instability.
- **SCR below about 2:** very weak; needs grid-forming inverters, synchronous condensers or a stronger connection.

Same 1,100 MW plant, three connection points



The plant is identical; only the grid behind it changes. At the strong bus the SCR is 13.6, at the weak bus 2.7: below 3, where standard inverters start to struggle.

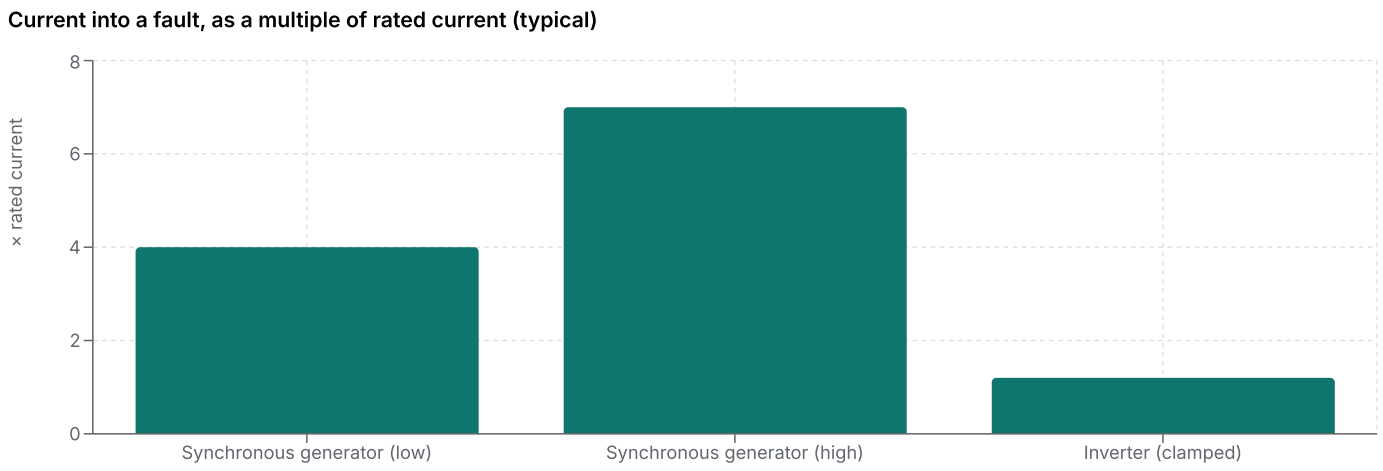
Grid-following vs grid-forming. Almost every PV inverter today is **grid-following**: it measures the grid's voltage with a control called a **phase-locked loop (PLL)** and injects current in step with it. It needs a stiff voltage to follow. A **grid-forming** inverter instead creates its own voltage waveform, like a generator does, so it can hold up a weak grid, give synthetic inertia and even restart a dead grid. It needs energy headroom, which is why it is mostly specified for **batteries**.

Current clamping. An inverter's power transistors are destroyed by overcurrent within microseconds. So its control **clamps** (limits) the output current at roughly **1.1 to 1.2 times** its rating, even during a short circuit. A spinning generator, by contrast, pushes **4 to 7 times** its rated current into a fault.

This has three practical consequences:

- **Protection.** Relays detect faults by their large current. In an area fed mostly by inverters the fault current is small, so relays may not see the fault, or may misjudge its direction. Protection settings must be checked with inverter models.
- **Priority.** During a voltage dip the inverter must choose what to spend its limited current on. Grid codes usually require **reactive current first**, to hold voltage up: $\Delta I_q = k \cdot \Delta V$, with k typically about 2.
- **Control behaviour.** When the current is clamped, the controllers are saturated; badly designed ones overshoot or oscillate when the fault clears. Grid-forming inverters can even lose synchronism while clamped.

Two other "clamps" you will hear about: **clipping**, the flat top of a PV plant's output at noon when the DC modules can give more than the inverter's AC rating (not a fault behaviour), and the **DC-link chopper**, a resistor that dumps surplus energy so the inverter's internal DC voltage stays safe during a fault.



A generator floods a fault with current, which is what protection relies on. An inverter clamps at about 1.2 times its rating, so a grid full of inverters has much weaker fault currents.

RMS and EMT studies

Grid studies simulate the network on a computer. There are two kinds of dynamic simulation:

- **RMS (phasor) studies** treat voltages and currents as smooth 50 Hz quantities and step in **milliseconds**. They are fast and cover the whole national grid over seconds to minutes: frequency after a generator trip, voltage stability, how plants ride through faults at system level. Tools: PSS/E, PowerFactory.
- **EMT (electromagnetic transient) studies** simulate the **actual waveforms**, all three phases, in steps of **1 to 50 microseconds**, including the inverter's real control code. They are slow, so they model a local area for a few seconds. Tools: PSCAD, EMTP, PowerFactory EMT, RTDS for real-time hardware testing.

EMT is needed when: the SCR is low; several inverter plants, STATCOMs or HVDC links are electrically close and could interact; ride-through must be verified with real current clamping and unbalanced faults; grid-forming behaviour must be proven; or for fast events such as transformer energisation (inrush), cable and capacitor switching, temporary overvoltages and harmonic resonance.

A memory picture: RMS is a normal video showing the smooth motion; EMT is a slow-motion camera at 50,000 frames per second showing every wiggle.

How a PM manages the studies

1. **1 Agree scope with MEWRE Transmission**
   Which studies, cases (summer peak, winter midday, N-1), SCR range, neighbouring plants
2. **2 Get the network model**
   Latest MEWRE base case under NDA
3. **3 Get validated plant models**
   Vendor RMS and EMT models, locked to the firmware version
4. **4 Run and review**
   Owner's engineer checks against a checklist
5. **5 Approve and freeze settings**
   Then order equipment and set protection
6. **6 Prove on site**
   Compliance tests must match the model; re-study if firmware changes

The PM does not run the simulations; the PM makes sure the right ones are run, with the right models, and that site tests confirm them.

✳ **Numbers you will hear**

- **SCR** thresholds about **5, 3, 2**. Module 6 rule: EMT required if SCR is below about 5.
- Example: 15,000 MVA at the 400 kV point of interconnection and a 1,100 MW plant give SCR ≈ 13.6 (strong).
- Inverter fault current about **1.1 to 1.2 pu**; generator **4 to 7 pu**.
- Reactive current injection $\Delta I_q = k\,\Delta V$, k about **2** (typical).
- RMS time step about **1 to 10 ms**; EMT about **1 to 50 µs**.
- Kuwait's own grid-code values were not found in the course sources: ask MEWRE Transmission.

Say it at MEWRE

- What is the minimum fault level at the connection point, and what SCR does that give with all the plants nearby?
- Has an EMT study been done with the inverter vendor's real, version-locked models?
- What is the inverter's maximum fault current, and is it reactive-priority during dips?
- Have the protection settings been checked against the low fault current from the inverters?
- Is the battery's power conversion system grid-forming capable?

The classic trap.

Assuming a big, strong 400 kV connection means no stability studies are needed. The grid can be weak at minimum generation on a winter night, several plants share the same node (use the weighted SCR), and inverters clamp their fault current, so protection and control still need checking with detailed models.

Check yourself (answers at the end of Part 3)

Q1.

A 500 MW solar plant connects where the fault level is 2,000 MVA. What is its short-circuit ratio?

Your answer: _____

Q2.

Why do protection engineers worry about areas fed mostly by inverters?

A. Inverters clamp their current at about 1.1 to 1.2 times rating, so fault currents are small and relays may not detect faults reliably

B. Inverters produce too much fault current and overload the breakers

C. Inverters cannot be switched off by breakers

D. Inverters only work on DC networks

Q3.

Which situation most clearly calls for an EMT study rather than only an RMS study?

A. Two large inverter plants and a STATCOM connected close together at a node with SCR around 2.5

B. Checking whether a 132 kV line overloads at summer peak

C. Estimating the annual energy yield of a PV plant

D. Comparing the levelised cost of two tariff bids

Lesson 19 · Part 3 · full lesson · about 7 min

A project's life from idea to handover

Why you care at MEWRE. In your first weeks you will be dropped into projects that are already somewhere in their life: a feasibility study under review, a tender in evaluation, a plant in commissioning. If you know the stages and which gate comes next, you know what decision is on the table and what to ask for.

The picture first

Think of building a family house.

- First you decide you **need** one and roughly what you can afford.
- You look at plots and check each one: is it really for sale, can you get a permit, is it on a flood path? (**feasibility**, **site and permits**)
- You pick a builder and sign a contract (**procurement and award**).
- The architect draws, the builder orders windows and steel, then builds (**design, procurement, construction**).
- You test everything before you move in: water, power, air conditioning (**testing and commissioning**).
- You get the keys (**handover**, or **COD** for a power plant), and the builder must fix anything that breaks in the first year (**defects period**).

Before each big payment, the family sits down and decides: go on, wait, or stop. Those meetings are **gates**. Changing your mind costs little while it is still on paper, and a lot once the concrete is poured.

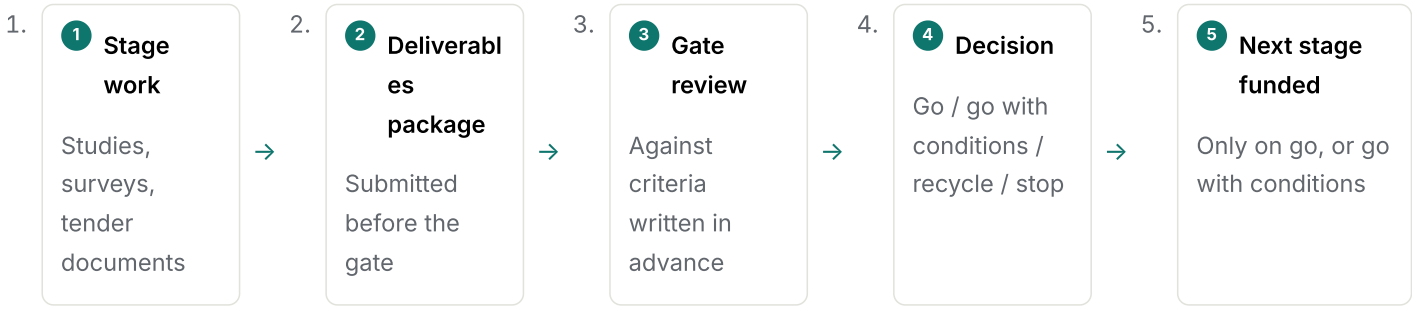
Where the analogy breaks: at MEWRE the "family" is a chain of committees, and the land, the grid connection and the permits often belong to other government bodies, not to you.

What is actually happening

A **stage** is a period of work that produces defined deliverables. A **gate** is a decision point where an accountable person or committee reviews those deliverables and decides one of four things:

- **Go**: release budget and authority for the next stage.
- **Go with conditions**: proceed, but named gaps must close by a date (for example, a geotechnical report still pending).
- **Recycle / hold**: redo or extend the current stage.
- **Stop**: kill the project and record the lessons.

What happens at every gate



The same loop repeats at G0 to G7. Notice that the criteria exist before the review, and that recycle and stop are real outcomes, not failures of the process.

Each gate answers a different question. A typical sequence for a PV or battery project:

Gate	Question	Stage before it typically takes
G0 Need	Is there a real need we are mandated to meet?	1 to 3 months
G1 Pre-feasibility	Worth studying? Which options survive?	2 to 4 months
G2 Feasibility	Which option, which site, which procurement route, what cost?	6 to 12 months
G3 Investment	Do we commit to tender?	3 to 6 months
G4 Award / financial close	Do we sign?	6 to 18 months
G5 Notice to proceed	Can construction start safely?	1 to 3 months
G6 COD / taking over	Does the plant meet the contract?	1 to 3 months
G7 Close-out	Is everything finished and learned?	end of the defects period

Along the way:

- **Feasibility** is the real engineering: resource, site, grid connection, technology options, cost, procurement route, business case, risks. A feasibility study that cannot say where the plant connects, at what voltage, and when that substation will be ready, is not finished.
- **Site, permits and EIA**: land allocation, no-objection letters (oil and gas, defence, aviation, utilities), surveys, flood and unexploded-ordnance checks, and the **environmental impact assessment (EIA)** with the Environment Public Authority.

- **Design, procurement, construction** overlap. Long-lead items such as HV transformers are ordered early.
- **Testing and commissioning** end at **COD** (commercial operation date) or **taking over**: the owner takes the plant, delay damages stop, and the **defects notification period (DNP)** starts.
- At the end of the DNP the **Performance Certificate** is issued; only then are the works accepted.

Project lifecycle — 1 GW PV IPP

Stages, gate decisions, key documents and indicative durations; click a stage

1. 01 Feasibility → 2. 02 Site & EIA → 3. 03 Procurement (RFQ/RFP) → 4. 04 Award & Financial Close → 5. 05 Design →
6. 06 Procurement & Manufacturing → 7. 07 Construction → 8. 08 Testing & Commissioning → 9. 09 COD → 10. 10 O&M / DNP



Bar widths ∝ midpoint durations (stages overlap in practice; O&M continues for 25+ years after DNP).

1. Feasibility

Typical duration: 6–12 months

Key documents:

- Pre-feasibility & feasibility study
- Resource assessment (satellite + ground data)
- Grid integration study
- PPP value-for-money / affordability analysis

Gate decision: Approval to proceed and procurement route (PPP via KAPP or public EPC via CAPT).

PM focus: Lock the business case assumptions you will be judged against later; document options rejected and why.

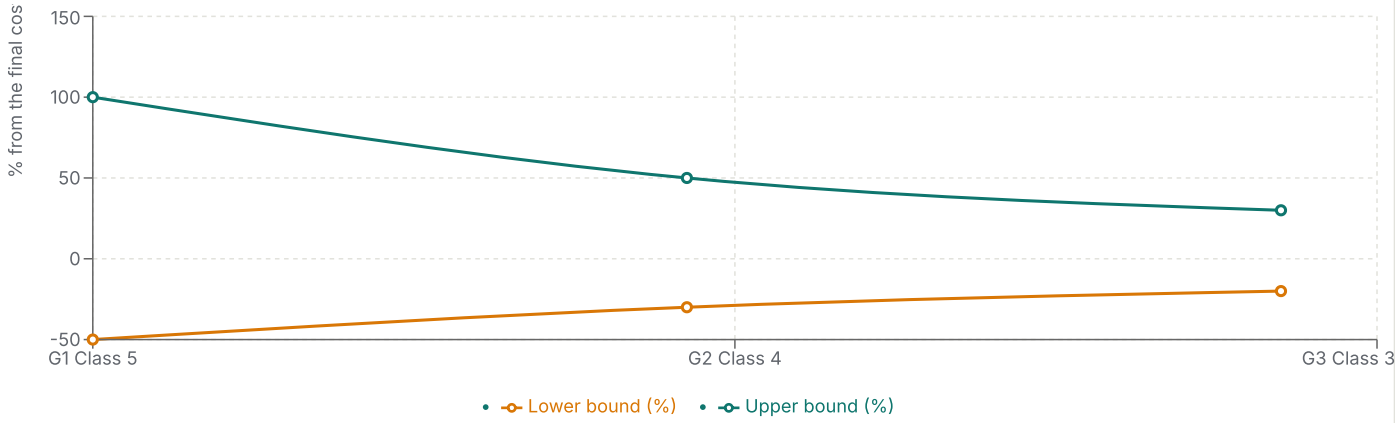
What to notice: years of development and procurement precede roughly 18–24 months of construction — most schedule risk is decided before NTP.

Durations are indicative for a large Gulf PV IPP and vary with procurement route, approvals and grid works.

Click each stage. Notice that the stages before construction add up to as long as building the plant, and that design, manufacturing and construction overlap.

The cost estimate gets sharper at each gate. A screening number at G1 can be off by half or double; a budget number at G3 should be within about −20 % to +30 %.

How wide the cost range is at each gate (AACE estimate classes, approximate)



The funnel narrows gate by gate, and the upper bound is always further out than the lower one: early estimates miss on the high side more than the low side.

Why it matters at MEWRE

On a KAPP IPP, MEWRE is the offtaker and the grid owner. Its own deliverables (the 400 kV line and bay, metering, SCADA, approvals) sit on the critical path of somebody else's revenue. The critical path of a GW plant almost always runs through **unique, single items**: HV transformers, the substation, the grid connection and approvals, not the repetitive arrays.

Gates fail when they become ceremonies: "we already spent two years on studies" (sunk cost), an announced target date nobody dares to move, or criteria never written down so every gate passes. Write the criteria **before** the gate.

The IPP route in Kuwait (as Track 0 gives it)

Typical PPP sequence for an IPP, supported by the project research file: entity request, KAPP feasibility, Higher Committee, RFQ, RFP, preferred bidder, Higher Committee, Audit Bureau, PPA signing, financial close. The exact EIA steps and review durations with the Environment Public Authority are flagged **unverified** in Track 0: confirm them with the EPA before you plan on them.

Numbers you will hear

- **Development spend**: about 1 to 3 % of capex across all stages before construction.
- **Estimate classes (AACE, approximate)**: Class 5 screening −50 % / +100 %; Class 4 feasibility −30 % / +50 %; Class 3 budget −20 % / +30 %.
- **Construction** 12 to 20 months, then **testing and commissioning** 2 to 4 months.
- **HV power transformer lead time**: 18 to 24+ months, often the longest item.
- **DNP**: commonly 12 to 24 months (FIDIC default 365 days); Performance Certificate within 28 days after it ends.

Say it at MEWRE

- Which gate are we at, and what are the written criteria for passing it?
- Is this cost a Class 5 screening number or a Class 3 budget number? What is the range?
- Where exactly does this plant connect, at what voltage, and when will that substation be ready?
- Have the HV transformers been ordered or factory slots reserved?
- When does the defects period end for each section, and who is running the inspections before then?

The classic trap.

Treating a gate as a formality and COD as the finish line. A gate only works with written criteria and someone allowed to say recycle or stop; money already spent is irrelevant to whether the next stage should be funded. And after COD the defects period is the owner's most valuable year: the contractor still has retention and performance security at risk, so that is when you inspect and notify defects.

Check yourself (answers at the end of Part 3)

Q1.

At the pre-feasibility gate a consultant presents a cost of USD 600 m (an AACE Class 5 estimate). How should you treat it?

A. As a screening figure that could be roughly half to double the final cost, not a budget

B. As the budget to request from the Ministry of Finance

C. As accurate to within ±5 %, since it comes from a consultant

D. As the contract price bidders will offer

Q2.

You chair a feasibility gate. The study, business case and grid statement are complete. Only the final geotechnical report is pending, due in 6 weeks, and the desk study shows no fatal flaw. What decision do you recommend?

A. Go with conditions: proceed, with the geotechnical report and any design consequences to close by a named date.

B. Recycle the whole feasibility stage.

C. Go with no conditions; the report is a formality.

D. Stop the project.

Q3.

What happens at taking over (COD)?

A. The owner takes care of the plant, delay damages stop accruing and the defects notification period starts

B. All retention is released and the contractor's obligations end

C. The Performance Certificate is issued, which accepts the works

D. The design basis is frozen

Lesson 20 · Part 3 · quick lesson · about 4 min

Testing, commissioning and COD

Why you care at MEWRE. COD is when MEWRE starts paying for energy (on an IPP) or takes over the plant (on an EPC), and the tests before it decide millions in payments and damages. You will witness these tests and be asked whether the plant is ready to sign off.

Quick lesson: the essentials only. The full version, with more drawings and worked numbers, is on the website under Trial week.

The picture first

You buy a new car.

- The factory checks it before it leaves (**factory acceptance test, FAT**).
- The dealer inspects it on arrival for transport damage (**site acceptance test, SAT**).

- With the engine off, you check oil, tyres, lights and wiring (**cold commissioning**).
- You start the engine and check every system works (**hot commissioning**).
- You drive a fixed route and compare fuel use with the brochure, corrected for weather and load (**performance test**).
- You drive it for a week without a breakdown (**reliability run**).
- You sign the handover papers and take the keys (**COD / taking over**). A short list of small faults is fixed later (**punch list**), and the warranty year starts (**defects period**).

Where the analogy breaks: a power plant's "road test" result depends heavily on the weather and on the measurement rules, so the rules must be agreed **before** the test, not after.

What is actually happening

The testing chain, in order:

- **FAT** at the factory: transformers, inverters, battery containers and switchgear meet spec before shipment.
- **Receipt inspection / SAT** on site: no transport damage, equipment works as installed.
- **Mechanical completion**: built per drawings, ready to test.
- **Cold (pre-)commissioning**, de-energised: insulation resistance, polarity, string open-circuit voltage and short-circuit current, earthing, cable tests, relay injection, SCADA point checks.
- **Energisation** under a switching programme and **permit to work**. From now on the site is a live electrical installation.
- **Hot commissioning**, energised: inverters, trackers, plant controller, SCADA and battery run correctly.
- **Grid compliance tests** at the connection point, witnessed by MEWRE's transmission and control centre: power setpoints, ramps, frequency response, reactive power and voltage control.
- **Performance tests**: **PR test** (performance ratio over days), **capacity test** (short regression test), battery capacity and efficiency.
- **Reliability run**: continuous operation without contractor-caused trips.
- **COD / taking over**.

From factory to taking over

1. **1 FAT**
   At the factory
2. **2 SAT / receipt**
   Transport damage, as installed
3. **3 Mechanical completion**
   Built per drawings
4. **4 Cold commissioning**
   De-energised checks
5. **5 Energisation**
   Switching programme, permit to work
6. **6 Hot commissioning**
   Systems running
7. **7 Grid compliance tests**
   Witnessed by MEWRE
8. **8 PR, capacity, battery tests**
   Against the guarantees
9. **9 Reliability run**
   No contractor-caused trips
10. **10 COD / taking over**
    Category A punch items closed

Each step must pass before the next one starts. MEWRE is directly involved at energisation and the grid tests, and the owner's engineer witnesses the rest.

✳ Numbers you will hear

- **PR test duration**: typically 7 to 15 consecutive days (10 is common); minimum valid data often 95 % of intervals.
- **PR values**: about 0.85 to 0.88 corrected to 25 °C; about 0.80 to 0.83 corrected to the annual average cell temperature. Always ask which.
- **Temperature effect**: module power falls about 0.3 to 0.4 % per °C.
- **Capacity test pass mark**: measured at least about 97 % of the guaranteed value at reporting conditions.
- **Reliability run**: e.g. 7 to 30 continuous days for the plant, 72 hours for subsystems.

Say it at MEWRE

- Is the test procedure, with data filters, reference temperature, soiling and curtailment rules, approved before the test starts?
- Which reference temperature is the PR guarantee based on, and does the test use the same one?
- When were the reference sensors calibrated, and are they cleaned daily during the test?
- Has the control centre witnessed and accepted the grid compliance tests?
- Which open punch items are Category A, and has each been closed and verified?

The classic trap.

Quoting a PR without its reference temperature. The same plant can show 0.88 corrected to 25 °C and 0.82 corrected to the annual average cell temperature; a guarantee from one basis tested on the other is a gift to one side. Also, commissioned is not taken over: COD needs every test, MEWRE's grid approval and all Category A items closed.

Check yourself (answers at the end of Part 3)

Q1.

A 50 MWp plant receives 65 kWh/m² on the modules during a 10-day test and delivers 2,665 MWh. What is its PR, in percent?

Your answer: _____ %

Q2.

Insulation resistance tests, string open-circuit voltage checks and relay secondary injection belong to which stage?

A. Cold (pre-)commissioning, de-energised

B. Factory acceptance test

C. Hot commissioning

D. Reliability run

Q3.

The EPC asks you to sign COD. PR and capacity tests passed and the reliability run is done, but a faulty protection relay at one MV block is still open and MEWRE's grid test acceptance letter has not been issued. What do you do?

A. Do not recommend COD yet: write back listing the two open conditions (a Category A item and the grid approval), with what is needed to close each.

B. Sign COD and move the relay to the Category B list.

C. Sign COD now and ask the control centre to send its letter later.

Lesson 21 · Part 3 · quick lesson · about 4 min

The schedule and the critical path

Why you care at MEWRE. Every month a contractor will hand you a programme and tell you the project is on track. If you can find the critical path, read float and spot the usual tricks, you will know whether to believe it, and you will see early when MEWRE's own work is about to delay somebody else.

Quick lesson: the essentials only. The full version, with more drawings and worked numbers, is on the website under Trial week.

The picture first

You are cooking dinner for guests arriving at 8 pm.

- Marinate the meat (2 hours), then grill it (25 minutes).
- Cook rice (40 minutes).
- Make a salad (10 minutes).

The meat chain is the longest: 2 h 25 min. It sets the earliest time dinner can be ready. That is the **critical path**. If the marinade starts 15 minutes late, dinner is 15 minutes late. The salad has lots of spare time (**float**): it can start late without anyone noticing. But if you also start the rice late, at some point the rice becomes the longest chain and it is now critical.

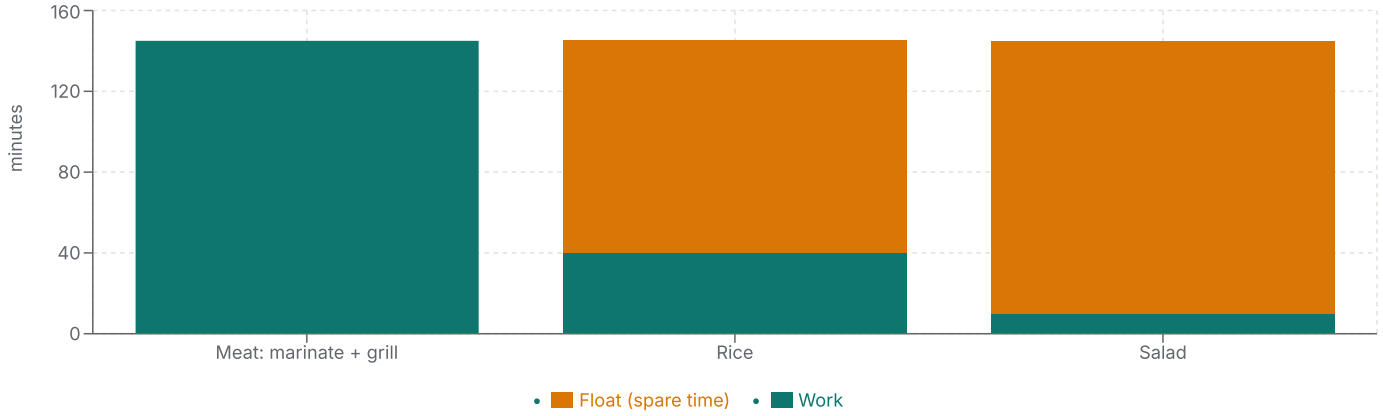
What is actually happening

A schedule is a network:

- **Activities**: a piece of work with a duration ("Piling Block 07, 12 weeks").
- **Dependencies** (logic): mostly **finish-to-start** (B starts when A finishes); also start-to-start and finish-to-finish, sometimes with a **lag** (a waiting time, e.g. concrete curing).
- **Forward pass**: earliest start and finish of every activity. **Backward pass** from the end date: latest start and finish.
- **Total float** = latest start − earliest start: how far an activity can slip without moving completion.
- **Critical path**: the longest chain through the network; its activities have zero (or the least) float. **Near-critical**: little float, e.g. under 10 working days.

Float belongs to the **path**, not the activity: use it upstream and everything downstream on that path loses it. If an activity slips by more than its float, completion moves by the difference.

Dinner at 8 pm: each chain's work and its spare time (minutes)



Every bar is 145 minutes tall because every chain must be ready by the same moment. The meat has no spare time: it is the critical path.

Numbers you will hear

- **Total float** $= LS - ES$. Completion delay = slip − total float (if positive).
- **DCMA thresholds**: missing logic ≤ 5 %; leads 0; hard constraints ≤ 5 %; finish-to-start ≥ 90 % of links; high float or high duration (over 44 working days) ≤ 5 %; negative float 0.
- **CPLI and BEI** (critical path length index, baseline execution index) ≥ 0.95.
- **FIDIC 2017 Sub-Clause 8.3**: programme within 28 days of the Commencement Date notice; Engineer responds within 21 days.
- **Kuwait calendar**: the summer midday outdoor work ban is commonly reported as about 11:00 to 16:00, June to August (**unverified** in Track 0: confirm before baselining).

Say it at MEWRE

- What is the longest path today, and how has it moved over the last three updates?
- Please send the native XER with every update, not only the PDF.
- Where are MEWRE's activities (approvals, grid works, outages, energisation permits), and with what durations?
- Which activities have kept the same remaining duration for two or more updates?
- Are there hard constraints or leads in the logic? How much float does the grid connection have?

The classic trap.

Believing that an activity with float can slip freely, or that zero float always means the logic works. Float belongs to the path: consume it upstream and every downstream activity loses it. And a hard constraint on the completion date can show zero float while hiding negative float, so trace the logic, not the colours.

Check yourself (answers at the end of Part 3)

Q1.

An activity has 8 weeks of total float and slips by 12 weeks. Everything else goes to plan. By how many weeks does completion move?

Your answer: _____ weeks

Q2.

What is the critical path?

A. The longest chain of dependent activities through the network; its activities have zero (or the least) total float

B. The list of the most expensive activities

C. The activities the contractor marks as important

D. The shortest route from start to finish

Q3.

In P6, what is the "as of" date of a schedule update, before which everything is actual and after which everything is forecast?

A. The data date

B. The baseline date

C. The longest path

D. The finish constraint

Lesson 22 · Part 3 · quick lesson · about 3 min

Earned value: are we on time and on budget?

Why you care at MEWRE. Contractor reports say 'progressing well'; leadership asks 'are we on time and on budget?'. Earned value turns the monthly numbers into two honest answers and a forecast, and lets you see through a green status that the data do not support.

Quick lesson: the essentials only. The full version, with more drawings and worked numbers, is on the website under Trial week.

The picture first

You hire a painter for 10 rooms: KWD 100 per room, KWD 1,000 in total, 10 days.

At the end of day 5:

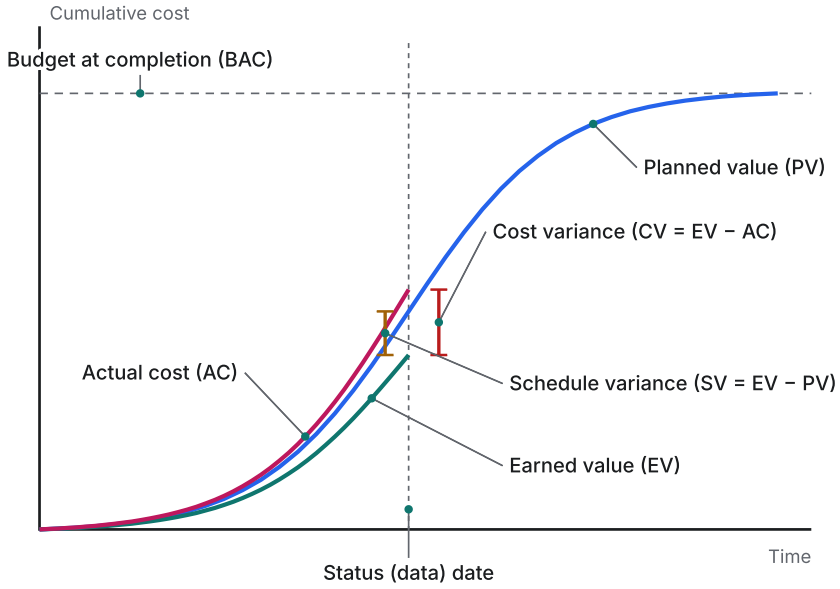
- The plan says 5 rooms should be done: **KWD 500 of work planned**.
- 4 rooms are actually finished: **KWD 400 of work earned**.
- You have paid KWD 480 for materials and labour: **KWD 480 actually spent**.

Looking only at spending (480 against a plan of 500) you would think you are under budget. You are not: you paid 480 for 400 worth of work (over cost), and you are one room behind (late).

What is actually happening

Earned value uses three cumulative numbers at the status date, all against the **budget at completion (BAC)**:

- **Planned value (PV)**: budget of the work the baseline says should be done by now.
- **Earned value (EV)**: budget of the work **actually done**. Not money spent: budget earned.
- **Actual cost (AC)**: what that work actually cost. For MEWRE's own budget, AC is typically what has been certified plus accrued.



Three cumulative curves at the status date: the gap from EV down to PV is the schedule variance, the gap from EV up to AC is the cost variance. Here EV is below both: late and over cost, like the painter.

Numbers you will hear

- **CPI and SPI**: 1.0 is on plan; below 1 is bad. Example RAG thresholds (set per project): CPI ≥ 0.97 green, 0.92 to 0.97 amber, below 0.92 red.
- **CPI stability**: cumulative CPI stabilises at about 20 % complete; the final CPI is rarely more than about 0.1 better.
- **TCPI credibility**: a TCPI more than about 0.05 to 0.10 above the current CPI is not credible.
- **Module supply rules of credit** (Track 0 example): approval 5 %, PO 5 %, FAT passed 15 %, shipped 15 %, delivered and inspected on site 60 %.

Say it at MEWRE

- What rules of credit are you using for this package, and when were they agreed?
- Your report says green, but CPI is 0.93 and SPI(t) is 0.88. What is the recovery plan?
- What EAC range do you forecast, and which method do you believe?
- What TCPI does finishing on budget require, and what will change to achieve it?
- Is 'installed' counted only once it is inspected and tested?

The classic trap.

Comparing spend with plan (AC against PV). Spending less than planned may only mean less work was done. Compare EV with AC for cost and EV with PV for schedule. And EV is budget earned for work done, not money paid: crediting progress at purchase order or for untested work inflates it.

Check yourself (answers at the end of Part 3)

Q1.

A KWD 200 m contract has EV = 60 and AC = 75 (KWD m). If current cost performance continues, what is the estimate at completion, in KWD m?

Your answer: _____ KWD m

Q2.

At month 6: PV = 50, EV = 40, AC = 38. Which statement is right?

A. Behind schedule (SPI 0.80) but under cost for the work done (CPI about 1.05)

B. On schedule and under budget, because AC is below PV

C. Over budget and behind schedule

D. Ahead of schedule (SPI 1.25)

Q3.

Module supply is measured with these rules of credit: approval 5 %, PO 5 %, FAT passed 15 %, shipped 15 %, on site and inspected 60 %. Approval and POs are done, nothing has been tested yet. The contractor claims 20 %. What has actually been earned?

A. 10 %

B. 20 %

C. 35 %

D. 0 %

Lesson 23 · Part 3 · quick lesson · about 4 min

Risks, changes and the monthly report

Why you care at MEWRE. Your seniors will judge you less on whether problems happen than on whether they heard about them early, with options. The risk register, change control and the monthly report are the three tools that make that possible.

Quick lesson: the essentials only. The full version, with more drawings and worked numbers, is on the website under Trial week.

The picture first

You are driving to the airport for an international flight.

- **Risks**: traffic on the motorway, a flat tyre, a long security queue. You list them and judge each: how likely, how bad (**risk register**, **probability × impact**).
- **Responses**: leave earlier (**mitigate**), buy travel insurance (**transfer** the cost, not the event), take the coastal road to skip roadworks (**avoid**).
- The map app says 40 minutes typically, but you allow 60 because you want to be fairly sure (**P50 against P80**).

- Someone asks you to stop at a relative's house on the way. Before saying yes, you check how much time it costs (**change control**).
- You message the family early: "we are 20 minutes late, should we skip lunch?" (**honest reporting**, with a decision attached).

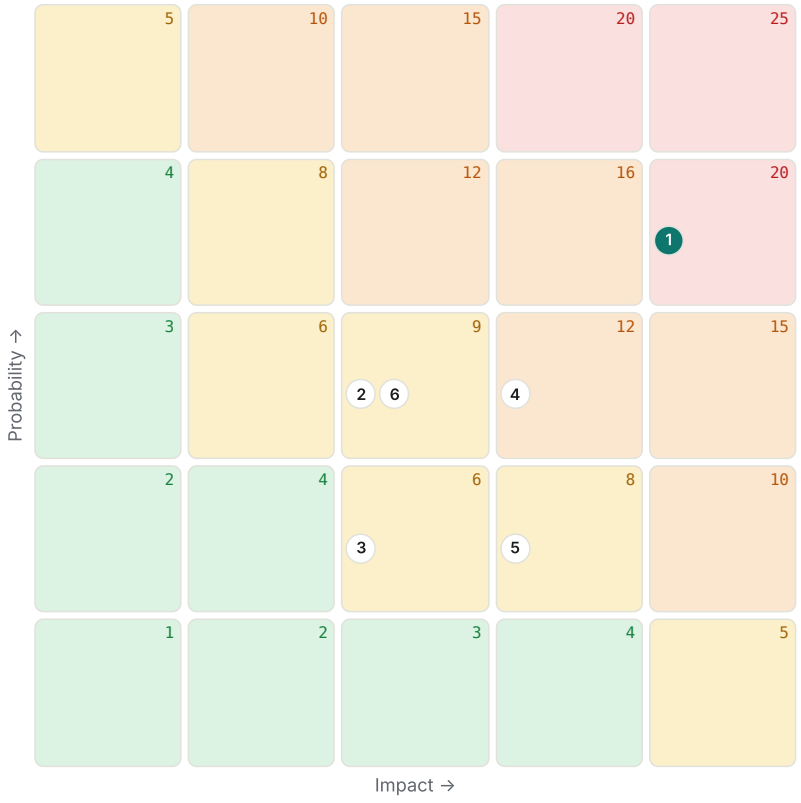
What is actually happening

Risk register. Write each risk as **cause, event, effect**: "Because the 400 kV route crosses an oil-sector corridor, right-of-way approval may take longer than planned, delaying energisation and exposing MEWRE to PPA compensation." Each risk has one **owner**, a **response** (avoid, transfer, mitigate, accept, escalate) with actions and dates, and a **score**.

Probability × impact. Score each from 1 to 5 and multiply (1 to 25). Typical bands: 15 to 25 red (response plan, steering committee sees it), 8 to 12 amber (owner manages), 1 to 6 green (monitor). An HSE impact of 5 is red regardless of probability. The score is a **ranking**, not a quantity: never add scores up. For money, use **expected monetary value (EMV)** = probability × cost impact (40 % × USD 6 m = USD 2.4 m).

5×5 probability–impact matrix

Select a risk, then click a cell to move it



1 Grid connection works late 20
2 Module price spike before order 9
3 Sandstorm damage during construction 6
4 Heat-stress HSE incident 12
5 EPC subcontractor insolvency 8
6 Permit / approval delay 9

Add a risk... Add

1. Grid connection works late

P = 4 (Likely) · I = 5 (Severe)

Score 20 — Extreme

Avoid or transfer: change scope/contract, escalate to sponsor; do not proceed without a funded response.

Remove risk

What to notice: score = probability × impact, but two risks with the same score can need very different responses — a rare catastrophic risk (grid failure, fatality) deserves more attention than its score suggests.

Move 'Grid connection works late' around: a MEWRE-side risk that is likely and severe lands in the red corner, and the widget suggests the response.

Numbers you will hear

- **Risk score** P × I on 1 to 5 scales: 15 to 25 red, 8 to 12 amber, 1 to 6 green.
- **Contingency**: P80 − base estimate (owner view); management reserve sits outside the baseline.
- **FIDIC 2017 Clause 20**: claim notice within 28 days of becoming aware; fully detailed claim within 84 days.
- **Example RAG for schedule**: 1 to 4 weeks late amber, more than 4 weeks (or a key date at risk) red.
- **Leadership report**: 2 to 4 pages, decisions on page 1.

Say it at MEWRE

- What is the cause, event and effect of this risk, who owns it, and what is the response with a date?
- Is our contingency set at P80, and how much did we draw down this month?
- Has this change been logged and assessed for time, or is it an informal request?
- Decision needed this month: here are the options and my recommendation.
- Which MEWRE-side interface deliverables are late, and by how much against float?

The classic trap.

Treating the risk score as a quantity, or keeping the status green to avoid bad news. P × I scores rank risks; they do not add up into a budget (use EMV or Monte Carlo). And 'watermelon' reporting (green outside, red inside) always ends with a sudden jump to red when the recovery options are gone; report amber early, with a plan.

Check yourself (answers at the end of Part 3)

Q1.

There is a 40 % chance that MEWRE's 400 kV line is late, which would cost about USD 6.0 m in compensation. What is the expected monetary value of this risk, in USD m?

Your answer: _____ USD m

Q2.

A Monte Carlo analysis gives COD at P50 = March and P80 = June. What should you commit to leadership?

A. Plan and fund to P80 (June): a P50 date is missed half the time, and overruns are larger than underruns

B. March, because it is the most likely date

C. The deterministic date from the contractor's programme

D. Any date between them, since both are estimates

Q3.

During a site visit a MEWRE department head asks the EPC to add extra CCTV cameras. The EPC agrees and says it will "sort the paperwork later". What do you do?

A. Stop it informally proceeding: log it as a change request, have it assessed for cost and time, decide at the right authority, and instruct formally only if approved.

B. Let it go ahead; it is a small item and the request came from a senior.

C. Refuse the request outright.

Answers: Part 3

Transformers and voltage levels: why 400 kV

Q1.

9.2 times

Current falls by $400/132 \approx 3.03$, and losses go with current squared: $3.03^2 \approx 9.2$. Losses fall to about a ninth.

Q2.

The IEC rating assumes at most 40 °C ambient; Kuwait exceeds it, so the unit will run hot and age faster unless derated or designed for site ambient

IEC 60076-2 rises assume 40 °C maximum, 30 °C hottest-month average, 20 °C yearly average. Kuwait exceeds all three, so rises must be reduced (site-specific design) or the unit derated.

Q3.

1588 A

$I = 1{,}100{,}000{,}000/(\sqrt{3} \times 400{,}000) \approx 1{,}588$ A. At 33 kV the same power would need about 19,200 A: that is why the plant steps up before leaving site.

Connecting a plant: the grid code and interconnection studies

Q1.

Stay connected through a short voltage dip caused by a grid fault, and inject reactive current to support the voltage

A transmission fault depresses voltage over a wide area for about 100 ms. If thousands of inverters tripped at once, a local fault would become a system-wide loss of generation. The code makes them stay on and help.

Q2.

131.5 Mvar

$Q = P\tan(\arccos 0.95) = 400 \times 0.3287 \approx 131.5$ Mvar. The inverters must deliver more than this, because transformers between them and the POI absorb reactive power, and they must do it at 50 °C.

Q3.

Add winter midday and the evening battery-discharge case, with N-1 outages and neighbouring plants included

Studies must cover the conditions that stress the network: summer peak, winter midday (low load, high PV, low inertia and fault level) and the evening peak with batteries discharging, in normal and N-1 cases, together with neighbouring plants.

Weak grids and inverters: SCR, current clamping and EMT studies

Q1.

4

SCR = 2,000 / 500 = 4: moderate. EMT studies are recommended, and the weighted SCR with any neighbouring plants could be lower.

Q2.

Inverters clamp their current at about 1.1 to 1.2 times rating, so fault currents are small and relays may not detect faults reliably

Generators give 4 to 7 times rated current into a fault; inverters clamp at about 1.2 times. Relays that rely on large fault currents must be checked with inverter models.

Q3.

Two large inverter plants and a STATCOM connected close together at a node with SCR around 2.5

Low SCR and several inverter-based devices close together create the risk of control interactions, which only EMT models with the real control code can show. Line loading is a load-flow question.

A project's life from idea to handover

Q1.

As a screening figure that could be roughly half to double the final cost, not a budget

Class 5 estimates are roughly −50 % / +100 %. Presenting one as a budget creates a future overrun. The range narrows at each gate (Class 3 at investment decision is about −20 % / +30 %).

Q2.

Go with conditions: proceed, with the geotechnical report and any design consequences to close by a named date.

Gate decisions are go, go with conditions, recycle or stop. Track 0 gives "geotech pending" as the classic trigger for go with conditions.

Q3.

The owner takes care of the plant, delay damages stop accruing and the defects notification period starts

Taking over transfers care and risk, stops delay damages and starts the DNP. Typically only the first half of retention is released. The Performance Certificate comes after the DNP.

Testing, commissioning and COD

Q1.

82 %

Final yield $= 2{,}665/50 = 53.3$ kWh/kWp. Reference yield $= 65/1 = 65$ h. $PR = 53.3/65 = 0.82$, i.e. 82 %.

Q2.

Cold (pre-)commissioning, de-energised

These checks are done before energisation, on de-energised equipment. Hot commissioning follows energisation and checks systems running.

Q3.

Do not recommend COD yet: write back listing the two open conditions (a Category A item and the grid approval), with what is needed to close each.

COD needs all tests passed, the grid operator's approval and all Category A punch items closed. Category B items can close during the defects period.

The schedule and the critical path

Q1.

4 weeks

Completion delay = slip − total float $= 12 - 8 = 4$ weeks. The first 8 weeks are absorbed by float; after that the activity is on the critical path.

Q2.

The longest chain of dependent activities through the network; its activities have zero (or the least) total float

The critical path is the longest path through the logic network. It sets the earliest possible completion; any delay on it delays completion.

Q3.

The data date

The data date separates actuals from forecast. The baseline is the frozen approved schedule you compare against.

Earned value: are we on time and on budget?

Q1.

250 KWD m

$CPI = 60/75 = 0.80$. $EAC = BAC/CPI = 200/0.80 = 250$ m, i.e. KWD 50 m over budget if nothing changes.

Q2.

Behind schedule (SPI 0.80) but under cost for the work done (CPI about 1.05)

$SPI = 40/50 = 0.80$; $CPI = 40/38 \approx 1.05$. Comparing AC (38) with PV (50) suggests "under budget", but that is mostly because less work was done.

Q3.

10 %

Only approval (5 %) and PO (5 %) are achieved: 10 %. Claiming more before FAT is progress inflation.

Risks, changes and the monthly report

Q1.

2.4 USD m

$EMV = 0.40 \times 6.0 = 2.4$ m. Because this risk is within MEWRE's control, spending well under 2.4 m on mitigation (early right-of-way work, more supervision) is good value.

Q2.

Plan and fund to P80 (June): a P50 date is missed half the time, and overruns are larger than underruns

Owners usually budget contingency and commit dates at P80. Going back for more time or money through Finance and the Audit Bureau costs months and credibility.

Q3.

Stop it informally proceeding: log it as a change request, have it assessed for cost and time, decide at the right authority, and instruct formally only if approved.

Change control: identify, log, assess, decide, instruct formally, re-baseline. Never let a variation run on an oral request.

PART 4 OF 4

Contracts, money, the ministry, and rehearsal

By the end you can: Explain FIDIC basics, bonds and delay damages, IPPs and PPAs, LCOE and NPV, how Kuwait tenders and how the ministry works, then rehearse with the final exam.

6 lessons · about 31 minutes

Lesson 24 · Part 4 · full lesson · about 7 min

Contract strategies and the FIDIC books

Why you care at MEWRE. Almost every contract you will touch at MEWRE is built on a FIDIC book with Kuwaiti particular conditions, and the first question in any meeting is "who designs, and who carries the risk?". Know the three main books and the three roles and you can read any contract's first page with confidence.

The picture first

You want a new house. You have three ways to get it:

- **Design-bid-build.** You pay an architect to draw every detail, then invite builders to bid on the drawings. The builder builds what is drawn. If the drawings are wrong, that is **your** problem.
- **Design-build.** You write a brief ("four bedrooms, energy class A, ready in 14 months") and one builder designs and builds it. Design errors are now the **builder's** problem.
- **Turnkey.** You agree a fixed price and a handover date, and you only come back to collect the keys. The builder carries almost everything, including surprises in the ground, and charges you for that comfort.

The more risk you push to the builder, the higher the price and the less you get to interfere. Where the house analogy breaks: on a turnkey power plant you still write a detailed performance specification, and you still test the plant hard before you accept it.

What is actually happening

A **contracting strategy** answers three questions: who designs, how many contracts (and who manages the gaps between them), and who finances and owns the asset. **FIDIC** (the International Federation of Consulting Engineers) publishes standard forms that match each answer. The 2017 editions are the ones you will meet:

- **Red Book** (Construction): the Employer designs, the Contractor builds. Paid by **measured quantities** (Bill of Quantities times rates). This is design-bid-build.
- **Yellow Book** (Plant and Design-Build): the Contractor designs to the **Employer's Requirements** and is paid a **lump sum**. This is design-build.
- **Silver Book** (EPC/Turnkey): the Contractor designs, procures and builds, for a **fixed lump sum**, and carries most of the risk, including **unforeseeable ground conditions**. This is EPC/turnkey, and the base of most IPP construction contracts.

	Red	Yellow	Silver
Who designs	Employer	Contractor, to the Employer's Requirements	Contractor
How it is paid	Measured quantities (BoQ times rates)	Lump sum	Fixed lump sum
Ground risk (4.12)	Employer	Employer	Contractor
Who administers	Engineer	Engineer	Employer's Representative
Delivery model	Design-bid-build	Design-build	EPC/turnkey

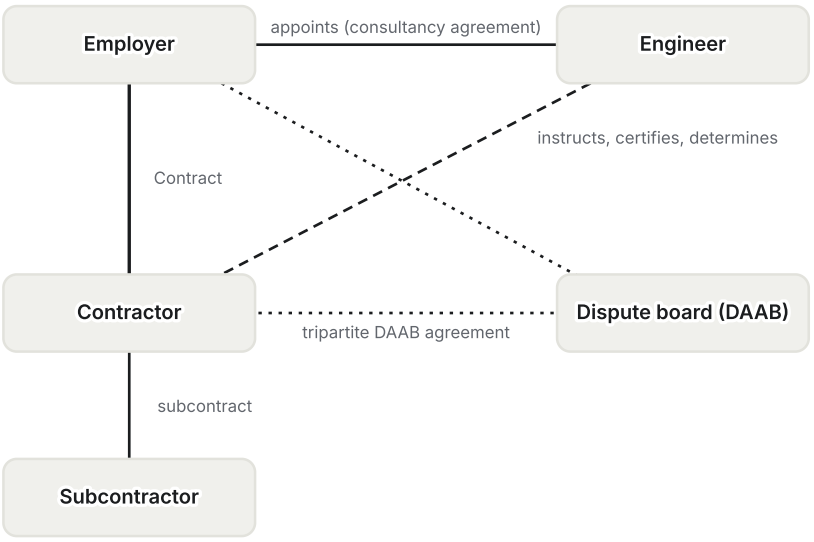
Read the table left to right: design, then price certainty, then ground risk move to the Contractor.

Each book has **General Conditions** (the standard text) and **Particular Conditions** (Part A: contract data; Part B: the Employer's amendments). Always read Part B before quoting a clause.

Three roles to keep straight:

- **Employer**: the owner who pays. On MEWRE's own substations or BESS, MEWRE is the Employer. On an IPP, the project company (SPV) is the Employer under its EPC contract; MEWRE is the **offtaker** under the PPA.
- **Contractor**: builds (and on Yellow and Silver, designs). Subcontractors have no contract with the Employer.
- **Engineer** (Red and Yellow): a professional firm appointed and paid by the Employer that instructs, certifies payments and, under **Sub-Clause 3.7**, agrees or determines claims **neutrally**. On Silver there is no Engineer: an **Employer's Representative** acts for the Employer.

A fourth body, the **DAAB** (Dispute Avoidance/Adjudication Board), is appointed jointly to help avoid and decide disputes.

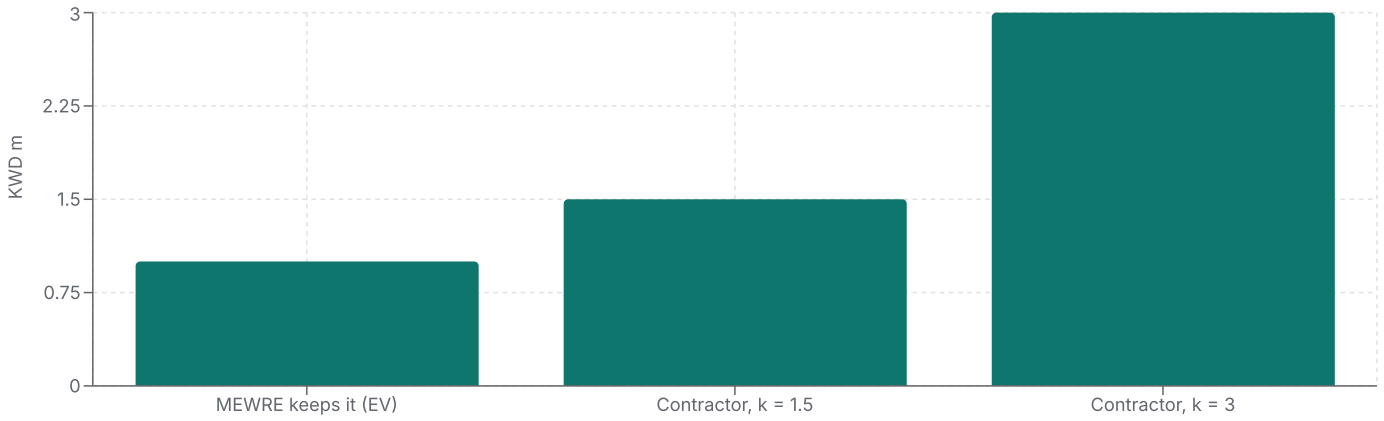


Who has a contract with whom: the Engineer is paid by the Employer but must determine claims neutrally, the DAAB is appointed jointly, and subcontractors sit only behind the Contractor.

Why it matters at MEWRE

The guiding rule is: put a risk with the party that can **control** it; if nobody controls it, with the party that can **bear** it most cheaply. Pushing a risk onto someone who can do neither does not remove it. It comes back as a **risk premium** in the price ($RP = k \cdot EV$, with k typically 1.5 for well-understood risks up to 3 or more for risks the contractor cannot control), plus a dispute when the event happens.

What one risk costs MEWRE: keep it, or pay a contractor to carry it



Same risk (20 % chance of KWD 5 m, EV = KWD 1 m): the contractor charges 1.5 to 3 times the expected cost, the most for risks it cannot control.

Two habits that cause trouble: MEWRE staff making the Engineer's determinations themselves on a Red or Yellow contract (it breaks the neutrality the contract promises), and reviewing every drawing on a Silver contract (it quietly turns turnkey into a hybrid and invites claims).

Contract risk allocation matrix

Who bears each risk under FIDIC 2017 books and a typical IPP/PPA structure — click a cell

Risk	FIDIC Red 2017 Employer designs; Contractor builds (re-measured)	FIDIC Yellow 2017 Contractor designs & builds (lump sum)	FIDIC Silver 2017 EPC/Turnkey — single-point, higher price certainty	IPP / PPA Offtaker ↔ SPV (project company) ↔ EPC
Unforeseeable ground conditions	Employer Sub-Clause 4.12	Employer Sub-Clause 4.12	Contractor Sub-Clause 4.12	SPV → EPC
Design errors	Employer	Contractor Sub-Clauses 4.1 & 5.1	Contractor Sub-Clauses 4.1 & 5.1	SPV → EPC
Errors in Employer's Requirements / specification	Employer	Shared Sub-Clause 1.9	Contractor	SPV / EPC
FX and inflation	Contractor (unless indexed) Sub-Clause 13.7	Contractor (unless indexed) Sub-Clause 13.7	Contractor	Offtaker ↔ SPV
Change in law	Employer Sub-Clause 13.6	Employer Sub-Clause 13.6	Employer Sub-Clause 13.6	Offtaker (mostly)
Grid connection delay	Employer Sub-Clause 8.5	Employer Sub-Clause 8.5	Employer (if its scope)	Offtaker
Force majeure / exceptional events	Shared Clause 18	Shared Clause 18	Shared Clause 18	Shared
Performance shortfall (output / PR)	Shared	Contractor Sub-Clause 4.1, Clauses 9 & 12	Contractor Sub-Clause 4.1, Clauses 9 & 12	SPV → EPC/O&M
Permits and approvals	Shared Sub-Clauses 1.13 & 2.2	Shared Sub-Clauses 1.13 & 2.2	Shared Sub-Clauses 1.13 & 2.2	Shared
Interface risk (multiple packages)	Employer Sub-Clause 4.6	Shared	Contractor	SPV

Click a cell to see who bears the risk, why, and the governing clause.

Allocation under the General Conditions only — Particular Conditions frequently amend them, and Gulf procurers often shift risk to the contractor. Always read the actual contract. IPP column describes common market practice, not a specific Kuwait PPA.

Click across a row: the same risk (ground, design, change in law, grid delay) moves from Employer to Contractor as you go from Red to Silver to an IPP EPC.

Numbers you will hear

- **FIDIC 2017** editions (reprinted with amendments in 2022): Red, Yellow, Silver are the main three; Gold (design-build-operate), White (consultant services, used for the Owner's Engineer), Green (short form).
- **Ground risk (Sub-Clause 4.12)**: Employer on Red and Yellow (time and cost); Contractor on Silver ("Unforeseeable Difficulties", price not adjusted).
- **Risk premium**: $RP = k \cdot EV$, k about 1.5 to 3+. A risk with a 20 % chance of costing KWD 5 m (EV = KWD 1 m) can add KWD 1.5 to 3 m to a price.
- **IPP chain**: MEWRE (offtaker) to SPV to EPC and O&M contractors, with the EPC usually on a Silver base, lump sum and date-certain.

- **MEWRE contracts**: often FIDIC-based with Kuwaiti particular conditions (Track 0, m11).

Say it at MEWRE

- Which FIDIC book is this based on, and what does Part B change in the risk allocation?
- Who carries ground risk here, and what site data did we give bidders to price it?
- On this Silver contract, are we reviewing drawings for compliance only, or are we approving design? We should be clear, because approval shifts risk back to us.
- Who is the Engineer, and how do we protect the neutrality of the 3.7 determinations?
- On the IPP, our leverage is the PPA and the grid connection, not the EPC: what does the PPA say about this risk?

The classic trap.

Thinking turnkey means "no risk for us". A Silver contract transfers risk at a price, and the Employer still carries what it controls: its grid connection, permits, late decisions and any changes. And on an IPP, MEWRE is not the Employer at all; it cannot instruct the EPC contractor and acts only through the PPA.

Check yourself (answers at the end of Part 4)

Q1.

MEWRE's consultant has fully designed a 132 kV substation extension and wants bidders to price a Bill of Quantities. Which FIDIC book fits?

A. Red Book (Construction)

B. Yellow Book (Plant and Design-Build)

C. Silver Book (EPC/Turnkey)

D. White Book (Client/Consultant)

Q2.

On a FIDIC Red Book contract where MEWRE is the Employer, a contractor's claim needs a determination under Sub-Clause 3.7. Who should make it?

A. The Engineer, acting neutrally, after consulting both Parties

B. MEWRE's project manager, because MEWRE pays

C. The DAAB, immediately

D. The Contractor's own planner

Q3.

A contractor estimates a 25 % chance that a grid-delay risk costs it KWD 4 m. It cannot control the risk and uses a loading factor $k = 2$. What risk premium (KWD m) does it add to its price if the contract makes it carry this risk?

Your answer: _____ KWD m

Lesson 25 · Part 4 · quick lesson · about 4 min

Bonds, guarantees and liquidated damages

Why you care at MEWRE. Bonds and liquidated damages are how MEWRE gets paid when a contractor fails, without first winning a two-year dispute. You will track bond expiries, check advance recoveries in payment certificates and, one day, have to decide whether to call a bond or deduct LDs.

Quick lesson: the essentials only. The full version, with more drawings and worked numbers, is on the website under Trial week.

The picture first

When you rent a flat you pay a **deposit**. You do not have to prove anything in court to keep part of it if the tenant wrecks the kitchen; the money is already in your hands. A **bank guarantee** is a deposit held by a bank instead of by you.

Now imagine the lease says: "for every day you stay past the end date, you pay KWD 50". Nobody argues afterwards about what the extra day really cost; the number was agreed up front. That is a **liquidated damage** (LD).

Where it breaks: an on-demand bond is easier to call than a deposit is to keep, and the Employer can still be liable later if it called without being entitled.

What is actually happening

Securities turn a contractor's promise into a claim on a creditworthy bank or parent company:

- **Performance security** (performance bond): commonly **10 %** of the contract price, delivered within 28 days of the Letter of Acceptance (FIDIC 2017, 4.2), kept until the **Performance Certificate**. Covers contractor default.
- **Advance payment guarantee (APG)**: equal to the advance (commonly 10 to 20 % of the price). The advance is recovered from each payment certificate (FIDIC default: **25 %** of each IPC once certified payments exceed 10 % of the price), and the APG reduces as it is recovered.
- **Retention**: money held back from each payment, for example **10 % of each IPC up to 5 %** of the price; **half released at Taking-Over**, half at the end of the defects period.
- Most Gulf bank guarantees are **on-demand**: the bank pays a compliant written demand without checking whether the contractor really defaulted. The contract still says the Employer may claim only what it is **entitled** to, and must indemnify the contractor for an unjustified call.

Which security is live when (FIDIC 2017 defaults)

1. **1 Bid** — Bid bond: covers a bidder withdrawing its bid
2. **2 Letter of Acceptance + 28 days** — Performance security, about 10 % of the price, delivered
3. **3 Advance paid** — Against an APG of the same amount
4. **4 Each payment certificate** — 25 % recovers the advance (APG shrinks); 10 % retention held, up to 5 % of the price
5. **5 Taking-Over** — Delay LDs stop; half the retention released
6. **6 Performance Certificate** — Rest of the retention released; performance security returned

Track every expiry date against this line: a bond that expires before the Performance Certificate protects nothing when you need it.

Numbers you will hear

- **Performance security**: about 10 % of the contract price. Kuwait public tenders: final guarantee of at least 10 % of contract value (Law 49/2016 Art. 65, as reported by a single academic source: confirm the article).
- **Retention**: e.g. 10 % per IPC up to 5 % of price; half at Taking-Over, half after the defects period. **Advance recovery**: 25 % of each IPC (FIDIC default).
- **Delay LD rate**: about **0.03 to 0.1 % of the contract price per day** for PV EPCs.
- **Caps**: delay LDs 10 to 20 %, performance LDs 10 to 15 %, aggregate 20 to 30 %, overall liability often 100 % of the price.
- **Kuwaiti law** on agreed damages (whether a court may reduce them) is not verified in Track 0: ask legal affairs before setting or enforcing LD rates.

Say it at MEWRE

- When does each bond expire, and is it at least 28 days after the expected Performance Certificate?
- Before we call this bond, what exact amount are we entitled to under the contract, and is it documented?
- How was the delay LD rate derived? We should keep the calculation on file to show it is a genuine pre-estimate.
- Are delay LDs applied after the EOT we have granted, and have we served our Notice of Claim?
- What happens when the delay cap is reached: do we have a termination right?

The classic trap.

Thinking higher LDs are always better for MEWRE. A rate far above the real loss makes every bidder add contingency to its price and invites a court to reduce it; a rate far below it lets a contractor rationally pay LDs rather than accelerate. The goal is an LD close to the genuine loss, with a sensible cap and a termination right.

Check yourself (answers at the end of Part 4)

Q1.

An EPC contract worth KWD 20 m has delay LDs of 0.1 % of the price per day, capped at 10 % of the price. The plant is taken over 130 days after the extended completion date. What delay LDs (KWD m) are payable?

Your answer: _____ KWD m

Q2.

What does an advance payment guarantee protect MEWRE against?

A. The contractor failing to repay the advance, for example if the contract ends before it is recovered

B. Defects found after Taking-Over

C. The plant failing its performance ratio test

D. A bidder withdrawing its bid

Q3.

You are the PM. A contractor is 40 days late on a MEWRE substation, with an EOT claim for 25 of those days still undetermined. A colleague wants to call the full 10 % performance bond today "to send a message". What do you do?

A. Ask the Engineer to determine the EOT, serve MEWRE's Notice of Claim for delay damages, and recover the entitled LDs through payment certificates; keep the bond for a real default.

B. Call the full bond now; it is on-demand so the bank must pay.

C. Deduct 40 days of LDs from the next payment without any notice.

D. Do nothing until the end of the project.

IPPs and PPAs: when a private company builds and sells power

Why you care at MEWRE. Kuwait's large solar plants, such as Shagaya Phase III Zones 1 and 2, are procured as IPPs by KAPP, with MEWRE signing a 30-year PPA as the buyer. As a renewable PM you sit on the offtaker side: you will not build these plants, but MEWRE pays for them, connects them and lives with their contracts for decades.

The picture first

A town needs a bakery but does not want to run one. It signs a 30-year deal with a baker: "Build the bakery with your own money, and we will buy all your bread at a fixed price per loaf." The baker borrows most of the money from a bank. The bank lends only because the town's purchase contract guarantees a steady income.

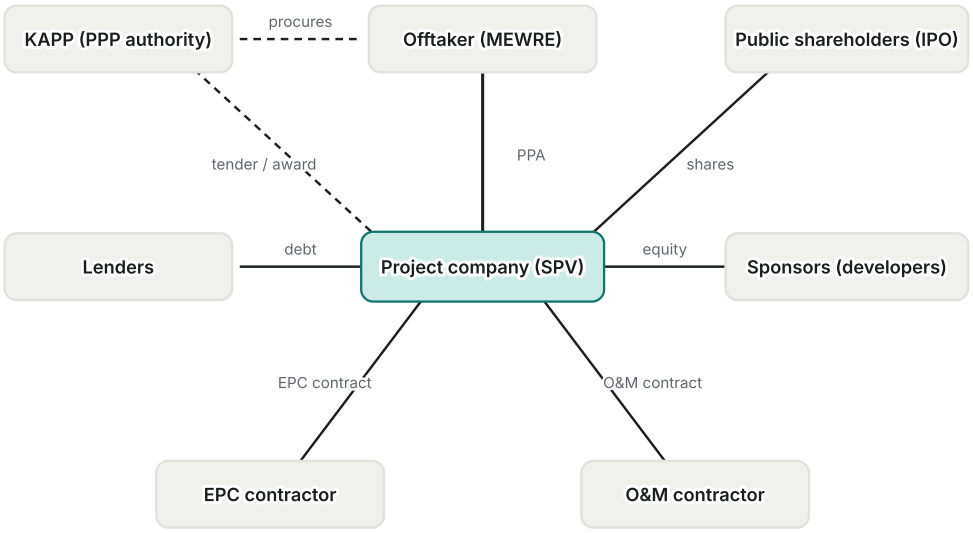
If the town closes the road to the bakery for a week, the bread goes stale. The deal says the town still pays for that week's bread. That is the logic of an IPP and its PPA.

Where it breaks: electricity cannot wait on a shelf, so the payment rules for energy that was not taken matter far more than for bread.

What is actually happening

An **Independent Power Producer (IPP)** is a privately financed company that builds and runs a plant and sells its output to a public buyer.

- **SPV** (special purpose vehicle, the project company): a new company created only for this plant. It owns the plant and signs all the contracts.
- **Sponsors**: the developers who put in **equity** and own the SPV (in Kuwait, alongside the state, see below).
- **Lenders**: banks that provide most of the money, typically **70 to 80 %** of capex, repaid only from the SPV's revenue (**non-recourse project finance**). They read every PPA clause asking "does this protect the debt?".
- **Offtaker**: the buyer. In Kuwait, **MEWRE**.
- **PPA** (Power Purchase Agreement): the 30-year sales contract. It sets the **tariff** (price per kWh), the commercial operation date (COD), grid connection, dispatch, metering, and termination.
- **EPC and O&M contractors**: the SPV hires them to build and operate, on terms back-to-back with the PPA.



Follow the money: MEWRE pays the tariff to the SPV, which pays its EPC and O&M contractors, services the lenders and then pays dividends to sponsors and shareholders. KAPP procures and holds the state's shares.

One year of money through the SPV, in order

1. **1 MEWRE pays the tariff**
   Energy delivered (plus any deemed energy) times the PPA tariff
2. **2 SPV pays operating costs**
   O&M contractor, insurance and other opex
3. **3 SPV pays the lenders**
   Interest plus principal on the 70 to 80 % of capex that was borrowed
4. **4 What is left goes to equity**
   Dividends to sponsors and the state's shareholding via KAPP

Lenders are paid before shareholders, which is why they read every PPA clause that could cut the first box.

Key payment ideas:

- **Energy-only tariff**: one price per MWh delivered. The norm for solar; the SPV carries sunshine, soiling and performance risk.
- **Take-or-pay / must-take**: the offtaker must accept all the energy delivered (or pay for a minimum).
- **Deemed energy**: if the plant could have generated but did not because of the offtaker's side (grid outage, curtailment beyond an annual allowance, late grid connection), the offtaker **pays as if the energy had been delivered**. MEWRE controls the grid, so it carries that risk; otherwise bidders would price it into 30 years of tariff.

What happened to the energy	Who loses	Does MEWRE pay?
Delivered to the grid	Nobody	Yes, at the tariff
Not generated: low sun, dust, plant fault	SPV	No
Curtailed, within the annual allowance	SPV	No
Curtailed beyond the allowance, or a MEWRE grid outage	MEWRE	Yes, as deemed energy
MEWRE's grid connection late	MEWRE	Deemed energy or a later COD, per the PPA wording

The dividing line is control: the SPV carries the sun and its own plant, MEWRE carries its grid.

Why it matters at MEWRE

KAPP runs the competition, but **MEWRE carries the 30-year obligations**: paying, connecting, dispatching and absorbing interface risk. For Shagaya Phase III Zone 1, the developer builds the 400 kV substation, while the Shagaya to Al-Wafra 400 kV line is MEWRE's own contract. If MEWRE's line is late, the PPA may deem the plant complete and MEWRE pays deemed energy, or COD is extended, depending on the PPA wording. Every RFP clause on grid delay, curtailment and deemed energy is a contingent liability on the ministry's budget.

Worked example (from Track 0, with an **assumed** tariff of 8 fils/kWh because the real one is not published): a MEWRE-side 400 kV outage prevents 18,000 MWh. Deemed payment $= 18{,}000 \times 1{,}000 \times 0.008 =$ KWD 144,000, for energy the ministry never received, and it still burns the fuel that energy would have displaced.

Numbers you will hear

- **KAPP**: Kuwait Authority for Partnership Projects, set up by **PPP Law 116/2014** (Executive Regulations **Decree 78/2015**), attached to the Minister of Finance; the **Higher Committee** (chaired by the Minister of Finance) approves projects and the selected investor. (Sourced, Track 0 m09/m11.)
- **Solar PPA term**: **30 years** with MEWRE as offtaker; Zone 1 is **1,100 MW**, Zone 2 **500 MW**. (Sourced.)
- **Shareholding** for PPPs above KD 60 m: 50 % IPO to Kuwaiti citizens, 6 to 24 % public entities, at least 26 % private (secondary legal guide; sources conflict, verify). In practice on IWPPs: developer **40 %**, state via KAPP **60 %**; KAPP warehouses the IPO shares until operation.
- **Not confirmed** in Track 0: PPA currency, government support, termination regime and the exact deemed-energy terms. Read the signed documents.
- **Lenders**: typically 70 to 80 % of capex as non-recourse debt (international practice).

Say it at MEWRE

- Under this PPA, what triggers deemed energy, and what is the annual curtailment allowance before MEWRE pays?
- Who builds which part of the grid connection, and what happens to COD and payments if our part is late?
- Which ministry-side conditions precedent (land, grid, permits) must we deliver before financial close, and who tracks them?
- Is the PPA term counted from COD or from a fixed date?
- Where do the lenders' requirements conflict with our positions on termination or step-in, and who is resolving that?

The classic trap.

Thinking the PPA is KAPP's document and MEWRE just pays the bills. KAPP procures, but MEWRE signs as offtaker and carries the grid, dispatch and curtailment obligations for 30 years. Promising grid dates MEWRE does not control is the fastest way to pay for energy it never receives.

Check yourself (answers at the end of Part 4)

Q1.

Who is the offtaker in a Kuwaiti solar IPP such as Shagaya Phase III Zone 1?

A. MEWRE

B. KAPP

C. The SPV's lenders

D. The EPC contractor

Q2.

A MEWRE transmission outage, outside the PPA allowances, prevents 25,000 MWh that the plant could have exported. With an assumed tariff of 8 fils/kWh, what deemed-energy payment (KWD) does MEWRE owe?

Your answer: _____ KWD

Q3.

Why does a solar PPA usually make the offtaker, not the SPV, carry the risk of grid unavailability?

A. MEWRE controls the grid; if the SPV carried the risk it would add a premium to 30 years of tariff and lenders would lend less

B. Because solar plants cannot measure their own output

C. Because the SPV owns the transmission lines

D. Because lenders are paid by MEWRE directly

Lesson 27 · Part 4 · quick lesson · about 4 min

Money over time: NPV, IRR and LCOE

Why you care at MEWRE. Consultants, bidders and the Ministry of Finance will throw NPV, IRR and LCOE at you from the first week, and every tariff MEWRE signs is a 30-year stream of payments. If you can say what drives these numbers, you can challenge a business case instead of just receiving it.

Quick lesson: the essentials only. The full version, with more drawings and worked numbers, is on the website under Trial week.

The picture first

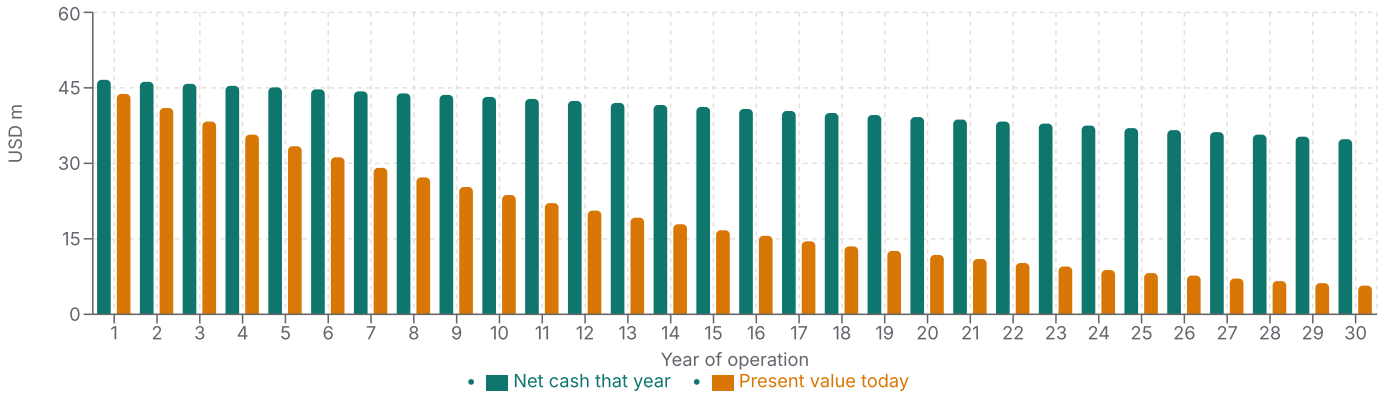
Would you rather have KWD 1,000 today or KWD 1,000 in ten years? Today, obviously: you could put it in a savings account and have more in ten years. So money in the future is worth **less** than money now.

Going the other way: how much would you need to put in the bank today to have KWD 1,000 in ten years? At 6.2 % a year, about KWD 548. That KWD 548 is the **present value** of KWD 1,000 received in year 10. Turning future money into today's money is called **discounting**, the savings account in reverse.

What is actually happening

- **Time value of money**: a dinar today beats a dinar later because it can earn a return. The **discount rate** r is that return (for a project, usually its cost of capital, the **WACC**).
- **Discount factor** for year t: $1/(1+r)^t$. At 6.2 %: year 10 is 0.548, year 20 is 0.300, year 30 is 0.165. A payment in year 30 counts for only a sixth of its face value today.

Track 0 base case (flat tariff 1.80 US¢/kWh): each year's cash, and its value today at 6.2 %



The plant earns roughly the same cash every year (about USD 47 m falling to 35 m), but the discounted bars shrink to about USD 6 m by year 30: distant money counts for little.

✳ **Numbers you will hear**

- **Track 0 base case** (illustrative, not Kuwait bid data): 1,100 MWac PV, capex USD 560 m, 3,003 GWh/yr, WACC 6.2 %: **LCOE about 1.76 US¢/kWh** (about 5.4 fils/kWh at 1 US¢ = 3.08 fils).
- **Gulf record tariffs** (reported, verify): MBR 5 about 1.69 (2019), Al Dhafra about 1.35 (2020), Sudair about 1.24 and Al Shuaibah about 1.04 US¢/kWh (2021). Kuwait's Zone 1 tariff is not published.
- **Discount factors at 6.2 %**: year 10 = 0.548, year 20 = 0.300, year 30 = 0.165.
- **Quick screening LCOE** $\approx (I_0 \times \mathrm{CRF} + O)/E$; CRF(6.2 %, 30 yr) = 0.0742. It ignores degradation and opex growth and understates by about 7 % in the base case.
- **Base case returns**: project IRR 6.45 %, equity IRR 8.97 % (with 80 % debt).

Say it at MEWRE

- Is this LCOE real or nominal, in AC or DC terms, and at what price date?
- Which discount rate did you use, and how does the answer change at plus or minus one point?
- What energy yield is behind this, P50 or P90, and what degradation?
- Does the capex include the grid connection and owner's costs, or are they outside the boundary?
- Show me the tornado: which three assumptions move this number most?

The classic trap.

Comparing a tariff or LCOE without checking its basis. A real LCOE and a nominal tariff, or a P50 yield and a P90 yield, can differ by 10 to 20 % on the same plant. Another classic: dividing discounted costs by undiscounted energy, which makes LCOE look far too low. Both costs and energy must be discounted.

Check yourself (answers at the end of Part 4)

Q1.

What is the present value (KWD thousand) of KWD 1,000,000 received in year 10, at a discount rate of 6.2 %?

Your answer: _____ KWD thousand

Q2.

A consultant says a project's IRR is 9 %. What does that mean?

A. The discount rate at which the project's NPV is exactly zero is 9 %

B. The project's NPV is 9 % of capex

C. The project returns 9 % of its cost every year in cash

D. The project's LCOE is 9 % below the tariff

Q3.

Screening estimate: a PV plant costs USD 100 m, the capital recovery factor is 0.08, opex is USD 2 m/yr and it produces 500 GWh/yr. Using LCOE $\approx (I_0 \times \mathrm{CRF} + O)/E$, what is the LCOE in US¢/kWh?

Your answer: _____ US¢/kWh

Lesson 28 · Part 4 · quick lesson · about 4 min

How Kuwait tenders a project

Why you care at MEWRE. Before anything gets built, it has to get through CAPT or KAPP, the State Audit Bureau and a series of committees, and this is where most Kuwaiti project time is lost. Know which route a purchase takes and what each gate needs, and you can plan approvals instead of waiting on them.

Quick lesson: the essentials only. The full version, with more drawings and worked numbers, is on the website under Trial week.

The picture first

Think of a household where one person knows what the family needs (the cook) and another holds the purse (the treasurer). The cook writes the shopping list and checks the quality of what arrives. The treasurer runs the purchase so that no shop is favoured. An auditor checks the receipts before big purchases are paid. It is slower than the cook just buying, but it is harder to cheat.

In Kuwait, MEWRE is the cook, **CAPT** (or **KAPP** for partnership projects) is the treasurer, and the **State Audit Bureau** is the auditor.

What is actually happening

Conventional tenders (CAPT). The **Public Tenders Law 49/2016** created the **Central Agency for Public Tenders (CAPT)** regime. CAPT oversees tenders above **KD 75,000** (Art. 19). The split of work:

- **MEWRE** (your sector) writes the need, budget certificate, specifications, Bill of Quantities and evaluation criteria, and runs the **technical evaluation**.
- **CAPT** owns the procedure: issues the tender, answers clarifications through addenda, receives bids with a **bid bond**, opens them publicly, and makes the award decision. The usual rule is the **lowest price once technical compliance is established**.
- The **State Audit Bureau** gives **prior approval** before signature for contracts above a reported **KD 100,000** (Law

30/1964). Signing without it is a violation.

- The winner provides a **final guarantee** of at least 10 % of contract value.

A MEWRE tender through CAPT (above the threshold)

1. **1 Need and budget certificate** — MEWRE sector, with cost estimate and annual aggregate for the same items →
2. **2 Tender documents** — MEWRE writes specs, BOQ, conditions (often FIDIC-based), evaluation criteria →
3. **3 CAPT issues the tender** — Clarifications answered in writing to all bidders; addenda →
4. **4 Bids and public opening** — Sealed bids with bid bond; prices read out →
5. **5 Technical evaluation** — MEWRE committee, often with the Owner's Engineer →
6. **6 CAPT award decision** — Lowest technically compliant price, after preferences →
7. **7 State Audit Bureau prior review** — Reported threshold KD 100,000; may approve with conditions →
8. **8 Contract and final guarantee** — At least 10 % of contract value (reported)

Each box is a place where a file can be returned with observations; most delay comes from returns and missed committee dates, not from the work itself.

Numbers you will hear

- **CAPT threshold**: KD 75,000 (Law 49/2016 Art. 19; medium to high confidence).
- **Law 94/2026**: direct contracting up to **KD 1 m per financial year** for the same items, effective about 27 Dec 2026; 30 % local products, 30 % local subcontracting; agents banned (medium confidence: single law firm plus cabinet-stage coverage; read the Arabic text).
- **Audit Bureau prior review**: contracts above **KD 100,000** (reported by a minister, unverified); one period counted **28 contracts worth KD 63 m** signed without approval.
- **Final guarantee**: at least 10 % of contract value (Art. 65, single academic source). Bid guarantee percentage: unknown.
- **Approval scheduling**: use three-point estimates; $P80 \approx \mu + 0.84\sigma$.

Say it at MEWRE

- Which regime does this purchase fall under, CAPT, direct, or KAPP, and is the legal basis recorded in the file?
- What is the annual total for the same items this financial year? We must not split it.
- Which version of the tenders law applies, given the announcement date?
- When is the next Higher Committee or Audit Bureau cut-off, and what would make them return the file?
- When do the bids' validity periods expire compared with our P80 approval date?

The classic trap.

Thinking the new KD 1 m limit means small projects skip all oversight. It removes only the CAPT step, per financial year for the same items; Audit Bureau review and internal approvals still apply, splitting is still a violation, and it is not in force until about 27 December 2026. Another trap: assuming KAPP projects follow CAPT rules; they follow the PPP law and their own RFPs.

Check yourself (answers at the end of Part 4)

Q1.

Which body procures a solar IPP such as Shagaya Phase III Zone 2?

A. KAPP, under PPP Law 116/2014

B. CAPT, under Law 49/2016

C. The State Audit Bureau

D. MEWRE directly, as a direct order

Q2.

After Law 94/2026 takes effect, MEWRE has already placed direct orders this financial year for the same PV spare parts of KD 300,000 and KD 450,000. What is the largest further direct order for the same parts (KD) it can place without CAPT?

Your answer: _____ KD

Q3.

You are the PM. A KD 4 m grid contract has been awarded by CAPT, and the contractor offers to mobilise next week "to save time" while the Audit Bureau review is still pending. What do you do?

A. Decline: no work or commitment before Audit Bureau approval. Pre-clear the file with the liaison, answer observations fast, and prepare mobilisation documents in parallel.

B. Accept and regularise the contract later.

C. Let the contractor mobilise at its own risk without any written instruction.

D. Re-tender to avoid the Audit Bureau.

Lesson 29 · Part 4 · full lesson · about 6 min

How a ministry works, and writing for it

Why you care at MEWRE. At MEWRE the critical path of a renewable project usually runs through memos, signatures and committees, not engineering. A PM who writes a one-page memo the Undersecretary can approve in two minutes moves projects that technically stronger but institutionally naive managers cannot.

The picture first

Picture a relay race where the baton is a **piece of paper**. Each runner (section head, director, Assistant Undersecretary, Undersecretary) reads it, writes a short note on it and passes it on. Nothing counts until the baton crosses the line with a signature. A great idea said in a corridor is a runner without a baton.

The structure feels slow to an engineer from the private sector. It is built for **accountability, not speed**: public money, assets committed for 25 to 30 years, and every decision open to later review by the State Audit Bureau, the National Assembly and the courts. Once you see that, you stop fighting the system and start writing paper that moves through it fast.

What is actually happening

The chain. Minister (الوزير), reached through the **Undersecretary** (وكيل الوزارة), who runs the ministry day to day and signs significant letters and memos. Below are **Assistant Undersecretaries** (الوكيل المساعد), each heading a sector; yours is your real internal client and budget owner. Then department directors, controllers and section heads, who review, draft and execute. A renewable IPP touches several sectors at once (planning, transmission, renewables, finance, legal); your job is to turn many partial authorities into one decision.

Formal vs practical authority. Formal authority is a signature under the **delegation of authority** (تفويض الصلاحيات). Practical authority belongs to the **drafters** (whoever writes the memo frames the options), the **gatekeepers** (office managers, committee secretaries, legal and financial reviewers) and trusted advisers.

The annotation is the decision. Each level writes a تأشيرة on the memo: موافق (approved), للعرض (for presentation upward), لاتخاذ اللازم (take the necessary action). A verbal "go ahead" in a meeting is agreement in principle only.

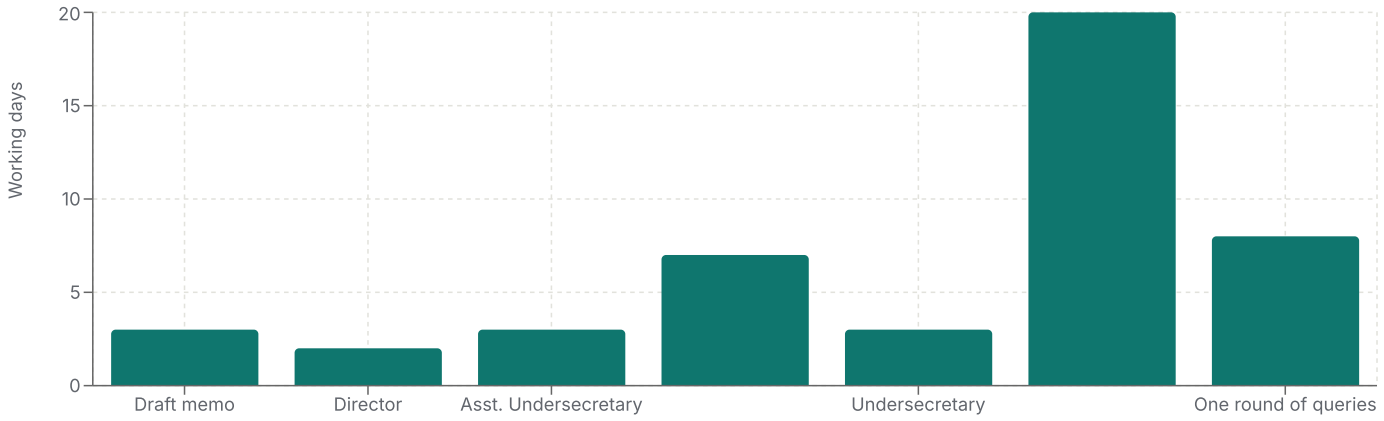
The baton: one memo going up the chain

1. **1 PM or section head drafts** — Frames the options: the drafter's quiet power →
2. **2 Department director** — Reviews, annotates للعرض (for presentation upward) →
3. **3 Assistant Undersecretary** — Your sector's head and budget owner →
4. **4 Financial and Legal Affairs** — Budget certification and legal review, in parallel →
5. **5 Undersecretary** — Annotates موافق: now it is a decision →
6. **6 State Audit Bureau** — Prior review, if the contract is above the reported threshold

Each arrow is a desk where the file can wait; your job is to make every desk's annotation quick and obvious.

How long the relay takes: Track 0's illustrative chain for a KD 450,000 contract extension needs about **46 working days**, and more than half of it is outside the ministry.

Where the 46 working days go (Track 0 illustrative critical path)



The two tall bars on the right are outside your control once the file leaves; a complete file that avoids the query round saves more than chasing every desk inside.

Decision culture, four forces: consensus across sectors (one silent sector can stall a file without ever saying no); **precedent** ("what did we do last time?"); **audit consciousness**; and **trust** (a memo from a PM the Undersecretary trusts is read first). For novel decisions, bring external legitimacy: a standard, a consultant's report, a GCC precedent.

Writing for seniors: BLUF (bottom line up front). The first paragraph says what decision is needed, by when, and what you recommend. If the reader stops there and writes موافق, the memo worked. Everything below is for the reviewers and the file.

The one-page decision memo (about 450 words)

1. **1 Decision required** — The ask, the deadline, your recommendation (2 to 4 lines). Seniors read this first, often only this. →
2. **2 Issue** — The problem in 1 or 2 sentences, with the number that matters →
3. **3 Background** — Only the facts needed to judge, with reference numbers and dates →
4. **4 Options** — 2 to 4 genuine options, always including do-nothing, each with cost, time, risk →
5. **5 Recommendation** — Which option, and the decisive reason →
6. **6 Risks** — Top 2 or 3 and how they are handled →
7. **7 Decision line** — Boxes the signatory can tick or annotate →
8. **8 Attachments** — Analysis, consultant summary, and the next letter drafted and ready to sign

Notice the order: conclusion first, evidence after. It is the reverse of an engineering report, which follows the order you discovered things.

A BLUF opening, as it appears on the memo

ENGLISH	العربية

To: The Undersecretary **Through:** The Assistant Undersecretary, [Sector] **Subject:** Approval to extend the Owner's Engineer contract No. [ ]

1. Decision required (by [date]) Approve a 6-month extension of the Owner's Engineer contract, estimated at KD 450,000, and its submission to the State Audit Bureau for prior review. Recommended.

إلى: السيد/ وكيل الوزارة المحترم **عن طريق:** السيد/ الوكيل المساعد لقطاع [ ] المحترم **الموضوع:** طلب الموافقة على تمديد عقد مهندس المالك رقم [ ]

تحية طيبة وبعد،،،

أولاً: المطلوب (قبل تاريخ [ ]) نأمل التكرم بالموافقة على تمديد عقد خدمات مهندس المالك لمدة ستة أشهر بقيمة تقديرية قدرها (450,000) د.ك، وإحالته إلى ديوان المحاسبة للرقابة المسبقة. ونوصي بالموافقة.

Formal letters in Arabic follow a fixed frame: reference number and date, addressee with title (السيد/ ... المحترم), subject, تحية طيبة وبعد،،،, a reference to prior correspondence by number and date, numbered points, a clear request with a deadline, the closing formula, signature, attachments and copies. Copy exact forms from your sector's recent outgoing letters. Firmness comes from facts and clause references, never adjectives.

🇰🇼 **Numbers you will hear**

- **Reading speed** for dense official text: about **200 words/min**. A 450-word memo takes about 2 minutes; a 1,400-word one about 7, and gets skimmed or deferred.
- **State Audit Bureau prior review**: contracts of **KD 100,000 and above**, as reported by a minister (verify against the current law and practice).
- **Law No. 94 of 2026** (gazetted 27 September 2026, effective about three months later): direct contracting without CAPT approval up to **KD 1,000,000 per financial year**; local and commission agents banned.
- **Approval chain**: the Track 0 example for a KD 450,000 extension takes about **46 working days**, including 20 for Audit Bureau review and 8 for one round of queries. Budget for it: the paper chain is usually the critical path.

Say it at MEWRE

- Which sectors must concur before this goes up, and has each of them seen the draft?
- Who signs this under the current delegation of authority, and is there a precedent we can cite?
- Here is the one-page memo with the decision in the first paragraph and the letter attached, ready for signature.
- Thank you for the agreement in principle; I will send the memo today so we have it on file.

The classic trap.

Treating a nod in a meeting as a decision, or writing the memo like an engineering report (background first, conclusion on page three). Nothing counts until the annotated memo is on file, and a senior who cannot find the ask in the first paragraph sends it back "for clarification", which costs a week.

Check yourself (answers at the end of Part 4)

Q1.

In a meeting, the Assistant Undersecretary says "fine, go ahead" to your proposal. What do you have?

A. Agreement in principle: the decision exists only when the annotated memo is signed and filed

B. A binding decision you can act on immediately

C. A decision, provided you write the minutes yourself

D. Nothing at all: verbal views are never relevant

Q2.

What must the **first paragraph** of a decision memo to the Undersecretary contain?

A. The decision required, the deadline and your recommendation

B. The project background since tender

C. A summary of the consultant's report

D. A list of everyone who attended the related meetings

Q3.

At about **200 words per minute**, how many minutes does a senior official need to read a **1,400-word** memo carefully?

Your answer: _____ min

Answers: Part 4

Contract strategies and the FIDIC books

Q1.

Red Book (Construction)

Employer design plus measured quantities is the Red Book: design-bid-build, administered by an Engineer. Yellow and Silver both put design on the Contractor; White is for consultant appointments.

Q2.

The Engineer, acting neutrally, after consulting both Parties

Under 3.7 the Engineer "shall not be deemed to act for the Employer" and must determine fairly. If MEWRE staff decide it themselves, the contractor gains a procedural argument. The DAAB comes in only after a Notice of Dissatisfaction.

Q3.

2 KWD m

$EV = 0.25 \times 4 = 1$ KWD m; $RP = k \cdot EV = 2 \times 1 = 2$ KWD m. If MEWRE kept the risk (it controls the grid), it would pay on average only the KWD 1 m expected cost.

Bonds, guarantees and liquidated damages

Q1.

2 KWD m

Daily rate $= 0.001 \times 20 = 0.02$ KWD m. Uncapped: $130 \times 0.02 = 2.6$ KWD m. Cap $= 0.10 \times 20 = 2$ KWD m. Payable $= \min(2.6, 2) = 2$ KWD m.

Q2.

The contractor failing to repay the advance, for example if the contract ends before it is recovered

The APG equals the advance and reduces as the advance is recovered from payment certificates. Defects are covered by retention and the performance security; bid withdrawal by the bid bond; underperformance by performance LDs.

Q3.

Ask the Engineer to determine the EOT, serve MEWRE's Notice of Claim for delay damages, and recover the entitled LDs through payment certificates; keep the bond for a real default.

On-demand does not mean unconditional entitlement. Decide the EOT first, then apply delay damages for the days beyond it through the claim procedure.

IPPs and PPAs: when a private company builds and sells power

Q1.

MEWRE

MEWRE signs the 30-year PPA and buys the output. KAPP procures the project and holds the state's shares; lenders finance the SPV; the EPC contractor builds for the SPV.

Q2.

200000 KWD

$25{,}000\ \mathrm{MWh} \times 1{,}000\ \mathrm{kWh/MWh} \times 0.008\ \mathrm{KWD/kWh} = 200{,}000$ KWD. (8 fils = 0.008 KWD. The tariff is an assumption; the real one is not published.)

Q3.

MEWRE controls the grid; if the SPV carried the risk it would add a premium to 30 years of tariff and lenders would lend less

Risk goes to the party that controls it. The SPV cannot prevent a 400 kV outage or a curtailment order, so deemed energy keeps the tariff low and the debt bankable, at the price of MEWRE having to deliver its grid works on time.

Money over time: NPV, IRR and LCOE

Q1.

548 KWD thousand

$PV = 1{,}000{,}000 / 1.062^{10} = 1{,}000{,}000 / 1.825 \approx 548{,}000$ KWD, i.e. 548 thousand. The discount factor at year 10 is 0.548.

Q2.

The discount rate at which the project's NPV is exactly zero is 9 %

IRR is the rate that makes NPV = 0. If the cost of capital is below 9 %, NPV is positive. It says nothing about size, so between options NPV is the better decision rule.

Q3.

2 US¢/kWh

Annual cost $= 100 \times 0.08 + 2 = 10$ USD m. $10\ \mathrm{USD\ m} / 500\ \mathrm{GWh} = 0.02$ USD/kWh $= 2.0$ US¢/kWh. Quick check: USD m divided by GWh gives USD/kWh.

How Kuwait tenders a project

Q1.

KAPP, under PPP Law 116/2014

PPPs are procured by KAPP with Higher Committee approvals, a different regime from CAPT tenders. The Audit Bureau reviews before PPA signing but does not procure.

Q2.

250000 KD

$1{,}000{,}000 - (300{,}000 + 450{,}000) = 250{,}000$ KD. Anything above goes through CAPT. Orders above the reported KD 100,000 may still need Audit Bureau prior review.

Q3.

Decline: no work or commitment before Audit Bureau approval. Pre-clear the file with the liaison, answer observations fast, and prepare mobilisation documents in parallel.

Prior review is mandatory above the reported threshold. The PM's lever is speed through preparation and follow-up, not skipping the gate.

How a ministry works, and writing for it

Q1.

Agreement in principle: the decision exists only when the annotated memo is signed and filed

Officials carry personal accountability for what they sign, so the rational pattern is to agree in principle in the room and decide on paper after Legal and Finance have commented. Send the complete memo quickly so "later" is short.

Q2.

The decision required, the deadline and your recommendation

Bottom line up front: if the senior reads only that paragraph and annotates موافق, the memo has done its job. Background and analysis go below or in attachments.

Q3.

7 min

$t = W/v = 1{,}400/200 = 7$ minutes. A one-page memo of about 450 words takes about 2 minutes, which is the time seniors actually give.

BEFORE YOU LAND

The trial-week playbook

Read this last. It turns what you learned into how you behave on the day.

Numbers to know cold

Say the number **with its source and confidence**: that habit alone signals seniority. H = official, M = trade press, L = weak source (from Module 0).

Item	Value	Conf.
Peak demand 2024 / 2025 / 2026	17,640 / 17,610 / 17,540 MW	M-H
Installed / reliably available	~21 / ~17 GW	M
Generation mix	~62% gas, ~36% oil, ~2% RE	M-L
Official RE target	15% by 2030	H
Stated MEWRE ambition	30% / 40% / 50% by 2030 / 2040 / 2050	M
Net zero	2060 national; 2050 oil and gas sector	H
Shagaya Phase I	70 MW (50 CSP + 10 PV + 10 wind)	H
Zone 1 PV	1,100 MW; Masdar + Fouad AlGhanim reported selected ~Aug 2026	M
Zone 2 PV	500 MW; bids 26 Jul 2026	M
BESS plan	1–1.5 GW / 4–6 GWh; first 500 MW summer 2027	M-L
Citizen tariff	2 fils/kWh (since 1966)	H
Fuel displaced by solar	≈ 26–37 fils/kWh (LNG–crude)	M
Fiscal year	1 April – 31 March	H
Cost of delay, 1.1 GW plant	≈ KWD 191k per day	M

Re-check every M/L item against the latest MEWRE, KAPP or press source before you quote it on the day.

Who's who in one minute

- **MEWRE:** policy, grid owner, offtaker (30-year PPAs), grid tenders.
- **KAPP:** procures PPP/IPP projects; holds 60% of IWPP project companies.
- **Higher Committee** (chaired by the Minister of Finance): approves the project, its stages and the investor.
- **CAPT:** public tenders above KD 75k (state-funded EPC route).
- **Audit Bureau:** prior review of large contracts (threshold: verify).
- **Fatwa & Legislation:** legal review of contract terms.
- **KISR:** Shagaya Phase I and resource measurement.

IPP chain: request (MEWRE) → feasibility (KAPP) → Higher Committee → RFQ → RFP → preferred bidder → Higher Committee → Audit Bureau → PPA → financial close.

Questions that show you've prepared

Ask two or three a day, at the right moment, and write down the answers. Good questions beat confident statements.

1. Which projects is this team working on this quarter, and which decision is next for each?
2. For Zone 1, what is the next milestone — PPA signature, financial close or construction start?
3. Where does the ministry sit in the approval chain for this project, and what is the file waiting on right now?
4. Which grid-code version applies to the new PV and BESS plants, and who runs the compliance tests?
5. How is BESS being procured — as part of the PV IPPs or as a separate tender?
6. Who is the Owner's Engineer on the current projects, and what do you rely on them for?
7. How do you report status to leadership — monthly report, dashboard, steering committee?
8. What has caused the longest delays on recent projects, and what helped unblock them?
9. Which risks keep the team awake: grid connection, land, approvals, supply chain, heat?
10. How are 400 kV transmission works coordinated with the solar projects' COD dates?
11. Is there a standard template for briefs and letters I should follow?
12. What would a useful contribution from me look like by the end of this week?

Your trial week, day by day

- **Day 1 — listen and map.** Learn names, roles and current projects. Keep notes in minutes format (decision, action, owner, date).
- **Day 2 — offer something small.** Send a short written summary of a meeting within 24 hours. Ask which documents you may read.
- **Day 3 — show judgement.** When something comes up (a delay, a contractor letter), ask what the contract or the approval chain says before giving an opinion.
- **Day 4 — produce.** Hand over one useful page: a pipeline tracker, a risk list or a bilingual brief on a topic they choose.
- **Day 5 — close well.** Thank people by name, summarise what you learned and what you would do in your first 90 days, in writing.

What hurts credibility

- Quoting MWp as if it were MW at the peak hour.
- Calling a G2G or framework deal "contracted" before anything is signed.
- Stating a number without its source or date.
- Criticising past decisions before you understand the approval history behind them.
- Pasting ministry documents or bidder data into public AI tools.
- Promising dates on behalf of other bodies (KAPP, CAPT, Audit Bureau).

Your 60-second introduction

Fill in the brackets and practise it until it sounds natural.

> I'm [name], a systems and energy engineer. For the past [n] years I've worked on [your background — e.g. automation and control of large systems, including battery and hybrid controls]. I've spent the last weeks preparing specifically for MEWRE's renewable programme — the Shagaya and Zone 1–2 PV projects, the BESS plans, and how projects move through KAPP, CAPT and the Audit Bureau. What I bring is [one strength — e.g. systems thinking and turning complex technical issues into clear decisions]. This week I'd like to understand how the team works and leave you with something useful in writing.